3

Physical Models for Long-Range Dependence and Self-Similarity

In this chapter, we present a number of physical models for long-range dependence and self-similarity. What do we mean by "physical models" for long-range dependence? First, models motivated by a real-life (physical) application. Second, though these models lead to long-range dependence, they are formulated by using other principles (than long-range dependence). One of these principles is often "responsible" for the emergence of long-range dependence. For example, the infinite source Poisson model discussed in Section 3.3 below, may model an aggregate traffic of data packets on a given link in the Internet. For this particular model, long-range dependence arises from the assumed heavy tails of workload distribution for individual arrivals. We consider the following models:

- Aggregation of short-range dependent series (Section 3.1)
- Mixture of correlated random walks (Section 3.2)
- Infinite source Poisson model with heavy tails (Section 3.3)
- Power-law shot noise model (Section 3.4)
- Hierarchical model (Section 3.5)
- Regime switching (Section 3.6)
- Elastic collision of particles (Section 3.7)
- Motion of a tagged particle in a simple symmetric exclusion model (Section 3.8)
- Power-law Pólya's urn (Section 3.9)
- Random walk in random scenery (Section 3.10)
- Two-dimensional Ising model (Section 3.11)
- Stochastic heat equation (Section 3.12)
- The Weierstrass function connection (Section 3.13)

Having physical models for long-range dependence is appealing and useful for a number of reasons. For example, in such applications, the use of long-range dependent models is then justified and thus more commonly accepted. In such applications, the parameters of long-range dependence models may carry a physical meaning. For example, in the infinite source Poisson model, the long-range dependence parameter is expressed through the exponent of a heavy-tailed distribution.

3.1 Aggregation of Short-Range Dependent Series

One way long-range dependence arises is through aggregation of short-range dependent (SRD) series. We focus here on aggregation of AR(1) series only, which is one of the simplest examples of SRD series and which has attracted most attention in the literature on aggregation.

Let a be a random variable supported on $(-1, 1)$ and having a distribution function F_a. Suppose that $a^{(j)}$, $j \geq 1$, are i.i.d. copies of a. Let also $\epsilon = \{\epsilon_n\}_{n\in\mathbb{Z}}$ be a sequence of i.i.d. random variables with $\mathbb{E}\epsilon_n = 0$, $\mathbb{E}\epsilon_n^2 = \sigma_\epsilon^2$, and $\epsilon^{(j)} = \{\epsilon_n^{(j)}\}_{n\in\mathbb{Z}}$, $j \geq 1$, be i.i.d. copies of ϵ. Suppose that $a^{(j)}$, $j \geq 1$, and $\epsilon^{(j)}$, $j \geq 1$, are independent.

Consider i.i.d. AR(1) series $Y^{(j)} = \{Y_n^{(j)}\}_{n\in\mathbb{Z}}$ with random parameters $a^{(j)}$ defined by

$$Y_n^{(j)} = a^{(j)} Y_{n-1}^{(j)} + \epsilon_n^{(j)}, \quad n \in \mathbb{Z}. \tag{3.1.1}$$

Define now a series $X^{(N)} = \{X_n^{(N)}\}_{n\in\mathbb{Z}}$ obtained by aggregating $Y^{(j)}$ as follows:

$$X_n^{(N)} = \frac{1}{N^{1/2}} \sum_{j=1}^{N} Y_n^{(j)}, \quad n \in \mathbb{Z}. \tag{3.1.2}$$

For example, many macroeconomic variables (such as aggregate U.S. consumption) can be considered as an aggregation of many micro-variables.

Since the sequences $\{Y_n^{(j)}, n \in \mathbb{Z}\}$, $j = 1, 2, \ldots$ are independent, the covariances and spectral density of $\{X_n^{(N)}, n \in \mathbb{Z}\}$ do not depend on N, so that

$$\gamma_X(h) = \mathrm{Cov}(X_{n+h}^{(N)}, X_n^{(N)}) = \mathrm{Cov}(Y_{n+h}^{(1)}, Y_n^{(1)})$$

and one can denote by $f_X(\lambda)$, $\lambda \in (-\pi, \pi]$, the corresponding spectral density, if it exists.

There are two complementary ways to look at the series $X^{(N)}$.

First, one can consider $X^{(N)}$ conditionally on the values of $a^{(j)}$, $j \geq 1$. Since by (2.3.11), the AR(1) series $Y_n = aY_{n-1} + \epsilon_n$ has a spectral density (conditionally on a) given by $f_{Y|a}(\lambda) = \sigma_\epsilon^2/(2\pi|1 - ae^{-i\lambda}|^2)$ and the spectral density of the sum of independent series is the sum of the corresponding spectral densities, the series $X^{(N)}$ has a spectral density conditionally on $a^{(j)}$, $j \geq 1$, given by

$$f_{N|a}(\lambda) = \frac{\sigma_\epsilon^2}{2\pi N} \sum_{j=1}^{N} \frac{1}{|1 - a^{(j)} e^{-i\lambda}|^2}. \tag{3.1.3}$$

Note that, by the Law of Large Numbers, for each $\lambda \in (-\pi, \pi]$,

$$f_{N|a}(\lambda) \to f_X(\lambda) = \frac{\sigma_\epsilon^2}{2\pi} \mathbb{E} \frac{1}{|1 - ae^{-i\lambda}|^2} = \frac{\sigma_\epsilon^2}{2\pi} \int_{-1}^{1} \frac{F_a(dx)}{|1 - xe^{-i\lambda}|^2}, \tag{3.1.4}$$

a.s. as $N \to \infty$, under the assumption that

$$\int_{-1}^{1} \frac{F_a(dx)}{|1 - xe^{-i\lambda}|^2} < \infty. \tag{3.1.5}$$

Alternatively, one can work in the time domain. Note that, by the independence of $Y^{(j)}$, $j \geq 1$, and (2.3.11),

$$\gamma_X(h) = \mathrm{Cov}(X_{n+h}^{(N)}, X_n^{(N)}) = \mathrm{Cov}(Y_{n+h}^{(1)}, Y_n^{(1)}) = \mathbb{E}\Big(\mathbb{E}(Y_{n+h}^{(1)} Y_n^{(1)} | a^{(1)})\Big)$$
$$= \mathbb{E}\Big(\frac{\sigma_\epsilon^2 a^{|h|}}{1-a^2}\Big) = \sigma_\epsilon^2 \int_{-1}^{1} \frac{x^{|h|} F_a(dx)}{1-x^2}. \tag{3.1.6}$$

Thus, the series $X^{(N)}$ is second-order stationary, if

$$\int_{-1}^{1} \frac{F_a(dx)}{1-x^2} < \infty. \tag{3.1.7}$$

To connect the time and spectral domain perspectives, note that

$$\frac{x^{|h|}}{1-x^2} = \frac{1}{2\pi}\int_{-\pi}^{\pi} \frac{e^{ih\lambda}}{|1-xe^{-i\lambda}|^2} d\lambda$$

(which is the usual relationship between the covariance and spectral density of an AR(1) series; see (2.3.11)). We thus have from (3.1.6), the Fubini theorem and using (3.1.7) that $X^{(N)}$ has covariance

$$\gamma_X(h) = \int_{-\pi}^{\pi} e^{ih\lambda}\Big\{\frac{\sigma_\epsilon^2}{2\pi}\int_{-1}^{1} \frac{F_a(dx)}{|1-xe^{-i\lambda}|^2}\Big\} d\lambda$$

and therefore, under (3.1.7), that $X^{(N)}$ has spectral density

$$f_X(\lambda) = \frac{\sigma_\epsilon^2}{2\pi}\int_{-1}^{1} \frac{F_a(dx)}{|1-xe^{-i\lambda}|^2}. \tag{3.1.8}$$

The series X corresponding to the spectral density $f_X(\lambda)$ in (3.1.4) and (3.1.8) is called the *aggregated series*. The next result shows that the aggregated series X is LRD in the sense (2.1.7) of condition IV if the distribution $F_a(dx)$ is suitably dense around $x = 1$.

Proposition 3.1.1 *Suppose that the distribution $F_a(dx)$ has a density*

$$f_a(x) = \begin{cases} (1-x)^{1-2d} l(1-x)\phi(x), & \text{if } x \in [0,1), \\ 0, & \text{if } x \in (-1,0), \end{cases} \tag{3.1.9}$$

where $d < 1/2$, $l(y)$ is a slowly varying function at $y = 0$ and $\phi(x)$ is a bounded function on $[0,1]$ and continuous at $x = 1$ with $\phi(1) \neq 0$. Then, as $\lambda \to 0$,

$$f_X(\lambda) \sim \lambda^{-2d} l(\lambda) \frac{\sigma_\epsilon^2 \phi(1)}{2\pi}\int_0^\infty \frac{s^{1-2d}}{1+s^2} ds, \tag{3.1.10}$$

where $f_X(\lambda)$ appears in (3.1.4) and (3.1.8).

Proof Observe that, by making the change of variables $x = \cos\lambda - u\sin\lambda$ so that $dx = -(\sin\lambda)du$, $x = 0$ and $x = 1$ become $u = 1/\tan\lambda$ and

$$u = \frac{\cos\lambda - 1}{\sin\lambda} = -\frac{2\sin^2\frac{\lambda}{2}}{2\sin\frac{\lambda}{2}\cos\frac{\lambda}{2}} = -\tan\frac{\lambda}{2}.$$

Then using the identities

$$1 - x = (1-\cos\lambda) + u\sin\lambda = 2\sin^2\frac{\lambda}{2} + u\, 2\sin\frac{\lambda}{2}\cos\frac{\lambda}{2} = 2\sin\frac{\lambda}{2}\Big(\sin\frac{\lambda}{2} + u\cos\frac{\lambda}{2}\Big)$$

and

$$1 - 2x\cos\lambda + x^2 = 1 - 2\cos^2\lambda + 2u\cos\lambda\sin\lambda + \cos^2\lambda - 2u\cos\lambda\sin\lambda + u^2\sin^2\lambda = (\sin\lambda)^2(1+u^2),$$

one gets by (3.1.8),

$$\frac{2\pi}{\sigma_\epsilon^2} f_X(\lambda) = \int_0^1 \frac{(1-x)^{1-2d} l(1-x)\phi(x)}{1-2x\cos\lambda + x^2} dx = \frac{(2\sin\frac{\lambda}{2})^{-2d} l(2\sin\frac{\lambda}{2})}{\cos\frac{\lambda}{2}}$$
$$\times \int_{-\tan\frac{\lambda}{2}}^{\frac{1}{\tan\lambda}} \frac{(\sin\frac{\lambda}{2} + u\cos\frac{\lambda}{2})^{1-2d}}{1+u^2} \frac{l(2\sin\frac{\lambda}{2}(\sin\frac{\lambda}{2} + u\cos\frac{\lambda}{2}))}{l(2\sin\frac{\lambda}{2})}$$
$$\times \phi(\cos\lambda - u\sin\lambda)du =: \frac{(2\sin\frac{\lambda}{2})^{-2d} l(2\sin\frac{\lambda}{2})}{\cos\frac{\lambda}{2}} \int_{\mathbb{R}} f_\lambda(u)du.$$

Focus now on $f_\lambda(u)$. For each $u \in \mathbb{R}$ and as $\lambda \to 0$,

$$f_\lambda(u) \to \frac{u^{1-2d}}{1+u^2}\phi(1)1_{[0,\infty)}(u)$$

and, by using Potter's bound (2.2.1) for the slowly varying function l and the boundedness of ϕ,

$$|f_\lambda(u)| \le C\frac{\max\{|\sin\frac{\lambda}{2} + u\cos\frac{\lambda}{2}|^{1-2d-\epsilon}, |\sin\frac{\lambda}{2} + u\cos\frac{\lambda}{2}|^{1-2d+\epsilon}\}}{1+u^2} \le C\frac{(1+|u|)^{1-2d+\epsilon}}{1+u^2},$$

where $\epsilon > 0$ is such that $1 - 2d \pm \epsilon > 0$ and $-2d \pm \epsilon < 0$. Since the bound above is integrable on $\mathbb{R}$, the dominated convergence theorem implies that, as $\lambda \to 0$,

$$\frac{2\pi}{\sigma_\epsilon^2} f_X(\lambda) \sim \lambda^{-2d} l(\lambda)\phi(1) \int_0^\infty \frac{u^{1-2d}du}{1+u^2}. \qquad \square$$

Example 3.1.2 (*Aggregation with beta-like distribution*) One of the examples of the distributions $F_a(dx)$ which leads to LRD and which was first considered in the aggregation literature, is that with the density

$$f_a(x) = \frac{2x^{2p-1}(1-x^2)^{1-2d}}{B(p, 2-2d)} 1_{(0,1)}(x), \tag{3.1.11}$$

where $p > 0$, $0 < d < 1/2$ and B is the beta function defined in (2.2.12). One has indeed

$$\int_0^1 f_a(x)dx = \frac{1}{B(p, 2-2d)} \int_0^1 y^{p-1}(1-y)^{1-2d}dy = 1.$$

Since

$$f_a(x) = (1-x)^{1-2d}\Big(\frac{2x^{2p-1}(1+x)^{1-2d}}{B(p, 2-2d)} 1_{(0,1)}(x)\Big),$$

we can set $l(y) \equiv 1$ and $\phi(x)$ to be equal to the contents of the bracket. Then the density (3.1.11) has the form (3.1.9) when $p \ge 1/2$. On the other hand, note also that with the density (3.1.11), it is easy to show LRD in the sense (2.1.5) in condition II directly from (3.1.6). Indeed, from (3.1.6), we get that, for $h \geq 0$,

$$\begin{aligned}\gamma_X(h) &= \sigma_\epsilon^2 \int_0^1 \frac{x^h f_a(x)}{1-x^2} dx = \frac{2\sigma_\epsilon^2}{B(p, 2-2d)} \int_0^1 x^{h+2p-1}(1-x^2)^{-2d} dx \\ &= \frac{\sigma_\epsilon^2}{B(p, 2-2d)} \int_0^1 y^{\frac{h}{2}+p-1}(1-y)^{-2d} dy = \frac{\sigma_\epsilon^2 B(p+\frac{h}{2}, 1-2d)}{B(p, 2-2d)} \\ &= \frac{\sigma_\epsilon^2 \Gamma(1-2d)\Gamma(p+\frac{h}{2})}{B(p, 2-2d)\Gamma(p+\frac{h}{2}+1-2d)} \sim \frac{\sigma_\epsilon^2 \Gamma(1-2d)}{B(p, 2-2d)2^{2d-1}} h^{2d-1}, \end{aligned} \tag{3.1.12}$$

as $h \to \infty$, by using (2.2.12) and (2.4.4).

We also note that the restriction to $x \in [0, 1)$ in (3.1.9) is not accidental, albeit simplifying the analysis. Exercise 3.2 considers the case where the support of $f_a(x)$ is on $x \in (-1, 0)$ along with assumptions similar to those in Proposition 3.1.1. The aggregated series X in this case is not LRD in the sense of condition IV but has instead a divergence at $\lambda = \pi$.

Another interesting direction related to aggregation is the following. Suppose X is a given LRD series with the spectral density $f_X(\lambda)$. In contrast to the preceding presentation, one could ask whether there is a distribution function $F_a(dx)$, called a *mixture distribution*, such that $f_X(\lambda)$ could be expressed as in (3.1.8). This is often referred to as a *disaggregation problem*.

Example 3.1.3 (*Disaggregation*) If X is a FARIMA$(0, d, 0)$ series with the spectral density

$$f_X(\lambda) = \frac{|1-e^{-i\lambda}|^{-2d}}{2\pi}, \tag{3.1.13}$$

it is known that it can be disaggregated with the mixing density

$$f_a(x) = C(d)x^{d-1}(1-x)^{d-1}(1+x)1_{[0,1)}(x), \tag{3.1.14}$$

where

$$C(d) = \frac{\Gamma(3-d)}{2\Gamma(d)\Gamma(2-2d)} = 2^{2d-2}\frac{\sin(d\pi)}{\sqrt{\pi}}\frac{\Gamma(3-d)}{\Gamma(\frac{3}{2}-d)}$$

(see Exercise 3.1 and Celov, Leipus, and Philippe [208], Proposition 5.1).

3.2 Mixture of Correlated Random Walks

A simple random walk $X_0 = 0$, $X_n = \sum_{k=1}^n \epsilon_k$, $n \in \mathbb{N}$ represents the position of a random walker at time n. Each step ϵ_k is either to the right ($\epsilon_k = 1$) with probability $1/2$ or to the left ($\epsilon_k = -1$) with probability $1/2$. X_n denotes the position at time n. In a usual random walk, the steps are independent. In correlated random walks (CRWs, in short), they are dependent.

We will assume here that if a step has been made, then the next step will be in the same direction with probability p and in the opposite direction with probability $1-p$. (The first step is taken into either direction with probability $1/2$.) If $p = 1/2$, we fall back on the simple random walk where the steps are independent. But if $p \in (1/2, 1)$, there is persistence and the steps are correlated (though they still have zero mean). As a consequence, the resulting correlated random walk (CRW) $\{X_n\}$ has zero mean and stationary increments which are dependent. Observe also that while $\{X_n\}$ is not Markovian, $\{X_n, \epsilon_{n-1}\}$ is Markovian. Since the steps are dependent, let us compute the covariance between two steps distant by h.

Lemma 3.2.1 *For $m \geq 1$ and $h \geq 0$,*

$$\mathbb{E}(\epsilon_{m+h}\epsilon_m) = (2p-1)^h. \tag{3.2.1}$$

Proof Note first that $\mathbb{E}\epsilon_m^2 = 1$. Suppose $n \geq 1$ and let $\mathcal{F}_n = \sigma\{X_0, \dots, X_n\} = \sigma\{X_0, \epsilon_1, \dots, \epsilon_n\}$. Then,

$$\mathbb{E}(\epsilon_{n+1}|\mathcal{F}_n)1_{\{\epsilon_n=1\}} = \big(p(+1) + (1-p)(-1)\big)1_{\{\epsilon_n=1\}} = (2p-1)1_{\{\epsilon_n=1\}},$$
$$\mathbb{E}(\epsilon_{n+1}|\mathcal{F}_n)1_{\{\epsilon_n=-1\}} = \big(p(-1) + (1-p)(+1)\big)1_{\{\epsilon_n=-1\}} = -(2p-1)1_{\{\epsilon_n=-1\}}.$$

Hence

$$\mathbb{E}(\epsilon_{n+1}|\mathcal{F}_n) = \mathbb{E}(\epsilon_{n+1}|\mathcal{F}_n)1_{\{\epsilon_n=1\}} + \mathbb{E}(\epsilon_{n+1}|\mathcal{F}_n)1_{\{\epsilon_n=-1\}} = (2p-1)\epsilon_n$$

and by taking $n = m + h$,

$$\mathbb{E}(\epsilon_{m+h+1}\epsilon_m|\mathcal{F}_{m+h}) = \epsilon_m\mathbb{E}(\epsilon_{m+h+1}|\mathcal{F}_{m+h}) = \epsilon_{m+h}\epsilon_m(2p-1).$$

Taking expectation on both sides, one gets

$$\mathbb{E}(\epsilon_{m+h+1}\epsilon_m) = (2p-1)\mathbb{E}(\epsilon_{m+h}\epsilon_m)$$

Since $\mathbb{E}(\epsilon_{m+1}\epsilon_m) = (2p-1)\mathbb{E}\epsilon_m^2 = 2p-1$, the result follows by induction on h. □

Since $\mathbb{E}\epsilon_m = 0$ and $\mathbb{E}\epsilon_m^2 = 1$, the relation (3.2.1) equals both the correlation and the covariance between ϵ_m and ϵ_{m+h}. Observe that this covariance is indeed 0 if $p = 1/2$. Observe also that this covariance decreases exponentially fast as $n \to \infty$ and hence the sequence $\{\epsilon_m\}$ is short-range dependent. In fact, the following result holds (Exercise 3.4).

Theorem 3.2.2 *Let $\{X_n^{(p)}\}_{n\geq 0}$ be the CRW introduced above. Then*

$$\frac{1}{\sqrt{N}}X_{[Nt]}^{(p)} = \frac{1}{\sqrt{N}}\sum_{n=1}^{[Nt]}\epsilon_n \overset{fdd}{\longrightarrow} \sqrt{\frac{p}{1-p}}B(t), \quad t \geq 0,$$

where $\{B(t)\}_{t\geq 0}$ is a standard Brownian motion.

To create long-range dependence and in the spirit of Section 3.1, we are going to make p random. We shall do so by choosing p according to a probability distribution μ, so that, in the corresponding enlarged probability space, for each $m \geq 1$, $h \geq 0$,

$$\mathbb{E}(\epsilon_{m+h}\epsilon_m) = \int_0^1 (2p-1)^h d\mu(p).$$

We shall also consider not just a single CRW but a sequence $X_{[Nt]}^{(p_j)}$, $j = 1, \dots, M$, of independent copies of the CRW. Each $X^{(p_j)}$ has its own p_j chosen independently from

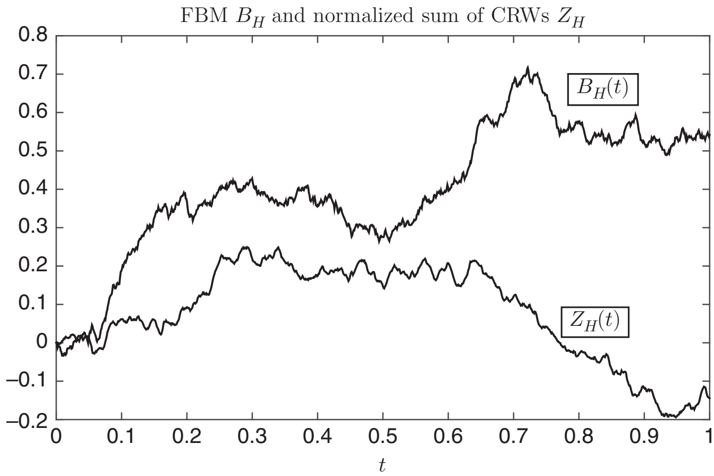

Figure 3.1 Normalized sum of CRWs $Z_H(t) = c_H(X_{[Nt]}^{(p_1)} + \cdots + X_{[Nt]}^{(p_M)})/(N^H M^{1/2})$ with $M = 100$ and $N = 1024$, and standard FBM $B_H(t)$ on the interval $[0, 1]$ when $H = 0.75$.

the distribution μ. The next theorem shows that by choosing the measure μ appropriately, $\sum_{j=1}^{M} X_{[Nt]}^{(p_j)}$ normalized, converges to FBM as $M \to \infty$ followed by $N \to \infty$. See also Figure 3.1.

Theorem 3.2.3 *Let $X_{[Nt]}^{(p_1)}, \ldots, X_{[Nt]}^{(p_M)}$ be independent samples from the population of CRWs with*

$$\frac{d\mu(p)}{dp} = C(H)(1-p)^{1-2H}, \quad 1/2 < p < 1, \tag{3.2.2}$$

where $1/2 < H < 1$, $C(H) = (1-H)2^{3-2H}$ and p_j, $j \geq 1$, are i.i.d. with the distribution μ. Then,

$$\lim_{N\to\infty} c_H \frac{1}{N^H} \lim_{M\to\infty} \frac{X_{[Nt]}^{(p_1)} + \cdots + X_{[Nt]}^{(p_M)}}{\sqrt{M}} = B_H(t), \quad t \geq 0, \tag{3.2.3}$$

where

$$c_H = \sqrt{H(2H-1)/\Gamma(3-2H)}, \tag{3.2.4}$$

B_H is a standard FBM with the self-similarity parameter H and the convergence is in the sense of finite-dimensional distributions.

Proof The central limit theorem yields

$$\lim_{M\to\infty} (X_m^{(p_1)} + \cdots + X_m^{(p_M)})/\sqrt{M} \xrightarrow{fdd} Y_m$$

in the sense of finite-dimensional distributions in m. Here, Y_m, $m \geq 1$, is a Gaussian process with stationary increments $\Delta Y_m = Y_m - Y_{m-1}$ that have zero mean and unit variance, and in view of (3.2.1) and (3.2.2), the autocovariances

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$$\gamma_{\Delta Y}(h) = \mathbb{E}(\Delta Y_{m+h}\Delta Y_m) = (1-H)2^{3-2H}\int_{1/2}^{1}(2p-1)^h(1-p)^{1-2H}dp$$

$$= (2-2H)\int_0^1 x^h(1-x)^{1-2H}dx = (2-2H)\frac{\Gamma(h+1)\Gamma(2-2H)}{\Gamma(h+3-2H)} \sim \Gamma(3-2H)h^{2H-2},$$

as $h\to\infty$, by (2.2.12) and (2.4.4). Thus, the series $\{\Delta Y_m\}_{m\geq 1}$ is LRD in the sense (2.1.5) in condition II with $d = H - 1/2$ and $L_2(u) \sim \Gamma(3-2H)$. By Corollary 2.2.7, the series $\{\Delta Y_m\}_{m\geq 1}$ also satisfies (2.1.8) in condition V with $L_5(u) \sim \Gamma(3-2H)/(H(2H-1)) = c_H^{-2}$ with c_H given in (3.2.4). Proposition 2.8.7 then yields

$$\frac{Y_{[Nt]}}{c_H^{-1}N^H} \xrightarrow{fdd} B_H(t), \quad t \geq 0,$$

where $\{B_H(t)\}$ is a standard FBM, showing the desired convergence (3.2.3). □

Observe that since $1-2H<0$, the density $C(H)(1-p)^{1-2H}$ explodes at $p=1$, making it likely that the random walker would tend to continue walking in the same direction.

3.3 Infinite Source Poisson Model with Heavy Tails

Long-range dependence is often associated with heavy tails. (Heavy-tailed random variables are discussed in Appendix A.4.) This is apparent in many models incorporating heavy tails. One such model, called the *infinite source Poisson model*, is considered here. For completeness and illustration's sake, we consider this model for a number of different parameter regimes leading to various self-similar processes with stationary increments. One such process turns out to be FBM with the self-similarity parameter $H > 1/2$, whose increments are long-range dependent.

3.3.1 Model Formulation

In the infinite source Poisson model, workload sessions arrive to a system at Poisson arrival times and each session carries workload at some rate (reward) for a random duration time. See Figure 3.2. The basic notation and assumptions are as follows.

Arrivals:

Workload sessions start according to a Poisson process on the real line with intensity $\lambda > 0$. The arrival times are denoted $\ldots S_j, S_{j+1}, \ldots$

Durations:

The session length distribution is represented by a random variable U with distribution function $F_U(u) = \mathbb{P}(U \leq u)$ and expected value

$$\nu = \mathbb{E}U < \infty.$$

Suppose that either

$$\mathbb{E}U^2 < \infty \tag{3.3.1}$$

or U is heavy-tailed in the sense that

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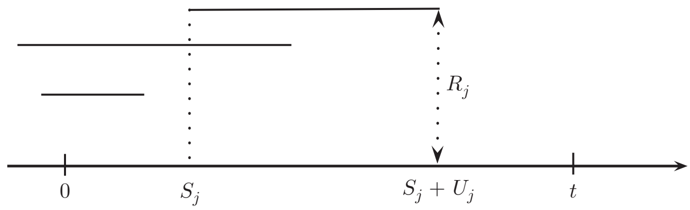

Figure 3.2 Sessions are denoted by horizontal lines. A typical session j starts at Poisson time S_j, has a workload rate R_j and lasts for a random time U_j. We are interested in the total workload from time 0 to time t.

$$\mathbb{P}(U > u) = L_U(u)\frac{u^{-\gamma}}{\gamma}, \tag{3.3.2}$$

as $u \to \infty$, where $1 < \gamma < 2$ and L_U is a slowly varying function at infinity (which is positive for all $u > 0$). We extend the parameter range $1 < \gamma < 2$ to $1 < \gamma \le 2$, by letting $\gamma = 2$ represent the case $\mathbb{E}U^2 < \infty$. Now let $\{U_j\}_{j\in\mathbb{Z}}$ be i.i.d. random variables with a common distribution F_U, with U_j representing session duration associated with the arrivals S_j, $j \in \mathbb{Z}$.

Rewards:

The workload rate valid during a session is given by a random variable $R > 0$ with $F_R(r) = \mathbb{P}(R \le r)$ and

$$\mathbb{E}R < \infty.$$

We suppose either

$$\mathbb{E}R^2 < \infty \tag{3.3.3}$$

or

$$\mathbb{P}(R > r) = L_R(r)\frac{r^{-\delta}}{\delta}, \tag{3.3.4}$$

as $r \to 0$, where $1 < \delta < 2$ and L_R is a slowly varying function at infinity (which is positive for all $r > 0$). We extend the parameter range to $1 < \delta \le 2$, by letting $\delta = 2$ represent the case $\mathbb{E}R^2 < \infty$. Suppose that $\{R_j\}_{j\in\mathbb{Z}}$ are i.i.d. variables with a common distribution F_R, with R_j representing the session rate associated with the arrivals S_j, $j \in \mathbb{Z}$. The workload rate is also called the *reward*.

We assume that the arrival times, durations and rewards are mutually independent.

In an application to internet traffic modeling, arrivals are associated with initiations of web sessions (file transfers), which have varying durations and workload rates. A particularly simple case is that of a fixed workload rate, say $R = 1$, $F_R(dr) = \delta_{\{1\}}(dr)$ and a duration U corresponding to the file size to be transmitted. With real data sets from Internet traffic, both file sizes and session durations are typically found to be heavy-tailed (e.g., Crovella and Bestavros [270], Guerin, Nyberg, Perrin, Resnick, Rootzén, and Stărică [438], López-Oliveros and Resnick [645]).

The cumulative workload $W^*_\lambda(t)$ from time 0 to time t is defined in Section 3.3.2 below. We are interested in the deviation of the cumulative workload from its mean $\mathbb{E}W^*_\lambda(t) = \lambda t\mathbb{E}(U)\mathbb{E}(R)$ (see (3.3.17) below), and more precisely, in the asymptotic behavior of

$$\frac{1}{b}(W_\lambda^*(at) - \mathbb{E}W_\lambda^*(at)),$$

as $\lambda, a, b \to \infty$, which describes what happens when the number of sessions increases ($\lambda \to \infty$) and when the time grows ($a \to \infty$). Different types of self-similar processes may appear in the limit depending on the respective values of the indices γ and δ, and on the respective rates at which λ, a, b tend to infinity. These are:

1) *Fractional Brownian motion* $B_H(t)$ with $1/2 < H < 1$. Recall that B_H is Gaussian with stationary dependent increments. The limit is B_H when

$$1 < \gamma < \delta = 2 \quad \text{and} \quad \frac{b}{a} \to \infty$$

with $H = \frac{3-\gamma}{2} \in (1/2, 1)$ (see Theorem 3.3.8, (a)).

2) *The Telecom process*. This is an infinite variance self-similar stable process $Z_{\gamma,\delta}(t)$ with stationary dependent increments. The limit appears when

$$1 < \gamma < \delta < 2 \quad \text{and} \quad \frac{b}{a} \to \infty$$

(Theorem 3.3.8, (b)). Because the variance is infinite, it is best to define the Telecom process through an integral representations. One can do this in two ways. First, in terms of an integral with respect to a Poisson random measure:

$$Z_{\gamma,\delta}(t) = \int_{-\infty}^{\infty}\int_0^{\infty}\int_0^{\infty} K_t(s,u)r(N(ds,du,dr) - ds u^{-(1+\gamma)}du r^{-(1+\delta)}dr), \qquad (3.3.5)$$

where the kernel K_t is defined in (3.3.10) below and where the Poisson random measure $N(ds,du,dr)$ has control measure $ds u^{-(1+\gamma)}du r^{-(1+\delta)}dr$.

The factor r in the kernel together with the presence of the measure $r^{-(1+\delta)}dr$ is characteristic of a stable process with index δ (see Appendix B.1.3) and therefore one can also represent the Telecom process by an integral with respect to a δ-stable random measure $M_\delta(ds,du)$, as

$$Z_{\gamma,\delta}(t) = c_\delta \int_{-\infty}^{\infty}\int_0^{\infty} K_t(s,u)M_\delta(ds,du), \qquad (3.3.6)$$

where $c_\delta > 0$ is a constant and $M_\delta(ds,du)$ has control measure $ds u^{-(1+\gamma)}du$. With the representation (3.3.6), the Telecom process is the same as that in Section 2.6.7. The Telecom process is self-similar with index

$$H = \frac{\delta + 1 - \gamma}{\delta} \in (\frac{1}{\delta}, 1).$$

As noted in Section 2.6.7, the Telecom process is different from LFSM, which is another stable self-similar process introduced in Section 2.6.5.

The second representation (3.3.6), involving $M_\delta(ds,du)$, continues to make sense if $\delta = 2$. In this case, $M_\delta(ds,du)$ is a Gaussian measure. Since $Z_{\gamma,2}(t)$ is Gaussian and $H-$ self-similar with stationary increments, it is necessarily fractional Brownian motion $B_H(t)$ (see Corollary 2.6.3).

3) *The stable Lévy process.* This is an infinite variance process with independent stationary increments (see Section 2.6.4). It appears when

$$1 < \gamma < \delta \leq 2 \quad \text{and} \quad \frac{b}{a} \to 0$$

(see Theorem 3.3.10) and when

$$1 < \delta < \gamma \leq 2$$

(see Theorem 3.3.17). For example, in the first case, it can be represented as

$$\Lambda_\gamma(t) = \int_{-\infty}^{\infty}\int_0^{\infty}\int_0^{\infty} 1_{\{0<s<t\}} ur(N(ds,du,dr) - dsu^{-(1+\gamma)}du F_R(dr)),$$

where F_R is the distribution of R given by (3.3.3) or (3.3.4). The Poisson random measure N has control measure $dsu^{-(1+\gamma)}du F_R(dr)$. Both u and r appear in the kernel but because $\gamma < \delta$, the process is γ-stable and can be represented as

$$\Lambda_\gamma(t) = c_\gamma \int_{-\infty}^{\infty}\int_0^{\infty} 1_{\{0<s<t\}} r M_\gamma(ds,dr),$$

where $c_\gamma > 0$ and $M_\gamma(ds,dr)$ is a γ-stable random measure with control measure $ds F_R(dr)$. The presence of the indicator function $1_{\{0<s<t\}}$ yields the independent stationary increments.

4) *The Intermediate Telecom process.* This is the process $Y_{\gamma,R}(t)$ having an integral representation

$$Y_{\gamma,R}(t) = \int_{-\infty}^{\infty}\int_0^{\infty}\int_0^{\infty} K_t(s,u) r \left(N(ds,du,dr) - ds\, u^{-(1+\gamma)}du\, F_R(dr)\right),$$

where the kernel K_t is defined in (3.3.10) below and where the Poisson random measure $N(ds,du,dr)$ has control measure $dsu^{-(1+\gamma)}F_R(dr)$. It appears in a boundary case $b = a$ (see Theorem 3.3.12 below). The limit process $Y_{\gamma,R}(t)$ is not self-similar but is second-order self-similar (see Remark 3.3.13). Note the important difference between this and the Telecom process. In the Telecom process, the distribution $F_R(dr)$ is replaced by $r^{-(1+\delta)}dr$, which is not a distribution. It is the presence of $r^{-(1+\delta)}dr$ which makes the Telecom process $Z_{\gamma,\delta}(t)$ a δ-stable process with infinite variance.

5) *Brownian motion.* The limit process is Brownian motion in the case

$$\gamma = \delta = 2$$

(see Theorem 3.3.16).

3.3.2 Workload Process and its Basic Properties

The focus now is on a workload process W^*_λ defined by

$$\begin{aligned} W^*_\lambda(t) &= \text{the aggregated workload in the time interval } [0,t] \\ &= \sum_{j:S_j\leq 0} (t \wedge (S_j + U_j)_+)R_j + \sum_{j:0<S_j\leq t} ((t-S_j)\wedge U_j)R_j, \end{aligned} \tag{3.3.7}$$

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where the first sum above represents the aggregated workload for the arrivals before time 0, and the second sum represents the aggregated workload due to the arrivals in $[0, t]$. The first sum involves the contributions of arrivals that occurred before time 0 and are still alive by time $t > 0$ and the second involves the arrivals that occurred in the time interval $(0, t]$. In the first case, the effective duration of the jth session is the smallest of $t > 0$ and $(S_j + U_j)_+$, the end-time of the jth session, whenever $S_j + U_j$ is nonnegative. In the second case, the effective duration of the jth session, up to time t, is the minimum of U_j and $t - S_j$, the amount of time lapsed since the arrival of the jth session.

For further analysis, it is convenient to express the process W_λ^* as an integral with respect to a Poisson random measure. Let $\mathbb{R}_+ = (0, \infty)$ and $N(ds, du, dr)$ denote a Poisson random measure on $\mathbb{R} \times \mathbb{R}_+ \times \mathbb{R}_+$ with intensity measure

$$n(ds, du, dr) = \lambda ds F_U(du) F_R(dr). \tag{3.3.8}$$

One can think of a realization of the point measure $N(ds, du, dr)$ as a collection of points $\{(S_j, U_j, R_j)\}_{j \in \mathbb{Z}}$ and the intensity measure

$$n(ds, du, dr) = \mathbb{E} N(ds, du, dr)$$

as the mean number of points in the box of size $dsdudr$ centered at s, u and r. The process W_λ^* can then be represented as

$$\begin{aligned} W_\lambda^*(t) &= \int_{-\infty}^0 \int_0^\infty \int_0^\infty (t \wedge (s+u)_+) r N(ds, du, dr) \\ &\quad + \int_0^t \int_0^\infty \int_0^\infty ((t-s) \wedge u) r N(ds, du, dr) \\ &= \int_{-\infty}^\infty \int_0^\infty \int_0^\infty ((t-s)_+ \wedge u - (0-s)_+ \wedge u) r N(ds, du, dr) \\ &=: \int_{-\infty}^\infty \int_0^\infty \int_0^\infty K_t(s, u) r N(ds, du, dr), \end{aligned} \tag{3.3.9}$$

where the kernel $K_t(s, u)$ is

$$K_t(s, u) = (t-s)_+ \wedge u - (-s)_+ \wedge u. \tag{3.3.10}$$

The kernel $K_t(s, u)$ is such that

$$0 \le K_t(s, u) = \begin{cases} 0 & \text{if } s + u \le 0 \text{ or } s \ge t \\ s + u & \text{if } s \le 0 \le s + u \le t \\ t & \text{if } s \le 0, t \le s + u \\ u & \text{if } 0 \le s, s + u \le t \\ t - s & \text{if } 0 \le s \le t \le s + u. \end{cases} \tag{3.3.11}$$

It is depicted in Figure 3.3.

The following lemma lists properties of the kernel $K_t(s, u)$ which will be used in the sequel.

Lemma 3.3.1 *Let $s \in \mathbb{R}$, $t, u \in \mathbb{R}_+$ and $a > 0$. Then,*

$$K_{at}(as, au) = a K_t(s, u), \tag{3.3.12}$$

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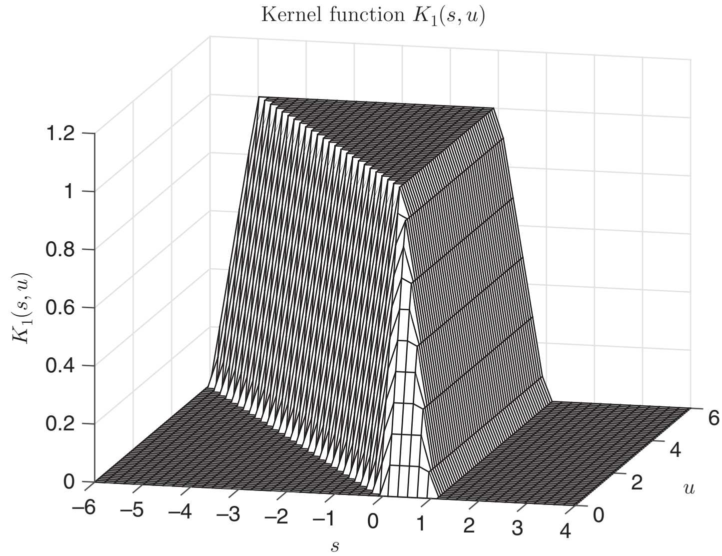

Figure 3.3 Kernel $K_t(s, u)$ in (3.3.10) with $t = 1$.

$$0 \le K_t(s, u) = |(0, t) \cap (s, s + u)| \le t \wedge u, \tag{3.3.13}$$

$$K_t(s, u) = \int_0^t 1_{\{s<y<s+u\}} dy = \int_0^u 1_{\{0<y+s<t\}} dy, \tag{3.3.14}$$

$$\int_{-\infty}^{\infty} K_t(s, u) ds = ut. \tag{3.3.15}$$

Moreover, $K_t(s, u)$ is differentiable with respect to the variable u and

$$\frac{\partial K_t}{\partial u}(s, u) = 1_{\{0<u+s<t\}}. \tag{3.3.16}$$

Proof Note that

$$K_t(s, u) = \int_{-s}^{t-s} 1_{\{0<y<u\}} dy = \int_0^t 1_{\{s<y<s+u\}} dy$$

so that $K_t(s, u) = |(0, t) \cap (s, s + u)| \le t \wedge u$. Moreover, for (3.3.15),

$$\int_{-\infty}^{\infty} K_t(s, u) ds = \int_0^t \Big(\int_{-\infty}^{\infty} 1_{\{s<y<s+u\}} ds \Big) dy = \int_0^t u ds = ut.$$

One also has the second representation in (3.3.14), from which the other statements follow. □

Note that the series (3.3.7) or the integral (3.3.9) is well-defined (absolutely convergent) with probability 1. Indeed, by Campbell's theorem (e.g., Kingman [558], Section 3.2), an integral $I(f) = \int_S f(x) N(dx)$, where $N(dx)$ is a Poisson random measure on a space S with intensity measure $n(dx)$, exists with probability 1 if and only if $\int_S (|f(x)| \wedge 1) n(dx)$

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$< \infty$. Moreover, if $\int_S |f(x)|n(dx) < \infty$ or f is nonnegative, then $\mathbb{E}I(f) = \int_S f(x)n(dx)$. See also Appendix B.1.4.

In the case of the integral (3.3.9), note that

$$\mathbb{E}W_\lambda^*(t) = (\mathbb{E}R)(I_1 + I_2),$$

where

$$I_1 = \int_{-\infty}^{0}\int_0^\infty \Big(t \wedge (s+u)_+\Big)\lambda ds F_U(du), \quad I_2 = \int_0^t \int_0^\infty \Big((t-s)\wedge u\Big)\lambda ds F_U(du).$$

By integration by parts,

$$\begin{aligned}
I_1 &= \lambda \int_0^\infty dz \int_0^\infty \Big(t \wedge (u-z)_+\Big) F_U(du) \\
&= \lambda \int_0^\infty dz \Big(\int_z^{z+t} (u-z)F_U(du) + \int_{z+t}^\infty tF_U(du)\Big) \\
&= \lambda \int_0^\infty dz \Big(\int_z^{z+t} uF_U(du) - \int_z^{z+t} zF_U(du) + \int_{z+t}^\infty tF_U(du)\Big) \\
&= \lambda \int_0^\infty dz \Big(-\int_z^{z+t} ud\mathbb{P}(U>u) - z\mathbb{P}(U>z) + (z+t)\mathbb{P}(U>z+t)\Big) \\
&= \lambda \int_0^\infty dz \int_z^{z+t} \mathbb{P}(U>u)du = \lambda \int_0^\infty dz \int_z^\infty \mathbb{P}(U>u)du \\
&\quad - \lambda \int_0^\infty dz \int_{z+t}^\infty \mathbb{P}(U>u)du \\
&= \lambda \int_0^\infty dz \int_z^\infty \mathbb{P}(U>u)du - \lambda \int_t^\infty dz \int_z^\infty \mathbb{P}(U>u)du \\
&= \lambda \int_0^t dz \int_z^\infty du\, \mathbb{P}(U>u)
\end{aligned}$$

and

$$\begin{aligned}
I_2 &= \lambda \int_0^t ds \Big(\int_0^{t-s} uF_U(du) + (t-s)\mathbb{P}(U>t-s)\Big) \\
&= \lambda \int_0^t ds \int_0^{t-s} \mathbb{P}(U>u)du = \lambda \int_0^t dz \int_0^z du\, \mathbb{P}(U>u).
\end{aligned}$$

It follows that $I_1 + I_2 = \lambda \int_0^t dz \int_0^\infty du\, \mathbb{P}(U>u) = \lambda(\mathbb{E}U)t$, so that:

Proposition 3.3.2 *We have*

$$\mathbb{E}W_\lambda^*(t) = \lambda\mathbb{E}(U)\mathbb{E}(R)t = \lambda\nu\mathbb{E}(R)t < \infty. \tag{3.3.17}$$

We shall now involve the intensity measure $n(ds,du,dr) = \lambda ds F_U(du)F_R(dr)$ introduced in (3.3.8) and define

$$\widetilde{N}(ds,du,dr) = N(ds,du,dr) - n(ds,du,dr),$$

the so-called *compensated Poisson measure* with intensity measure $n(ds, du, dr)$. In view of (3.3.17), it is convenient to express the process $W_\lambda^*(t)$ as

$$W_\lambda^*(t) = \lambda \nu \mathbb{E}(R)t + \int_{-\infty}^{\infty} \int_0^{\infty} \int_0^{\infty} K_t(s, u) r \widetilde{N}(ds, du, dr), \qquad (3.3.18)$$

where the last integral has now zero mean.

It is now instructive to discuss briefly several other properties of the process W_λ^*.

Proposition 3.3.3 *One can write*

$$W_\lambda^*(t) = \int_0^t W_\lambda(y) dy, \qquad (3.3.19)$$

where

$$W_\lambda(y) = \int_{-\infty}^{\infty} \int_0^{\infty} \int_0^{\infty} 1_{\{s<y<s+u\}} r N(ds, du, dr). \qquad (3.3.20)$$

The process W_λ is stationary, and hence W_λ^* has stationary increments.

Proof See Exercise 3.6. □

Another important property is stated in the next proposition.

Proposition 3.3.4 *Suppose that $\mathbb{E}R^2 < \infty$. Then, $\mathbb{E}W_\lambda(y)^2 < \infty$ and*

$$\mathrm{Cov}(W_\lambda(0), W_\lambda(y)) \sim \frac{\lambda \mathbb{E}R^2 L_U(y) y^{1-\gamma}}{\gamma(\gamma - 1)}, \qquad (3.3.21)$$

as $y \to \infty$. As a consequence, $\mathbb{E}W_\lambda^*(t)^2 < \infty$ and

$$\mathrm{Var}(W_\lambda^*(t)) \sim \frac{2\lambda \mathbb{E}R^2 L_U(y) t^{3-\gamma}}{\gamma(\gamma - 1)(2 - \gamma)(3 - \gamma)}, \qquad (3.3.22)$$

as $t \to \infty$.

Proof By using the representation (3.3.20) and Campbell's theorem (e.g., Kingman [558], Section 3.2), which states that if $\int_S f(x)^2 n(dx) < \infty$ and $\int_S g(x)^2 n(dx) < \infty$, then $\mathrm{Cov}(I(f), I(g)) = \int_S f(x) g(x) n(dx) < \infty$ for $I(f) = \int_S f(x) N(dx)$ and $I(g) = \int_S g(x) N(dx)$, we have

$$\begin{aligned}
\mathrm{Var}(W_\lambda(y)) &= \int_{-\infty}^{\infty} \int_0^{\infty} \int_0^{\infty} 1_{\{s<y<s+u\}}^2 r^2 n(ds, du, dr) \\
&= \int_{-\infty}^{\infty} \int_0^{\infty} \int_0^{\infty} 1_{\{s<y<s+u\}} r^2 \lambda ds F_U(du) F_R(dr) \\
&= \lambda \mathbb{E}R^2 \int_0^{\infty} u F_U(du) = \lambda \nu \mathbb{E}R^2 < \infty,
\end{aligned}$$

since $\int_{-\infty}^{\infty} 1_{\{s<y<s+u\}} ds = \int_{-\infty}^{\infty} 1_{\{y-u<s<y\}} ds = u$ and $\mathbb{E}R^2 < \infty$ by assumption. By Campbell's theorem again, for $y \geq 0$,

$$\begin{aligned}\operatorname{Cov}(W_\lambda(0), W_\lambda(y)) &= \int_{-\infty}^{\infty}\int_0^{\infty}\int_0^{\infty} 1_{\{s<0<s+u\}} 1_{\{s<y<s+u\}} r^2 \lambda ds F_U(du) F_R(dr)\\ &= \lambda \mathbb{E}R^2 \int_0^{\infty} \Big(\int_{-\infty}^{0} 1_{\{0<y<s+u\}} ds \Big) F_U(du)\\ &= \lambda \mathbb{E}R^2 \int_0^{\infty}\int_0^{\infty} 1_{\{0<y<u-s\}} ds F_U(du)\\ &= \lambda \mathbb{E}R^2 \int_y^{\infty} (u-y) F_U(du). \end{aligned} \tag{3.3.23}$$

By using integration by parts, (3.3.2) and Proposition 2.2.2, we further get that

$$\operatorname{Cov}(W_\lambda(0), W_\lambda(y)) = \lambda \mathbb{E}R^2 \int_y^{\infty} \mathbb{P}(U > u) du \sim \frac{\lambda \mathbb{E}R^2 L_U(y) y^{1-\gamma}}{\gamma(\gamma-1)},$$

as $y \to \infty$. The statements concerning W_λ^* follow easily from those for W_λ, since $\operatorname{Var}(W_\lambda^*(t)) = \int_0^t \int_0^t \operatorname{Cov}(W_\lambda(u), W_\lambda(v)) du dv$. □

Remark 3.3.5 In view of (3.3.21), the stationary process W_λ underlying W_λ^* is LRD in the sense (2.1.5) in condition II with

$$d = \frac{2-\gamma}{2} \in (0, 1/2) \tag{3.3.24}$$

(see also Remark 2.1.6). This is an important example of how heavy tails can induce long-range dependence.

The relation (3.3.21) yields

$$\operatorname{Var}(W_\lambda^*(t)) \sim C L_U(t) t^{2H},$$

as $t \to \infty$, where

$$H = \frac{3-\gamma}{2} \in (1/2, 1). \tag{3.3.25}$$

The parameter H in (3.3.25) will appear below as a self-similarity parameter of Gaussian processes arising as suitable limits of scaled workload process W_λ^*.

3.3.3 Input Rate Regimes

We shall now focus on the limiting behavior of the scaled workload process

$$\frac{1}{b}\Big(W_\lambda^*(at) - \mathbb{E}W_\lambda^*(at) \Big) = \frac{1}{b}\Big(W_\lambda^*(at) - \lambda \nu \mathbb{E}(R) at \Big), \tag{3.3.26}$$

as

$$\lambda, a, b \to \infty. \tag{3.3.27}$$

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As will be seen below, a number of different limiting processes can be obtained depending on the speeds at which λ and a tend to infinity in (3.3.27) (the normalizing factor b will be a function of λ and a). The following three regimes will play a key role:

$$\begin{aligned} \text{fast:} \quad & \frac{\lambda L_U(a)}{a^{\gamma-1}} \to \infty, \\ \text{slow:} \quad & \frac{\lambda L_U(a)}{a^{\gamma-1}} \to 0, \\ \text{intermediate:} \quad & \frac{\lambda L_U(a)}{a^{\gamma-1}} \to c, \quad 0 < c < \infty. \end{aligned} \tag{3.3.28}$$

We write

$$\overline{F}_U(a) = 1 - F_U(a).$$

Since, by (3.3.2), we have

$$\frac{\lambda L_U(a)}{a^{\gamma-1}} = \gamma\lambda a \overline{F}_U(a), \tag{3.3.29}$$

we can reexpress the three regimes as follows:

$$\begin{aligned} \text{fast:} \quad & \gamma\lambda a\overline{F}_U(a) \to \infty, \\ \text{slow:} \quad & \gamma\lambda a\overline{F}_U(a) \to 0, \\ \text{intermediate:} \quad & \gamma\lambda a\overline{F}_U(a) \to c, \quad 0 < c < \infty. \end{aligned} \tag{3.3.30}$$

The parameter λ determines the *input rate* of the Poisson arrivals. Viewing $\lambda = \lambda(a)$ as a function of a, the three regimes (3.3.28) thus impose conditions on how fast the input rate $\lambda(a)$ grows as the time scale $a \to \infty$. It will sometimes be convenient to write these three regimes as one condition,

$$\frac{\lambda L_U(a)}{a^{\gamma-1}} \to c, \tag{3.3.31}$$

where $c \in [0, \infty]$.

We shall use the following notation: the (generalized) inverse function of a non-decreasing function $a(x)$ is the function

$$a^{\leftarrow}(t) = \inf\{x : a(x) \geq t\}.$$

For properties of $a^{\leftarrow}(t)$, see for example Resnick [847], Section 2.1.2. If $a(x)$ is continuous and strictly monotone, then one gets $a^{\leftarrow}(t)$ by solving $t = a(x)$. For example, if $a(x) = x^3$, then $a^{\leftarrow}(t) = t^{1/3}$.

There are several equivalent ways of writing the regimes (3.3.28), and several interpretations.

Proposition 3.3.6 *The condition (3.3.31) with $c \in [0, \infty]$ is equivalent to any of the following conditions (i), (ii) or (iii):*

(i) with

$$g(t) = \left(\frac{1}{\overline{F}_U}\right)^{\leftarrow}(t) = \inf\left\{x : \frac{1}{\overline{F}_U(x)} = \frac{1}{1 - F_U(x)} \geq t\right\}, \quad t > 0, \tag{3.3.32}$$

we have

$$\frac{g(\lambda a)}{a} \to \frac{c^{1/\gamma}}{\gamma^{1/\gamma}}. \tag{3.3.33}$$

(ii) with a stationary process $W_\lambda(y)$ defined in (3.3.20) and in the case $\mathbb{E}R^2 < \infty$, we have

$$\mathrm{Cov}(W_\lambda(0), W_\lambda(a)) \to \frac{\mathbb{E}R^2}{\gamma(\gamma-1)}c. \tag{3.3.34}$$

(iii) with $X_{\lambda,a}$ defined in (3.3.42) and denoting the number of sessions active at both times 0 and a, we have

$$\mathbb{E}X_{\lambda,a} \to \frac{c}{\gamma(\gamma-1)}. \tag{3.3.35}$$

Proof Consider first the equivalence of (3.3.31) and (i) when $0 < c < \infty$. Using (3.3.29), write (3.3.31) as

$$\gamma\lambda a\overline{F}_U(a) \sim c \tag{3.3.36}$$

or

$$\frac{1}{\overline{F}_U(a)} \sim \frac{\gamma\lambda a}{c}.$$

On the other hand, $\overline{F}_U(a) \sim L_U(u)\gamma^{-1}u^{-\gamma}$ as $u \to \infty$. By (3.3.2), using Theorem 1.5.12 in Bingham et al. [157], the function g in (3.3.32), namely the inverse of $f = 1/\overline{F}_U$, can be expressed as

$$g(t) = L_g(t)t^{1/\gamma}. \tag{3.3.37}$$

It is regularly varying with index $1/\gamma$ and slowly varying function L_g. It is, moreover, an "asymptotic inverse" of the function f; that is,

$$g(f(x)) \sim x, \quad \text{as } x \to \infty. \tag{3.3.38}$$

Then,

$$\begin{aligned} a \sim g\Big(\frac{1}{\overline{F}_U(a)}\Big) \sim g\Big(\frac{\gamma\lambda a}{c}\Big) &= L_g\Big(\frac{\gamma\lambda a}{c}\Big)\Big(\frac{\gamma\lambda a}{c}\Big)^{1/\gamma} \\ &= L_g\Big(\frac{\gamma\lambda a}{c}\Big)\frac{1}{L_g(\lambda a)}\frac{\gamma^{1/\gamma}}{c^{1/\gamma}}L_g(\lambda a)(\lambda a)^{1/\gamma} \sim \frac{\gamma^{1/\gamma}}{c^{1/\gamma}}g(\lambda a) \end{aligned} \tag{3.3.39}$$

and thus

$$\frac{g(\lambda a)}{a} \sim \frac{c^{1/\gamma}}{\gamma^{1/\gamma}},$$

proving (3.3.33) when $0 < c < \infty$.

Consider next the case (i) with $c = 0$. We shall choose $c > 0$ and then let $c \to 0$. Then, the relation (3.3.39) becomes

$$a \sim \frac{L_g\Big(\frac{\gamma\lambda a}{c}\Big)}{L_g(\lambda a)}\frac{\gamma^{1/\gamma}}{c^{1/\gamma}}g(\lambda a)$$

or

$$\frac{c^{1/\gamma}}{\gamma^{1/\gamma}} \frac{L_g(\lambda a)}{L_g\left(\frac{\gamma\lambda a}{c}\right)} \sim \frac{g(\lambda a)}{a}. \tag{3.3.40}$$

By using Potter's bound (2.2.1), for $\epsilon < 1/\gamma$,

$$\frac{L_g(\lambda a)}{L_g\left(\frac{\gamma\lambda a}{c}\right)} \leq 2\left(\frac{c}{\gamma}\right)^{-\epsilon}$$

and (3.3.40) shows that $g(\lambda a)/a \to 0$.

In the case *(i)* with $c = \infty$, write as above

$$\frac{a}{g(\lambda a)} \sim \frac{\gamma^{1/\gamma}}{c^{1/\gamma}} \frac{L_g\left(\frac{\gamma\lambda a}{c}\right)}{L_g(\lambda a)}. \tag{3.3.41}$$

Applying Potter's bound (2.2.1) again, we get that $a/g(\lambda a) \to 0$.

The equivalence of (3.3.31) and *(ii)* follows directly from (3.3.29) and (3.3.21).

To see the equivalence of (3.3.31) and *(iii)*, observe that the number of sessions $X_{\lambda,a}$ active at both 0 and a can be represented as

$$X_{\lambda,a} = \int_{-\infty}^{0} \int_{0}^{\infty} \int_{0}^{\infty} 1_{\{s<0<a\leq s+u\}} N(ds, du, dr). \tag{3.3.42}$$

Then, as for (3.3.23), by integration by parts,

$$\mathbb{E}X_{\lambda,a} = \lambda \int_a^\infty (u-a) F_U(du) = \lambda \int_a^\infty \mathbb{P}(U > u) du \sim \frac{\lambda L_U(a) a^{1-\gamma}}{\gamma(\gamma-1)}$$

and the equivalence follows. □

The equivalence *(i)* in Proposition 3.3.6 will be useful in the results below when the limit is a stable process. The equivalences *(ii)* and *(iii)* provide physical interpretations of the regimes. For example, the fast regime, namely one where $c \to 0$, is the regime where the covariance in the underlying stationary workload process grows (case *(ii)*), and the average number of active sessions throughout $[0, a]$ increases (case *(iii)*).

3.3.4 Limiting Behavior of the Scaled Workload Process

We turn now to the behavior of scaled workload process under various regimes. We will work with characteristic functions of integrals with respect to Poisson random measures. In this regard, recall from Appendix B.1.4 that the characteristic function of $\int f(x)(N(dx) - n(dx))$ is well defined if $\int (f(x)^2 \wedge |f(x)|)\, n(dx) < \infty$, in which case it is given by

$$\log \mathbb{E} \exp\left\{i \int f(x)(N(dx) - n(dx))\right\} = \int (e^{if(x)} - 1 - if(x))\, n(dx). \tag{3.3.43}$$

The following lemma about slowly varying functions and their inverse will be handy.

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Lemma 3.3.7 *For a slowly varying function L and a number $\alpha > 0$, there is a slowly varying function L_α^* satisfying: for $x > 0$,*

$$L_\alpha^*(u)^{-\alpha} L\Big(u^{1/\alpha} L_\alpha^*(u) x\Big) \to 1, \quad \text{as } u \to \infty. \tag{3.3.44}$$

Moreover, L_α^* is unique within asymptotic equivalence.

Proof The existence of the slowly varying function L_α^* is proved following (A.4.9) in Appendix A.4 after setting $h = L$ and $h_\alpha = (L_\alpha^*)^\alpha$. The uniqueness part is also mentioned following (A.4.9). □

We shall use Lemma 3.3.7 below with $L = L_U, \alpha = \gamma$ and $L = L_R, \alpha = \delta$, in which case the corresponding slowly varying function L_α^* will be denoted $L_{U,\gamma}^*$ and $L_{R,\delta}^*$, respectively. One can use Lemma 3.3.7 also to conclude that the function g in (3.3.32) satisfies

$$g(t) \sim \gamma^{-1/\gamma} t^{1/\gamma} L_{U,\gamma}^*(t), \quad \text{as } t \to \infty. \tag{3.3.45}$$

Indeed, the function g satisfies $f(g(x)) \sim x$ in (3.3.38), where $f(x) = (1/\overline{F}_U)(x) = \gamma x^\gamma L_U(x)^{-1}$. Setting $g(t) = \gamma^{-1/\gamma} t^{1/\gamma} L_0(t)$, we deduce that $L_0(t)$ satisfies $L_0(x)^\gamma L_U(\gamma^{-1/\gamma} x^{1/\gamma} L_0(x))^{-1} \sim 1$ and hence that $L_0(x) \sim L_{U,\gamma}^*(x)$ by Lemma 3.3.7.

The following theorem concerns the behavior of a scaled workload process in the fast regime. It involves the index γ characterizing asymptotic behavior of the tail distribution of the session length (see (3.3.2)) and the index δ characterizing the asymptotic behavior of the tail distribution of the rewards (see (3.3.4)). Recall that $\delta = 2$ corresponds to finite variance rewards. The integrals with respect to Gaussian and stable measures appearing below are defined in Appendices B.1.2 and B.1.3.

Theorem 3.3.8 *Suppose that the fast regime in (3.3.28) holds, namely $\lambda L_U(a)/a^{\gamma-1} \to \infty$ as λ and $a \to \infty$.*

(a) In the case (3.3.3) of finite variance rewards,

$$1 < \gamma < \delta = 2,$$

set

$$b = \Big(\gamma \lambda a^3 \overline{F}_U(a)\Big)^{1/2} = \Big(\lambda a^{3-\gamma} L_U(a)\Big)^{1/2} \tag{3.3.46}$$

for the normalization in the scaled workload process (3.3.26). Then, as $a \to \infty$,

$$\frac{b}{a} \to \infty \tag{3.3.47}$$

and

$$\frac{1}{b}(W_\lambda^*(at) - \lambda \nu \mathbb{E}(R) at) \xrightarrow{fdd} (\mathbb{E}R^2)^{1/2} \sigma B_H(t), \tag{3.3.48}$$

where $\xrightarrow{fdd}$ denotes convergence in the sense of finite-dimensional distributions, B_H is a standard FBM with the self-similarity parameter

$$H = \frac{3-\gamma}{2} \in \Big(\frac{1}{2}, 1\Big),$$

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and where

$$\sigma^2 = \int_{-\infty}^{\infty}\int_0^{\infty} K_1(s,u)^2\, u^{-(\gamma+1)} ds du = \frac{2}{\gamma(\gamma-1)(2-\gamma)(3-\gamma)}. \tag{3.3.49}$$

Alternatively, the limit process can be represented as

$$(\mathbb{E}R^2)^{1/2} \int_{-\infty}^{\infty}\int_0^{\infty} K_t(s,u)\, M_2(ds,du), \tag{3.3.50}$$

where $K_t(s,u)$ is the kernel defined in (3.3.10) and $M_2(ds,du)$ is a Gaussian random measure with control measure

$$ds\, u^{-(1+\gamma)} du.$$

(b) In the case (3.3.4) of infinite variance rewards, and

$$1 < \gamma < \delta < 2,$$

set

$$b = \lambda^{1/\delta} a^{(1+\delta-\gamma)/\delta} L_U(a)^{1/\delta} L^*_{R,\delta}\Big(\lambda a^{1-\gamma} L_U(a)\Big) \tag{3.3.51}$$

for the normalization. Then, as $a \to \infty$,

$$\frac{b}{a} \to \infty \tag{3.3.52}$$

and

$$\frac{1}{b}(W^*_\lambda(at) - \lambda\nu\mathbb{E}(R)at) \xrightarrow{fdd} Z_{\gamma,\delta}(t), \tag{3.3.53}$$

where

$$Z_{\gamma,\delta}(t) = \int_{-\infty}^{\infty}\int_0^{\infty}\int_0^{\infty} K_t(s,u) r \left(N(ds,du,dr) - ds\, u^{-(1+\gamma)} du\, r^{-(1+\delta)} dr\right) \tag{3.3.54}$$

$$\stackrel{d}{=} c_\delta \int_{-\infty}^{\infty}\int_0^{\infty} K_t(s,u)\, M_\delta(ds,du), \tag{3.3.55}$$

the random measure $M_\delta(ds,du)$ is δ-stable, totally skewed to the right and has the control measure

$$ds\, u^{-(\gamma+1)} du,$$

and where

$$c_\delta = \left(\frac{2\Gamma(2-\delta)}{\delta(1-\delta)}\cos(\frac{\delta\pi}{2})\right)^{1/\delta}. \tag{3.3.56}$$

Remark 3.3.9 The process $Z_{\gamma,\delta}$ in part (b) of the theorem is a δ-stable process, called the *Telecom process*, introduced in (3.3.5). It is H–self-similar with

$$H = \frac{\delta+1-\gamma}{\delta} \in \Big(\frac{1}{\delta}, 1\Big).$$

The stationary increments of $Z_{\gamma,\delta}$ are then LRD in the sense of condition A in Section 2.9 since

$$H - \frac{1}{\delta} = \frac{\delta - \gamma}{\gamma} > 0.$$

The stationary increments of FBM in part (a) are also LRD since $H > 1/2$.

Proof summary The idea of the proof of part (a) is as follows. Use the logarithm of the characteristic function (3.3.58) of the rescaled workload process. It is expressed as an integral. Because $a/b \to 0$, the integrand converges to the simpler form (3.3.61). Then apply the dominated convergence theorem and show that the integral can be expressed as (3.3.62) which is the logarithm of the characteristic function of a Gaussian process.

For part (b), express the logarithm of the characteristic function as (3.3.66) and compare it to (3.3.67) which turns out to be the logarithm of the characteristic function of the limit process (3.3.54). We shall now develop the proof in detail. □

Proof *We first show part* (a). By (3.3.2) and the fast regime condition in (3.3.28), we have

$$\left(\frac{b}{a}\right)^2 = \gamma\lambda a \overline{F}_U(a) \sim \lambda a a^{-\gamma} L_U(a) = \frac{\lambda L_U(a)}{a^{\gamma-1}} \to \infty, \tag{3.3.57}$$

proving (3.3.47). We now prove (3.3.48). Let $J \geq 1$ and $\theta_1, \ldots, \theta_J$ be arbitrary real numbers. By using the representation (3.3.9) of $W^*_\lambda(t)$ and (3.3.43), observe that

$$\begin{aligned}
I_\lambda &:= \log \mathbb{E} \exp\left\{ i \sum_{j=1}^{J} \theta_j \frac{W^*_\lambda(at_j) - \lambda\nu(\mathbb{E}R)at_j}{b} \right\} \qquad (3.3.58)\\
&= \log \mathbb{E} \exp\left\{ i \int_{-\infty}^{\infty} \int_0^{\infty} \int_0^{\infty} \sum_{j=1}^{J} \frac{\theta_j K_{at_j}(s,u) r}{b} \widetilde{N}(ds, du, dr) \right\}\\
&= \int_{-\infty}^{\infty} \int_0^{\infty} \int_0^{\infty} \Bigg(\exp\left\{ i \sum_{j=1}^{J} \frac{\theta_j K_{at_j}(s,u) r}{b} \right\} - 1\\
&\quad - i \sum_{j=1}^{J} \frac{\theta_j K_{at_j}(s,u) r}{b} \Bigg) \lambda ds F_U(du) F_R(dr)\\
&= \int_{-\infty}^{\infty} \int_0^{\infty} \int_0^{\infty} \Bigg(\exp\left\{ i \sum_{j=1}^{J} \theta_j K_{t_j}(s,u) (\frac{ra}{b}) \right\} - 1\\
&\quad - i \sum_{j=1}^{J} \theta_j K_{t_j}(s,u) (\frac{ra}{b}) \Bigg) \lambda a ds F_U(d(au)) F_R(dr),
\end{aligned}$$

where we used the scaling property (3.3.12) of K_t. By using (3.3.16), the kernel $K_t(s,u)$ is differentiable with respect to the variable u and $\frac{\partial K_t}{\partial u}(s,u) = 1_{\{0<u+s<t\}}$. We shall now perform integration by parts on the variable u using the relation

$$\frac{d}{dx}(e^{icx} - 1 - icx) = (e^{icx} - 1)ic$$

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for constant c. Since $d\mathbb{P}(U > au) = -d(F_U(au))$ and $K_t(s,0) = 0$, we get

$$I_\lambda = i \int_{-\infty}^{\infty} \int_0^{\infty} \int_0^{\infty} \Big(\exp\Big\{ i \sum_{j=1}^{J} \theta_j K_{t_j}(s,u)(\frac{ra}{b}) \Big\} - 1 \Big) \times \sum_{j=1}^{J} \theta_j \frac{\partial K_{t_j}}{\partial u}(s,u) \Big(\frac{ra}{b}\Big) \mathbb{P}(U > au) \lambda a ds du F_R(dr). \tag{3.3.59}$$

We showed that, in the fast regime,

$$\frac{a}{b} = \Big(\frac{a^2}{\gamma \lambda a^3 \overline{F}_U(a)} \Big)^{1/2} = \Big(\frac{1}{\gamma \lambda a \overline{F}_U(a)} \Big)^{1/2} \to 0$$

by (3.3.30), and that

$$\Big(\frac{a}{b}\Big)^2 \mathbb{P}(U > au) \lambda a = \frac{a^2}{\gamma \lambda a^3 \overline{F}_U(a)} \overline{F}_U(au) \lambda a = \frac{u^{-\gamma}}{\gamma} \frac{L_U(au)}{L_U(a)} \to \frac{u^{-\gamma}}{\gamma}, \tag{3.3.60}$$

as $a \to \infty$, for any $u > 0$. In view of these relations and since $e^z - 1 \sim z$ as $z \to 0$, we obtain that the integrand $I_\lambda(s,u,r)$ of the integral I_λ converges,

$$I_\lambda(s,u,r) := i \Big(\exp\Big\{ i \sum_{j=1}^{J} \theta_j K_{t_j}(s,u)(\frac{ra}{b}) \Big\} - 1 \Big) \sum_{j=1}^{J} \theta_j \frac{\partial K_{t_j}}{\partial u}(s,u)(\frac{ra}{b}) \mathbb{P}(U > au) \lambda a$$
$$\to -\sum_{j=1}^{J} \theta_j K_{t_j}(s,u) \sum_{j=1}^{J} \theta_j \frac{\partial K_{t_j}}{\partial u}(s,u)\, r^2 \frac{u^{-\gamma}}{\gamma}, \tag{3.3.61}$$

as $\lambda, a, b \to \infty$, for any $s \in \mathbb{R}$, $u, r > 0$. We now want to apply the dominated convergence theorem to conclude the convergence of integrals.

To show that the dominated convergence theorem applies, we need to split the integration domain into $\{au \le a_0\}$ and $\{au > a_0\}$, where a_0 is fixed. For $\{au \le a_0\}$, by using the relation $|e^{ix} - 1| \le |x|$, (3.3.13) and the expression for $\frac{\partial K_t}{\partial u}(s,u)$ above, we bound the integrand as

$$|I_\lambda(s,u,r)| \le C \frac{a^3 \lambda}{b^2} 1_{\{0<u+s<T\}} u r^2 \mathbb{P}(U > au),$$

where $T > 0$ is fixed and $C > 0$ is a constant which may change from line to line. By using $\mathbb{P}(U > au) \le \mathbb{E}U/(au)$ from Markov's inequality and $u \le 1$ on $\{au \le a_0\}$ for large enough a, we get that, on $\{au \le a_0\}$, and using (3.3.57),

$$|I_\lambda(s,u,r)| \le C \frac{a^2}{b^2} \lambda r^2 1_{\{0<u\le 1\}} 1_{\{-1<s\le T\}} \le C_1 \frac{r^2}{a \overline{F}_U(a)} 1_{\{0<u\le 1\}} 1_{\{-1<s\le T\}}.$$

By using (2.2.3) and for large enough a, $1/(a\overline{F}_U(a)) = \gamma a^{\gamma-1}/L_U(a) \le C a^{\gamma-1+\epsilon}$, where ϵ is arbitrarily small. On $\{au \le a_0\}$, this yields

$$|I_\lambda(s,u,r)| \le C r^2 u^{1-\gamma-\epsilon} 1_{\{0<u\le 1\}} 1_{\{-1<s\le T\}}.$$

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Note that the function in the bound is integrable on $(-\infty, \infty) \times (0, \infty) \times (0, \infty)$ with respect to $ds du F_R(dr)$ since $2 - \gamma - \epsilon > 0$ for small enough ϵ. On the other hand, for $\{au \geq a_0\}$, we bound the integrand as

$$|I_\lambda(s, u, r)| \leq C \sum_{j=1}^{J} K_{t_j}(s, u) \sum_{j=1}^{J} \frac{\partial K_{t_j}}{\partial u}(s, u)\, r^2 \Big(\frac{a}{b}\Big)^2 \mathbb{P}(U > au)\lambda a,$$

where we used the fact that $|e^{ix} - 1| \leq |x|$ and $K_t(s, u)$, $\frac{\partial K_t}{\partial u}(s, u) \geq 0$. By using (3.3.60),

$$\Big(\frac{a}{b}\Big)^2 \mathbb{P}(U > au)\lambda a = \frac{a^2}{\gamma \lambda a^3 \overline{F}_U(a)} \overline{F}_U(au)\lambda a = \frac{u^{-\gamma}}{\gamma} \frac{L_U(au)}{L_U(a)}$$

and applying Potter's bound (2.2.1) for large enough fixed a_0, we obtain for $au \geq a_0$ that

$$|I_\lambda(s, u, r)| \leq C \sum_{j=1}^{J} K_{t_j}(s, u) \sum_{j=1}^{J} \frac{\partial K_{t_j}}{\partial u}(s, u)\, r^2 u^{-\gamma \pm \epsilon},$$

where ϵ is arbitrarily small. By using the bound $K_t(s, u) \leq t \wedge u$ (see (3.3.13)), we further get that

$$|I_\lambda(s, u, r)| \leq C(T \wedge u) 1_{\{0<s+u<T\}} r^2 u^{-\gamma \pm \epsilon},$$

where T is fixed. The function in the bound is integrable on $(-\infty, \infty) \times (0, \infty) \times (0, \infty)$ with respect to $ds du F_R(dr)$.

Applying the dominated convergence theorem now yields

$$\begin{aligned}
I_\lambda &\to -\mathbb{E}R^2 \int_{-\infty}^{\infty} \int_0^{\infty} \sum_{j=1}^{J} \theta_j K_{t_j}(s, u) \sum_{j=1}^{J} \theta_j \frac{\partial K_{t_j}}{\partial u}(s, u) \frac{u^{-\gamma}}{\gamma} ds du \\
&= -\frac{\mathbb{E}R^2}{2} \int_{-\infty}^{\infty} \int_0^{\infty} \frac{\partial}{\partial u} \Big[\Big(\sum_{j=1}^{J} \theta_j K_{t_j}(s, u) \Big)^2 \Big] \frac{u^{-\gamma}}{\gamma} ds du \\
&= -\frac{\mathbb{E}R^2}{2} \int_{-\infty}^{\infty} \int_0^{\infty} \Big(\sum_{j=1}^{J} \theta_j K_{t_j}(s, u) \Big)^2 u^{-\gamma-1} ds du \qquad (3.3.62)
\end{aligned}$$

after integration by parts, observing that as $u \to 0$, $K_t(s, u)^2 u^{-\gamma} \leq u^2 u^{-\gamma} \to 0$. Since I_λ is the logarithm of the characteristic function, we see that the limit is a Gaussian process. This process can be represented as in (3.3.50). It is easily seen that it has stationary increments (use (3.3.14)) and is H–self-similar with parameter $H = (3 - \gamma)/2$. Hence, by Corollary 2.6.3, it is FBM.

To obtain the form of the scaling constant σ^2 in (3.3.49), observe that, by using (3.3.14) and making the change of variables $u = (y_2 - y_1)v$ in (3.3.64) below,

$$\begin{aligned}
\sigma^2 &= \int_{-\infty}^{\infty} \int_0^{\infty} K_1(s, u)^2 u^{-(\gamma+1)} ds du \\
&= \int_{-\infty}^{\infty} \int_0^{\infty} \int_0^1 \int_0^1 1_{\{s<y_1<s+u\}} 1_{\{s<y_2<s+u\}} u^{-(\gamma+1)} dy_1 dy_2 ds du
\end{aligned}$$

$$
\begin{aligned}
&= 2\int_{-\infty}^{\infty}\int_{0}^{\infty}\int_{0<y_1<y_2<1} 1_{\{s<y_1<s+u\}}1_{\{s<y_2<s+u\}}u^{-(\gamma+1)}dy_1dy_2dsdu \\
&= 2\int_{0<y_1<y_2<1}\int_{y_2-y_1}^{\infty}\Big(\int_{-\infty}^{\infty}1_{\{y_2-u<s<y_1\}}ds\Big)u^{-(\gamma+1)}dudy_1dy_2 \qquad (3.3.63)\\
&= 2\int_{0<y_1<y_2<1}\int_{y_2-y_1}^{\infty}(u-(y_2-y_1))u^{-(\gamma+1)}dudy_1dy_2 \\
&= 2\int_{0<y_1<y_2<1}(y_2-y_1)^{1-\gamma}dy_1dy_2\int_1^{\infty}(v-1)v^{-(\gamma+1)}dv \qquad (3.3.64)\\
&= 2\frac{1}{(2-\gamma)(3-\gamma)}\Big(\frac{1}{\gamma-1}-\frac{1}{\gamma}\Big) = \frac{2}{\gamma(\gamma-1)(2-\gamma)(3-\gamma)},
\end{aligned}
$$

by using, for (3.3.63),

$$
1_{\{s<y_1<s+u\}}1_{\{s<y_2<s+u\}} = 1_{\{y_1-u<s<y_1\}}1_{\{y_2-u<s<y_2\}} = \begin{cases} 0, & \text{if } u \le y_2-y_1, \\ 1_{\{y_2-u<s<y_1\}}, & \text{if } u > y_2-y_1. \end{cases}
$$

We now turn to part (b). By (3.3.51) and (3.3.44) and the fast regime condition (3.3.28),

$$
\Big(\frac{b}{a}\Big)^{\delta} = \lambda a^{1-\gamma}L_U(a)\Big(L^{*}_{R,\delta}(\lambda a^{1-\gamma}L_U(a))\Big)^{\delta} \to \infty, \qquad (3.3.65)
$$

proving (3.3.52).

We now prove the convergence of the finite-dimensional distributions (3.3.53). Using the notation I_λ introduced in (3.3.58) and arguing as above, write

$$
I_\lambda = \int_{-\infty}^{\infty}\int_0^{\infty}\int_0^{\infty} h(s,u,r)\lambda a ds F_U(d(au))F_R(d\frac{b}{a}r), \qquad (3.3.66)
$$

where

$$
h(s,u,r) = \exp\Big\{i\sum_{j=1}^{J}\theta_j K_{t_j}(s,u)r\Big\} - 1 - i\sum_{j=1}^{J}\theta_j K_{t_j}(s,u)r.
$$

On the other hand, the logarithm of the characteristic function of the limit process $Z_{\gamma,\delta}(t)$ in (3.3.54) is

$$
I = \int_{-\infty}^{\infty}\int_0^{\infty}\int_0^{\infty} h(s,u,r)dsu^{-\gamma-1}dur^{-\delta-1}dr. \qquad (3.3.67)
$$

Write the difference between I_λ and I as

$$
I_\lambda - I = I_\epsilon^1 + I_\epsilon^2 + I_\epsilon^3,
$$

where, for $k = 1, 2, 3$ and $\epsilon > 0$,

$$
I_\epsilon^k = \int_{-\infty}^{\infty}\iint_{A_\epsilon^k} h(s,u,r)\Big(\lambda a ds F_U(d(au))F_R(d\frac{b}{a}r) - dsu^{-\gamma-1}dur^{-\delta-1}dr\Big)
$$

with the integration domains $A_\epsilon^1 = \{u > \epsilon, r > \epsilon\}$, $A_\epsilon^2 = \{u < \epsilon < r\}$ and $A_\epsilon^3 = \{r < \epsilon\}$, involving u and r but not involving s.

We show next that, as $\lambda, a, b \to \infty$, for fixed ϵ, $I_\epsilon^1 \to 0$ and that $I_\epsilon^2, I_\epsilon^3$ are arbitrarily small for small enough ϵ. This will establish the desired convergence to the limit process given by (3.3.54).

Step 1: For I_ϵ^1, let

$$H(u,r) = \frac{1}{ur}\int_{-\infty}^{\infty} h(s,u,r)ds$$

and write

$$\begin{aligned} I_\epsilon^1 &= \iint\limits_{A_\epsilon^1} H(u,r)\mu_\lambda(du,dr) - \iint\limits_{A_\epsilon^1} H(u,r)\mu(du,dr) \\ &= \mu_\lambda(A_\epsilon^1)\iint\limits_{A_\epsilon^1} H(u,r)\widetilde{\mu}_\lambda(du,dr) - \mu(A_\epsilon^1)\iint\limits_{A_\epsilon^1} H(u,r)\widetilde{\mu}(du,dr), \end{aligned}$$

where

$$\begin{aligned} \mu_\lambda(du,dr) &= \lambda a\, u F_U(dau)\, r F_R(d(b/a)r), \\ \mu(du,dr) &= u^{-\gamma}du\, r^{-\delta}dr \end{aligned}$$

and $\widetilde{\mu}_\lambda$ and $\widetilde{\mu}$ are corresponding (normalized) probability measures. By using the inequality $|e^{ix}-1| \le |x|$, note that $|h(s,u,r)|$ is bounded by $2\sum_{j=1}^J |\theta_j| K_{t_j}(s,u)r$. This implies that $H(u,r)$ is bounded by using (3.3.15). Note also that $H(u,r)$ is continuous. Then, to show that $I_\epsilon^1 \to 0$ as $\lambda, a, b \to \infty$, it is enough to prove that the normalizations converge

$$\mu_\lambda(A_\epsilon^1) \to \mu(A_\epsilon^1), \tag{3.3.68}$$

and that the normalized probability measures converge weakly

$$\widetilde{\mu}_\lambda \xrightarrow{d} \widetilde{\mu}. \tag{3.3.69}$$

For the convergence (3.3.68) of the normalizations, observe that

$$\begin{aligned} \mu_\lambda(A_\epsilon^1) &= \lambda a \int_\epsilon^\infty u F_U(dau)\int_\epsilon^\infty r F_R(d\frac{b}{a}r) = \frac{\lambda a}{b}\int_{a\epsilon}^\infty u F_U(du)\int_{b\epsilon/a}^\infty r F_R(dr) \\ &= \frac{\lambda a}{b}\Big(a\epsilon\overline{F}_U(a\epsilon) + \int_{a\epsilon}^\infty \overline{F}_U(u)du\Big)\Big(\frac{b}{a}\epsilon\overline{F}_R(\frac{b}{a}\epsilon) + \int_{b\epsilon/a}^\infty \overline{F}_R(r)dr\Big) \\ &\sim \frac{\lambda a}{b}\Big(a\epsilon\overline{F}_U(a\epsilon)\Big(1-\frac{1}{-\gamma+1}\Big)\Big)\Big(\frac{b}{a}\epsilon\overline{F}_R(\frac{b}{a}\epsilon)\Big(1-\frac{1}{-\delta+1}\Big)\Big) \\ &= \frac{\lambda a}{b}\Big(\frac{(a\epsilon)^{1-\gamma}}{\gamma}L_U(a\epsilon)\frac{\gamma}{\gamma-1}\Big)\Big(\frac{1}{\delta}\Big(\frac{b}{a}\epsilon\Big)^{1-\delta}L_R(\frac{b}{a}\epsilon)\frac{\delta}{\delta-1}\Big) \\ &= \frac{\epsilon^{1-\delta}}{\delta-1}\frac{\epsilon^{1-\gamma}}{\gamma-1}\frac{L_U(a\epsilon)}{L_U(a)}\lambda a^{1+\delta-\gamma}b^{-\delta}L_U(a)L_R(\frac{b}{a}\epsilon) \end{aligned}$$

by integration by parts and by using (3.3.2) and (3.3.4) for the regularly varying functions $\overline{F}_U, \overline{F}_R$. Note that

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$$\lambda a^{1+\delta-\gamma} b^{-\delta} L_U(a) L_R(\frac{b}{a}\epsilon) = L^*_{R,\delta}(\lambda a^{1-\gamma} L_U(a))^{-\delta} \times L_R\Big(\lambda^{1/\delta} a^{(1-\gamma)/\delta} L_U(a)^{1/\delta} L^*_{R,\delta}(\lambda a^{1-\gamma} L_U(a))\epsilon\Big) \sim 1, \tag{3.3.70}$$

by using the expression for b in (3.3.51). This is of the form $L^*_{R,\delta}(y)^{-\delta} L_R(y^{1/\delta} L^*_{R,\delta}(y)\epsilon)$ where $y = \lambda a^{1-\gamma} L_U(a)$, and hence by (3.3.44), this tends to 1 as $y \to \infty$, since under the fast regime $\lambda, a, b \to \infty$ imply $y \to \infty$ by (3.3.65). It follows that

$$\mu_\lambda(A^1_\epsilon) \to \frac{\epsilon^{1-\delta}}{\delta-1}\frac{\epsilon^{1-\gamma}}{\gamma-1} = \int_\epsilon^\infty \int_\epsilon^\infty \mu(du, dr) = \mu(A^1_\epsilon).$$

The convergence (3.3.69) of the probability measures can be proved similarly since it is enough to show the convergence of $\widetilde{\mu}_\lambda((u_0,\infty)\times(r_0,\infty))$ to $\widetilde{\mu}((u_0,\infty)\times(r_0,\infty))$.

Step 2: Consider now I^2_ϵ. By using the inequalities $|e^{ix} - 1 - ix| \le C|x|^{1+\kappa}$ for fixed $\kappa \in [0,1]$ to be chosen later, $K_t(s,u) \le u$ and the fact that $\int_{-\infty}^\infty K_t(s,u)ds = ut$ (see (3.3.15)), we have

$$\Big|\int_{-\infty}^\infty \int\int_{A^2_\epsilon} h(s,u,r)\lambda a ds F_U(d(au)) F_R(d\frac{b}{a}r)\Big| \le C\lambda a \int_\epsilon^\infty r^{1+\kappa} F_R(d\frac{b}{a}r) \int_0^\epsilon u^{1+\kappa} F_U(dau). \tag{3.3.71}$$

We need $1+\kappa < \delta$ for the last integral in r to be well-defined. The limits of integration in the last integral in u result from $A^2_\epsilon = \{u < \epsilon < r\}$; this integral converges. The bound (3.3.71) behaves asymptotically as

$$C\lambda a\epsilon^{1+\kappa}\overline{F}_R(\frac{b}{a}\epsilon)\epsilon^{1+\kappa}\overline{F}_U(a\epsilon) = C\epsilon^{2(1+\kappa)-\gamma-\delta}\frac{L_U(a\epsilon)}{L_U(a)}$$

$$\times\ \lambda a^{1+\delta-\gamma} b^{-\delta} L_U(a) L_R(\frac{b}{a}\epsilon) \sim C\epsilon^{2(1+\kappa)-\gamma-\delta}, \tag{3.3.72}$$

by using (3.3.70). This is arbitrarily small for small enough ϵ if $1+\kappa > (\gamma+\delta)/2$. One deals in a similar way with the corresponding term

$$\int_{-\infty}^\infty \int\int_{A^2_\epsilon} h(s,u,r) ds u^{-\gamma-1} du r^{-\delta-1} dr$$

for the limiting process. The computations are in fact easier and one can show that this term is arbitrarily small for small enough ϵ.

Step 3: Finally, consider I^3_ϵ. By using the inequalities $|e^{ix} - 1 - ix| \le C|x|^2$, $K_t(s,u) \le u \wedge t$ and the fact that $\int_{-\infty}^\infty K_t(s,u)ds = ut$ (see (3.3.15)), we have with $A^3_\epsilon = \{r < \epsilon\}$

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$$\Big| \int_{-\infty}^{\infty} \int\int_{A_\epsilon^3} h(s,u,r)\lambda a ds F_U(d(au)) F_R(d\frac{b}{a}r) \Big|$$
$$\le C\lambda a \int_0^\epsilon r^2 F_R(d\frac{b}{a}r) \int_0^\infty u(u\wedge T) F_U(dau),$$

where $t_1, \dots, t_J \le T$. Arguing as for (3.3.72), the bound behaves asymptotically as

$$C\lambda a \epsilon^2 \overline{F}_R(\frac{b}{a}\epsilon) T^2 \overline{F}_U(aT) \sim C'\epsilon^{2-\delta}.$$

This is arbitrarily small for small enough ϵ. Similarly and in an easier way, one can show that

$$\int_{-\infty}^{\infty} \int\int_{A_\epsilon^3} h(s,u,r) ds u^{-\gamma-1} du r^{-\delta-1} dr$$

is arbitrarily small for small enough ϵ.

Steps 1, 2 and 3 yield

$$I_\lambda \to I, \quad \text{as } \lambda, a, b \to \infty.$$

By using (3.3.67), we can write I in (3.3.67) as

$$I = \int_{-\infty}^{\infty} \int_0^\infty \int_0^\infty \Big(e^{i \sum_{j=1}^J \theta_j K_{t_j}(s,u) r} - 1 - i \sum_{j=1}^J \theta_j K_{t_j}(s,u) r \Big) ds u^{-\gamma-1} du r^{-\delta-1} dr. \tag{3.3.73}$$

This is the logarithm of the characteristic function of the limit process (3.3.54). The process (3.3.54) can also be represented as (3.3.55) by using (B.1.39) in Appendix B.1.4. □

The next theorem concerns the behavior of a scaled workload process in the slow regime.

Theorem 3.3.10 *Suppose that the slow regime in (3.3.28) holds. Suppose*

$$1 < \gamma < \delta \le 2$$

and set

$$b = (\lambda a)^{1/\gamma} L^*_{U,\gamma}(\lambda a) \tag{3.3.74}$$

in the scaled workload process (3.3.26), for the normalization. Then, as $a \to \infty$,

$$\frac{b}{a} \to 0 \tag{3.3.75}$$

and

$$\frac{1}{b}(W^*_\lambda(at) - \lambda\nu\mathbb{E}(R)at) \xrightarrow{fdd} (\mathbb{E}R^\gamma)^{1/\gamma} \Lambda_\gamma(t), \tag{3.3.76}$$

where Λ_γ is a γ-stable Lévy process, totally skewed to the right. The limit process can be represented as

$$(\mathbb{E}R^\gamma)^{1/\gamma}\,\Lambda_\gamma(t) = \int_{-\infty}^{\infty}\int_0^{\infty}\int_0^{\infty} 1_{\{0<s<t\}} ur \times \Big(N(ds,du,dr) - ds\,u^{-(1+\gamma)}du\,F_R(dr)\Big) \tag{3.3.77}$$

$$\stackrel{d}{=} c_\gamma \int_{-\infty}^{\infty}\int_0^{\infty} 1_{\{0<s<t\}} r M_\gamma(ds,dr) \tag{3.3.78}$$

where c_γ is defined in (3.3.56) (with $\delta = \gamma$) and the random measure $M_\gamma(ds, dr)$ is γ-stable, totally skewed to the right with control measure $ds\,F_R(dr)$.

Remark 3.3.11 Let $g(t) = (1/\overline{F}_U)^{\leftarrow}(t)$ as in (3.3.32). Then, by (3.3.45),

$$g(t) \sim \gamma^{-1/\gamma} t^{1/\gamma} L^*_{U,\gamma}(t), \quad \text{as } t \to \infty,$$

where, by (3.3.44), $L^*_{U,\gamma}(t)^{-\gamma} L(t^{1/\gamma} L^*_{U,\gamma}(t)x) \to 1$ for $x > 0$. Therefore, the normalization constant b defined in (3.3.74) satisfies

$$b = (\lambda a)^{1/\gamma} L^*_{U,\gamma}(\lambda a) \sim \gamma^{1/\gamma} g(\lambda a), \quad \text{as } a \to \infty. \tag{3.3.79}$$

Proof summary The idea of the proof is as follows. Express the logarithm of the characteristic function of the rescaled workload process as an integral (3.3.81) with integrand (3.3.83). Because $a/b \to \infty$, the integrand converges as in (3.3.84). Then apply the dominated convergence theorem to get (3.3.87). It can be expressed as (3.3.88), which is the logarithm of the characteristic function of the process in (3.3.77). □

Proof By (3.3.74) and (3.3.79),

$$\frac{b}{a} \sim \frac{\gamma^{1/\gamma} g(\lambda a)}{a} \to c^{1/\gamma} = 0, \tag{3.3.80}$$

under the slow regime ($c = 0$), in view of Proposition 3.3.6, (i). Thus, (3.3.75) holds.

Let us now prove (3.3.76). This time, we will have to make a/b appear since $a/b \to \infty$. By using (3.3.43) and integration by parts, observe that

$$\begin{aligned} I_\lambda &:= \log \mathbb{E} \exp\Big\{ i \sum_{j=1}^J \theta_j \frac{W^*_\lambda(at_j) - \lambda\nu(\mathbb{E}R)at_j}{b} \Big\} \\ &= \log \mathbb{E} \exp\Big\{ i \int_{-\infty}^{\infty}\int_0^{\infty}\int_0^{\infty} \sum_{j=1}^J \frac{\theta_j K_{at_j}(s,u) r}{b} \widetilde{N}(ds,du,dr) \Big\} \\ &= \int_{-\infty}^{\infty}\int_0^{\infty}\int_0^{\infty} \Big(\exp\Big\{ i \sum_{j=1}^J \frac{\theta_j K_{at_j}(s,u) r}{b} \Big\} - 1 \\ &\quad - i \sum_{j=1}^J \frac{\theta_j K_{at_j}(s,u) r}{b} \Big) \lambda ds F_U(du) F_R(dr) \end{aligned}$$

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$$= i\int_{-\infty}^{\infty}\int_0^{\infty}\int_0^{\infty}\Big(\exp\Big\{i\sum_{j=1}^{J}\frac{\theta_j K_{at_j}(s,u)r}{b}\Big\}-1\Big)$$
$$\times\sum_{j=1}^{J}\theta_j\frac{\partial K_{at_j}}{\partial u}(s,u)\frac{r}{b}\lambda\mathbb{P}(U>u)dsduF_R(dr),$$

as was done in (3.3.59). We can now make the change of variables $s \to bs$ and $u \to bu$. Then, $\frac{1}{b}K_{at}(s,u)$ becomes $\frac{1}{b}K_{b\frac{a}{b}t}(bs,bu) = K_{\frac{a}{b}t}(s,u)$ by (3.3.12), and $\frac{\partial K_{at}}{\partial u}(s,u) = 1_{\{0<s+u<at\}}$ by (3.3.14) becomes $1_{\{0<bs+bu<at\}} = 1_{\{0<s+u<\frac{a}{b}t\}}$. With the further change of variables $s \to \frac{a}{b}s - u$, $K_{\frac{a}{b}t}(s,u)$ becomes $K_{\frac{a}{b}t}(\frac{a}{b}s-u,u)$ and $1_{\{0<s+u<\frac{a}{b}t\}}$ becomes $1_{\{0<s<t\}}$. Therefore,

$$I_\lambda = i\int_{-\infty}^{\infty}\int_0^{\infty}\int_0^{\infty}\Big(\exp\Big\{i\sum_{j=1}^{J}\theta_j K_{\frac{a}{b}t_j}(\frac{a}{b}s-u,u)r\Big\}-1\Big)$$
$$\times\sum_{j=1}^{J}\theta_j 1_{\{0<s<t_j\}}r\,\lambda a\mathbb{P}(U>bu)dsduF_R(dr)$$
$$=: \int_{-\infty}^{\infty}\int_0^{\infty}\int_0^{\infty} I_\lambda(s,u,r)dsduF_R(dr). \tag{3.3.81}$$

By using (3.3.74) for b, we can write[1]

$$\lambda a\mathbb{P}(U>bu) = \frac{u^{-\gamma}}{\gamma}\lambda ab^{-\gamma}L_U(bu) = \frac{u^{-\gamma}}{\gamma}\lambda a(\lambda a)^{-1}(L^*_{U,\gamma}(\lambda a))^{-\gamma}L_U(bu)$$
$$= \frac{u^{-\gamma}}{\gamma}(L^*_{U,\gamma}(\lambda a))^{-\gamma}L_U\Big((\lambda a)^{1/\gamma}L^*_{U,\gamma}(\lambda a)u\Big) =: \frac{u^{-\gamma}}{\gamma}f_\lambda(u) \tag{3.3.82}$$

and therefore

$$I_\lambda(s,u,r) = i\Big(\exp\Big\{i\sum_{j=1}^{J}\theta_j K_{\frac{a}{b}t_j}(\frac{a}{b}s-u,u)r\Big\}-1\Big)\sum_{j=1}^{J}\theta_j 1_{\{0<s<t_j\}}r\,\frac{u^{-\gamma}}{\gamma}f_\lambda(u). \tag{3.3.83}$$

Note that, by (3.3.14),

$$K_{zt}(zs-u,u) = \int_0^u 1_{\{0<y+zs-u<zt\}}dy \to \int_0^u 1_{\{0<s<t\}}dy = u1_{\{0<s<t\}},$$

as $z \to \infty$. Then, since $a/b \to \infty$ in the slow growth regime, and since $f_\lambda(u) \to 1$ as $\lambda \to \infty$ by using (3.3.44),

$$I_\lambda(s,u,r) \to i\Big(\exp\Big\{i\sum_{j=1}^{J}\theta_j 1_{\{0<s<t_j\}}ur\Big\}-1\Big)\sum_{j=1}^{J}\theta_j 1_{\{0<s<t_j\}}r\,\frac{u^{-\gamma}}{\gamma} =: I(s,u,r) \tag{3.3.84}$$

[1] Do not confuse the integral I_λ with its integrand $I_\lambda(s,u,r)$.

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for all $s \in \mathbb{R}$, $u > 0$, $r > 0$. We want to conclude next that the convergence also takes place for the corresponding integrals; that is,

$$I_\lambda \to \int_{-\infty}^{\infty}\int_0^{\infty}\int_0^{\infty} I(s,u,r)dsduF_R(dr) =: I. \tag{3.3.85}$$

To show (3.3.85), it is enough to prove that $|I_\lambda(s,u,r)|$ is dominated by an integrable function. For this, express $f_\lambda(u)$, defined in (3.3.82), as

$$f_\lambda(u) = (L^*_{U,\gamma}(\lambda a))^{-\gamma} L_U\Big((\lambda a)^{1/\gamma}L^*_{U,\gamma}(\lambda a)\Big)\frac{L_U\Big((\lambda a)^{1/\gamma}L^*_{U,\gamma}(\lambda a)u\Big)}{L_U\Big((\lambda a)^{1/\gamma}L^*_{U,\gamma}(\lambda a)\Big)}$$

and consider $I_\lambda(s,u,r)$ defined in (3.3.83).

For $(\lambda a)^{1/\gamma}L^*_{U,\gamma}(\lambda a)u \geq u_0$ with large enough fixed u_0 and large enough λa, by using (3.3.44) and Potter's bound (2.2.1), we get that

$$f_\lambda(u) \leq C\max\{u^{-\epsilon}, u^{\epsilon}\},$$

where ϵ is arbitrarily small and C is a constant. By using $|e^{ix}-1| \leq C(|x|^\kappa \wedge 1)$ with fixed $0 < \kappa < 1$ and the inequality $K_t(s,u) \leq u$, we get that

$$\left|\exp\left\{i\sum_{j=1}^{J}\theta_j K_{\frac{a}{b}t_j}(\frac{a}{b}s-u,u)r\right\}-1\right| \leq C'(u^\kappa r^\kappa \wedge 1). \tag{3.3.86}$$

Then, for $(\lambda a)^{1/\gamma}L^*_{U,\gamma}(\lambda a)u \geq u_0$ and large enough λa,

$$|I_\lambda(s,u,r)| \leq C1_{\{0<s<T\}}r(u^\kappa r^\kappa \wedge 1)u^{-\gamma}\max\{u^{-\epsilon},u^{\epsilon}\} =: Cg_1(s,u,r).$$

Note that, since $(u^\kappa r^\kappa \wedge 1) \leq u^\kappa r^\kappa$ and $(u^\kappa r^\kappa \wedge 1) \leq 1$,

$$\int_{-\infty}^{\infty}\int_0^{\infty}\int_0^{\infty} g_1(s,u,r)dsduF_R(dr) \leq T\mathbb{E}(R^{1+\kappa})\int_0^{u_1} u^{\kappa-\gamma}\max\{u^{-\epsilon},u^{\epsilon}\}du$$
$$+\,T\mathbb{E}(R)\int_{u_1}^{\infty} u^{-\gamma}\max\{u^{-\epsilon},u^{\epsilon}\}du$$

for fixed u_1. The bound is finite if we choose $1+\kappa < \delta$, $\kappa-\gamma-\epsilon+1 > 0$ and $-\gamma+\epsilon+1 < 0$ or, equivalently, $\gamma+\epsilon < 1+\kappa < \delta$ and $1+\epsilon < \gamma$.

On $(\lambda a)^{1/\gamma}L^*_{U,\gamma}(\lambda a)u \leq u_0$, we have $u \leq u_1$ for some fixed u_1 since $(\lambda a)^{1/\gamma}L^*_{U,\gamma}(\lambda a) \to \infty$ as $a \to \infty$. Then, by using (3.3.83) and (3.3.86), we have

$$|I_\lambda(s,u,r)| \leq C(u^\kappa r^\kappa \wedge 1)1_{\{0<s<T\}}1_{\{u\leq u_1\}}ru^{-\gamma}f_\lambda(u) \leq C1_{\{0<s<T\}}1_{\{u\leq u_1\}}u^{\kappa-\gamma}r^{1+\kappa}f_\lambda(u).$$

We need to estimate $f_\lambda(u)$ which is defined in (3.3.82). In particular, we need to bound $(L^*_{U,\gamma}(\lambda a))^{-\gamma}$ by some function of u. Since $1/L^*_{U,\gamma}$ is a slowly varying function, we have $(\lambda a)^{1/\gamma}u \leq u_0/L^*_{U,\gamma}(\lambda a) \leq C(\lambda a)^{\delta_0}u_0$ where $\delta_0 > 0$ is arbitrarily small or $\lambda a \leq C'u^{-\gamma/(1-\gamma\delta_0)} = C'u^{-\gamma-\delta_1}$ where $\delta_1 > 0$ is arbitrarily small. This also implies that $(L^*_{U,\gamma}(\lambda a))^{-\gamma} \leq C(\lambda a)^{\delta_0\gamma} \leq C'u^{-\epsilon}$ where $\delta_0 > 0$ and $\epsilon > 0$ are arbitrarily small. The other term of $f_\lambda(u)$ to bound is $L_U((\lambda a)^{1/\gamma}L^*_{U,\gamma}(\lambda a)u)$. On $(\lambda a)^{1/\gamma}L^*_{U,\gamma}(\lambda a)u \leq u_0$, this term is bounded by a constant since this is the case for the function $L_U(v) = v^\gamma \mathbb{P}(U > v)$ on $v \in [0, u_0]$. Then, combining the two bounds, on $(\lambda a)^{1/\gamma} L^*_{U,\gamma}(\lambda a)u \le u_0$,

$$|I_\lambda(s,u,r)| \le C 1_{\{0<s<T\}} 1_{\{u\le u_1\}} u^{\kappa-\gamma-\epsilon} r^{1+\kappa} =: g_2(s,u,r),$$

where u_1 is fixed and ϵ is arbitrarily small. The function $g_2(s,u,r)$ is integrable with respect to $dsduF_R(dr)$ as long as $\kappa-\gamma-\epsilon+1 > 0$ and $1+\kappa < \delta$ or, equivalently, $\gamma+\epsilon < 1+\kappa < \delta$.

This establishes the convergence in (3.3.85). The limit (3.3.85) can be written as

$$I = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\int_0^{\infty} \Big(\exp\Big\{\sum_{j=1}^J \theta_j 1_{\{0<s<t_j\}} ur\Big\} - 1\Big)\Big(i\sum_{j=1}^J \theta_j 1_{\{0<s<t_j\}} r\Big)\frac{u^{-\gamma}}{\gamma} dsduF_R(dr). \tag{3.3.87}$$

After integration by parts on the variable u, one gets

$$I = \int_{-\infty}^{\infty}\int_0^{\infty}\int_0^{\infty} \Big(\exp\Big\{i\sum_{j=1}^J \theta_j 1_{\{0<s<t_j\}} ur\Big\} - 1 - i\sum_{j=1}^J \theta_j 1_{\{0<s<t_j\}} ur\Big) u^{-\gamma-1} dsduF_R(dr), \tag{3.3.88}$$

which is the logarithm of the characteristic function of the process in (3.3.77). The equality (3.3.78) follows by using (B.1.39) in Appendix B.1.4. □

The third theorem concerns the behavior of a scaled workload process in the intermediate regime.

Theorem 3.3.12 *Suppose the intermediate regime in (3.3.28) holds. Suppose*

$$1 < \gamma < \delta \le 2$$

and set

$$b = a \tag{3.3.89}$$

in the scaled workload process (3.3.26). Then, as $a \to \infty$,

$$\frac{1}{b}(W^*_\lambda(at) - \lambda\nu\mathbb{E}(R)at) \xrightarrow{fdd} c^{1/(\gamma-1)} Y_{\gamma,R}\Big(\frac{t}{c^{1/(\gamma-1)}}\Big),$$

where c appears in (3.3.28), and

$$Y_{\gamma,R}(t) = \int_{-\infty}^{\infty}\int_0^{\infty}\int_0^{\infty} K_t(s,u) r \left(N(ds,du,dr) - ds\, u^{-(1+\gamma)} du\, F_R(dr)\right) \tag{3.3.90}$$

$$= \int_{-\infty}^{\infty}\int_0^{\infty} K_t(s,u)\Big(\int_0^{\infty} r\, N(ds,du,dr) - \mathbb{E}(R)\, ds\, u^{-(1+\gamma)} du\Big). \tag{3.3.91}$$

Remark 3.3.13 The limit process $Y_{\gamma,R}$ is not self-similar, because N does not have the scaling properties that a Gaussian or a stable process has. However, if we assume that the reward distribution $F_R(dr)$ has finite variance, that is $\mathbb{E}R^2 < \infty$, then

$$\mathrm{Var}(Y_{\gamma,R}(t)) = \mathbb{E}(R^2)\int_{-\infty}^{\infty}\int_0^{\infty} K_t(s,u)^2\, ds\, u^{-(1+\gamma)} du = \mathbb{E}(R^2)\,\sigma^2\, t^{2H}, \quad H = \frac{3-\gamma}{2},$$

where σ^2 is given in (3.3.49). Thus, in this case $Y_{\gamma,R}$ is *second-order self-similar* with Hurst index H.

Remark 3.3.14 Benassi, Jaffard, and Roux [113] introduced the notion of *local asymptotic self-similarity* as another means of generalizing the class of self-similar processes. See also Section 2.6.8. It is shown in Gaigalas and Kaj [380] and with a proof more adapted to the present setting in Gaigalas [379], that the process $Y_{\gamma,R}$ is locally asymptotically self-similar with index H and with fractional Brownian motion as tangent process, in the sense that

$$\left\{\frac{Y_{\gamma,R}(t+\lambda u)-Y_{\gamma,R}(t)}{\lambda^H},\ u\in\mathbb{R}\right\}\stackrel{fdd}{\longrightarrow}\{B_H(u),\ u\in\mathbb{R}\},\quad\text{as}\quad\lambda\to 0.$$

Benassi, Cohen, and Istas [115] defined a stochastic process $X(t)$ to be *asymptotically self-similar at infinity* with index H if there exists a process $V(t)$ such that

$$\lambda^{-H}X(\lambda t)\stackrel{fdd}{\longrightarrow}V(t),\quad\text{as }\lambda\to\infty.$$

The intermediate limit process $Y_{\gamma,R}(t)$ is asymptotically self-similar at infinity with index $H=1/\gamma$ and the asymptotic process $V(t)$ is given by a γ-stable Lévy process, totally skewed to the right, see Gaigalas [379].

Remark 3.3.15 In the special case of fixed rewards $R\equiv 1$, $F_R(dr)$ is the delta function at $r=1$ and the limit process (3.3.91) becomes

$$Y_\gamma(t)=\int_{-\infty}^{\infty}\int_0^{\infty}K_t(s,u)\Big(N(ds,du)-ds\,u^{-(1+\gamma)}du\Big).$$

Proof of Theorem 3.3.12 Let $c_0=c^{1/(\gamma-1)}$. As in the proof of Theorem 3.3.10, we have[2]

$$\begin{aligned}
I_\lambda &:= \log\mathbb{E}\exp\left\{i\sum_{j=1}^{J}\theta_j\frac{W_\lambda^*(at_j)-\lambda\nu(\mathbb{E}R)at_j}{a}\right\}\\
&= i\int_{-\infty}^{\infty}\int_0^{\infty}\int_0^{\infty}\left(\exp\left\{i\sum_{j=1}^{J}\theta_jK_{at_j}(s,u)\frac{r}{a}\right\}-1\right)\\
&\quad\times\sum_{j=1}^{J}\theta_j\frac{\partial K_{at_j}}{\partial u}(s,u)\frac{r}{a}\,\lambda\mathbb{P}(U>u)dsduF_R(dr).
\end{aligned}$$

We now make the change of variables $s\to c_0as$ and $u\to c_0au$. Then, $K_{at}(s,u)=K_{c_0a\frac{t}{c_0}}(s,u)$ becomes $K_{c_0a\frac{t}{c_0}}(c_0as,c_0au)=c_0aK_{\frac{t}{c_0}}(s,u)$ by the scaling property (3.3.12). Then,

$$\begin{aligned}
I_\lambda &= i\int_{-\infty}^{\infty}\int_0^{\infty}\int_0^{\infty}\left(\exp\left\{i\sum_{j=1}^{J}\theta_jc_0K_{t_j/c_0}(s,u)r\right\}-1\right)\sum_{j=1}^{J}\theta_jc_0\frac{\partial K_{t_j/c_0}}{\partial u}(s,u)\\
&\quad\times r\,\lambda c_0a\mathbb{P}(U>c_0au)dsduF_R(dr)\\
&=: \int_{-\infty}^{\infty}\int_0^{\infty}\int_0^{\infty}I_\lambda(s,u,r)dsduF_R(dr).
\end{aligned}$$

[2] Do not confuse I_λ with the integrand $I_\lambda(s,u,r)$ defined below.

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By using (3.3.28), note that $\lambda L_U(u)a^{1-\gamma} \sim c \sim c_0^{\gamma-1}$ and therefore

$$\lambda c_0 a\mathbb{P}(U > c_0 au) \sim c_0^{\gamma} \frac{\mathbb{P}(U > c_0 au)}{a^{-\gamma} L_U(a)} = \frac{c_0^{\gamma}}{\gamma} \frac{\mathbb{P}(U > c_0 au)}{\mathbb{P}(U > a)} \sim \frac{c_0^{\gamma}}{\gamma}(c_0 u)^{-\gamma} = \frac{u^{-\gamma}}{\gamma}.$$

Then, as $a \to \infty$,

$$I_\lambda(s,u,r) \to i\Big(\exp\Big\{i\sum_{j=1}^{J}\theta_j c_0 K_{t_j/c_0}(s,u)r\Big\} - 1\Big)$$
$$\times \sum_{j=1}^{J}\theta_j c_0 \frac{\partial K_{t_j/c_0}}{\partial u}(s,u)r\, \frac{u^{-\gamma}}{\gamma} =: I(s,u,r).$$

To prove that

$$I_\lambda \to \int_{-\infty}^{\infty}\int_0^{\infty}\int_0^{\infty} I(s,u,r)dsduF_R(dr) =: I,$$

it is enough to show that $|I_\lambda(s,u,r)|$ is dominated by an integrable function. This can be shown as in the proofs of Theorems 3.3.8, (a), and 3.3.10, and is left as Exercise 3.7. After an integration by parts, the limit I can be written as

$$I = \int_{-\infty}^{\infty}\int_0^{\infty}\int_0^{\infty}\Big(e^{i\sum_{j=1}^{J}\theta_j c_0 K_{t_j/c_0}(s,u)r} - 1 - i\sum_{j=1}^{J}\theta_j c_0 K_{t_j/c_0}(s,u)r\Big)u^{-\gamma-1}dsduF_R(dr),$$

which is the logarithm of the characteristic function of the process (3.3.90). □

The final two theorems concern the behavior of a scaled workload process in the remaining cases of parameter values, except the case $1 < \gamma = \delta < 2$ which is discussed in Remark 3.3.18 below. The distinction between fast, slow and intermediate regimes is now irrelevant. In the first theorem, we consider the case $\gamma = \delta = 2$.

Theorem 3.3.16 *Suppose*

$$\gamma = \delta = 2$$

and set

$$b = (\lambda a)^{1/2} \tag{3.3.92}$$

in the scaled workload process (3.3.26). Then, as $\lambda \to \infty$, $a \to \infty$ or $a \to \infty$, $b \to \infty$ in any arbitrary way,

$$\frac{1}{b}(W_\lambda^*(at) - \lambda\nu\mathbb{E}(R)at) \xrightarrow{fdd} (\mathbb{E}U^2\mathbb{E}R^2)^{1/2}B(t), \tag{3.3.93}$$

where $\{B(t)\}$ is a standard Brownian motion.

Proof We only give idea of the proof. As in (3.3.58), the logarithm of the characteristic function of the scaled workload process (3.3.26) can be expressed as

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$$
\begin{aligned}
I_\lambda &:= \log \mathbb{E} \exp \left\{ i \sum_{j=1}^{J} \theta_j \frac{W_\lambda^*(at_j) - \lambda\nu(\mathbb{E}R)at_j}{b} \right\} \\
&= \int_{-\infty}^{\infty} \int_0^{\infty} \int_0^{\infty} \Big(\exp\left\{ i \sum_{j=1}^{J} \frac{\theta_j K_{at_j}(as,u)r}{b} \right\} \\
&\quad - 1 - i \sum_{j=1}^{J} \frac{\theta_j K_{at_j}(as,u)r}{b} \Big) \lambda a ds F_U(du) F_R(dr) \\
&= \int_{-\infty}^{\infty} \int_0^{\infty} \int_0^{\infty} I_\lambda(s,u,r) ds F_U(du) F_R(dr),
\end{aligned}
$$

where

$$
I_\lambda(s,u,r) = \Big(\exp\left\{ i \sum_{j=1}^{J} \frac{\theta_j K_{at_j}(as,u)r}{b} \right\} - 1 - i \sum_{j=1}^{J} \frac{\theta_j K_{at_j}(as,u)r}{b} \Big) \lambda a.
$$

Since

$$
K_{at}(as,u) = \int_0^u 1_{\{0<y+as<at\}} dy \to u 1_{\{0\le s<t\}}, \tag{3.3.94}
$$

as $a \to \infty$, and hence $K_{at}(as,u)/b \to 0$, as $a, b \to \infty$, we have, using $e^{ix} - 1 - ix \sim -x^2/2$, as $x \to 0$, and (3.3.92) that

$$
I_\lambda(s,u,r) \sim -\frac{1}{2} \Big(\sum_{j=1}^{J} \theta_j K_{at_j}(as,u) r \Big)^2 \frac{\lambda a}{b^2} \to -\frac{1}{2} \Big(\sum_{j=1}^{J} \theta_j 1_{\{0\le s<t_j\}} \Big)^2 u^2 r^2,
$$

as $a, b \to \infty$. This yields

$$
\begin{aligned}
I_\lambda &\to -\frac{1}{2} \int_{-\infty}^{\infty} \int_0^{\infty} \int_0^{\infty} \Big(\sum_{j=1}^{J} \theta_j 1_{\{0\le s<t_j\}} \Big)^2 u^2 r^2 ds F_U(du) F_R(dr) \\
&= -\frac{1}{2} \mathbb{E}(U^2) \mathbb{E}(R^2) \int_{-\infty}^{\infty} \Big(\sum_{j=1}^{J} \theta_j 1_{\{0\le s<t_j\}} \Big)^2 ds,
\end{aligned}
$$

which is the logarithm of the characteristic function of the limit process in (3.3.93). □

In the next theorem, we consider the case $1 < \delta < \gamma \le 2$.

Theorem 3.3.17 *Suppose*

$$
1 < \delta < \gamma \le 2
$$

and set

$$
b = (\lambda a)^{1/\delta} L_{R,\delta}^*(\lambda a) \tag{3.3.95}
$$

in the scaled workload process (3.3.26). Then, as $\lambda \to \infty$, $a \to \infty$ or $a \to \infty, b \to \infty$ in any arbitrary way,

$$
\frac{1}{b}(W_\lambda^*(at) - \lambda\nu\mathbb{E}(R)at) \xrightarrow{fdd} (\mathbb{E}U^\delta)^{1/\delta} \Lambda_\delta(t), \tag{3.3.96}
$$

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where $\{\Lambda_\delta(t)\}$ is a δ-stable Lévy-stable process, totally skewed to the right. The limit process can be represented as

$$(\mathbb{E}U^\delta)^{1/\delta}\,\Lambda_\delta(t) = \int_{-\infty}^{\infty}\int_0^{\infty}\int_0^{\infty} 1_{\{0<s<t\}}ur\Big(N(ds,du,dr) - ds\,F_U(du)\,r^{-(1+\delta)}dr\Big) \tag{3.3.97}$$

$$\stackrel{d}{=} c_\delta \int_{-\infty}^{\infty}\int_0^{\infty} 1_{\{0<s<t\}}uM_\gamma(ds,du) \tag{3.3.98}$$

where c_δ is defined in (3.3.56) and the random measure $M_\delta(ds,du)$ is δ-stable, totally skewed to the right with control measure $ds\,F_U(du)$.

Proof We only give idea of the proof, which is similar to that of Theorem 3.3.10, after reversing the roles of R and U. As in (3.3.58), the logarithm of the characteristic function of the scaled workload process (3.3.26) can be expressed as

$$\begin{aligned}
I_\lambda &:= \log\mathbb{E}\exp\Big\{i\sum_{j=1}^J \theta_j \frac{W_\lambda^*(at_j) - \lambda\nu(\mathbb{E}R)at_j}{b}\Big\} \\
&= \int_{-\infty}^{\infty}\int_0^{\infty}\int_0^{\infty}\Big(\exp\Big\{i\sum_{j=1}^J \frac{\theta_j K_{at_j}(as,u)r}{b}\Big\} \\
&\qquad - 1 - i\sum_{j=1}^J \frac{\theta_j K_{at_j}(as,u)r}{b}\Big)\lambda a ds F_U(du)F_R(dr).
\end{aligned}$$

By integration by parts on the variable r and making the change of variables $r \to br$, we get

$$\begin{aligned}
I_\lambda &= \int_{-\infty}^{\infty}\int_0^{\infty}\int_0^{\infty}\Big(\exp\Big\{i\sum_{j=1}^J \frac{\theta_j K_{at_j}(as,u)r}{b}\Big\} - 1\Big)\Big(i\sum_{j=1}^J \frac{\theta_j K_{at_j}(as,u)}{b}\Big) \\
&\quad\times\ \lambda a\mathbb{P}(R>r)\,ds F_U(du)dr \\
&= \int_{-\infty}^{\infty}\int_0^{\infty}\int_0^{\infty}\Big(\exp\Big\{i\sum_{j=1}^J \theta_j K_{at_j}(as,u)r\Big\} - 1\Big)\Big(i\sum_{j=1}^J \theta_j K_{at_j}(as,u)\Big) \\
&\quad\times\ \lambda a\mathbb{P}(R>br)\,ds F_U(du)dr \\
&= \int_{-\infty}^{\infty}\int_0^{\infty}\int_0^{\infty} I_\lambda(s,u,r)ds F_U(du)dr,
\end{aligned}$$

where

$$I_\lambda(s,u,r) = \Big(\exp\Big\{i\sum_{j=1}^J \theta_j K_{at_j}(as,u)r\Big\} - 1\Big)\Big(i\sum_{j=1}^J \theta_j K_{at_j}(as,u)\Big)\lambda a\mathbb{P}(R>br).$$

Note that, using (3.3.95) and (3.3.44),

$$\lambda a\mathbb{P}(R>br) = \frac{r^{-\delta}}{\delta}(L_{R,\delta}^*(\lambda a))^{-\delta}L_R\Big((\lambda a)^{1/\delta}L_{R,\delta}^*(\lambda a)r\Big) \to \frac{r^{-\delta}}{\delta},$$

as $\lambda a \to \infty$. Then, using (3.3.94),

$$I_\lambda(s,u,r) \to \left(\exp\left\{i\sum_{j=1}^{J}\theta_j 1_{\{0\le s<t_j\}}ur\right\} - 1\right)\left(i\sum_{j=1}^{J}\theta_j 1_{\{0\le s<t_j\}}u\right)\frac{r^{-\delta}}{\delta}$$

and hence

$$I_\lambda \to \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\int_{0}^{\infty}\left(\exp\left\{\sum_{j=1}^{J}\theta_j 1_{\{0\le s<t_j\}}ur\right\} - 1\right)$$
$$\times\left(i\sum_{j=1}^{J}\theta_j 1_{\{0\le s<t_j\}}u\right)\frac{r^{-\delta}}{\delta}dsF_U(du)dr.$$

After integration by parts on the variable r, one gets

$$I = \int_{-\infty}^{\infty}\int_{0}^{\infty}\int_{0}^{\infty}\left(\exp\left\{i\sum_{j=1}^{J}\theta_j 1_{\{0<s<t_j\}}ur\right\}\right.$$
$$\left. - 1 - i\sum_{j=1}^{J}\theta_j 1_{\{0<s<t_j\}}ur\right)r^{-\delta-1}dsF_U(du)dr,$$

which is the logarithm of the characteristic function of the process in (3.3.97). The equality (3.3.98) follows by using (B.1.39) in Appendix B.1.4. □

Remark 3.3.18 As shown in Theorem 4, (ii), in Kaj and Taqqu [546], Theorem 3.3.17 holds in the case $1 < \gamma = \delta < 2$ provided $\mathbb{E}U^{\delta} < \infty$.

3.4 Power-Law Shot Noise Model

The workload process $W^*_\lambda(t)$ in (3.3.7) and the underlying stationary process $W_\lambda(y)$ in (3.3.19) can also be expressed as

$$W_\lambda(y) = \sum_{j=-\infty}^{\infty} 1_{\{S_j<y<S_j+U_j\}}R_j = \sum_{j=-\infty}^{\infty} g(y-S_j,U_j)R_j, \tag{3.4.1}$$

$$W^*_\lambda(t) = \int_0^t W_\lambda(y)dy = \sum_{j=-\infty}^{\infty}(g^*(t-S_j,U_j) - g^*(-S_j,U_j))R_j, \tag{3.4.2}$$

where S_j are Poisson arrivals, U_j, R_j, $j\in\mathbb{Z}$, are independent variables, and

$$g(y,u) = 1_{\{0<y<u\}},\quad g^*(x,u) = \int_0^x g(y,u)dy = x_+\wedge u. \tag{3.4.3}$$

The processes $W_\lambda(y)$ and $W^*_\lambda(t)$ are particular cases of the following more general processes, called shot noise processes.

Definition 3.4.1 Let $X_j = \{X_j(y)\}_{y\in\mathbb{R}}$, $j \in \mathbb{Z}$, and $X_j^* = \{X_j^*(y)\}_{y\in\mathbb{R}}$, $j \in \mathbb{Z}$, be two sets of i.i.d. processes, satisfying $X_j(y) = 0$, $y < 0$, and $X_j^*(t) = 0$, $t < 0$, and where, as above, $X_j^*(t) = \int_0^t X_j(y)dy$. Let S_j, $j \in \mathbb{Z}$, be Poisson arrivals with rate λ. Then, the processes

$$S_\lambda(y) = \sum_{j=-\infty}^{\infty} X_j(y - S_j) \tag{3.4.4}$$

and

$$S_\lambda^*(t) = \int_0^t S_\lambda(y)dy = \sum_{j=-\infty}^{\infty} \Big(X_j^*(t - S_j) - X_j^*(-S_j)\Big) \tag{3.4.5}$$

are called *shot noise processes*. The processes X_j and X_j^* are known as *shots* associated with S_λ and S_λ^*, respectively.

The idea behind a shot noise process is as follows: the impact of each Poisson arrival S_j lingers for a while according to a pattern X_j, such that at time $y > S_j$, it has an intensity value of $X_j(y - S_j)$. The total intensity at time $y > 0$ due to all the previous arrivals is then $S_\lambda(y) = \sum_{j=-\infty}^{\infty} X_j(y - S_j) = \sum_{j:S_j\le y} X_j(y - S_j)$.

Under suitable assumptions, the process S_λ is expected to be stationary and the process S_λ^* to have stationary increments. Observe, in particular, that $X_j^*(y - S_j) - X_j^*(-S_j) = X_j^*(y - S_j) - X_j^*(0 - S_j)$ so that $S_\lambda^*(0) = 0$. For example, in the case of W_λ in (3.4.1), the pattern is

$$X_j(y) = g(y, U_j)R_j = 1_{\{0<y<U_j\}}R_j. \tag{3.4.6}$$

This is therefore a shot which takes a random value R_j and lasts for a random time U_j. Taking U_j to be heavy-tailed in (3.3.2) is what made W_λ to be long-range dependent (Proposition 3.3.4) and made the scaled workload process to converge to fractional Brownian motion with the self-similarity parameter bigger than $1/2$ (Theorem 3.3.8, (a)).

The model (3.4.6) and the corresponding heavy tails, however, is not the only mechanism leading to long-range dependence. Consider, for example, the following alternate shot noise model.

Definition 3.4.2 Suppose that the i.i.d. shots $X_j = \{X_j(y)\}_{y\in\mathbb{R}}$ have a power-law pattern in the sense that

$$X_j(y) = g(y)R_j, \tag{3.4.7}$$

where R_j, $j \in \mathbb{Z}$, are i.i.d. variables with $\mathbb{E}R_j = 0$, $0 < \mathbb{E}R_j^2 < \infty$,

$$g(y) = 0,\ \ y < 0, \quad \text{and} \quad g(y) = y^{-\beta}L_g(y), \tag{3.4.8}$$

where L_g is a slowly varying function at infinity which is positive, and locally bounded away from 0 and infinity in $[0, \infty)$, and

$$1/2 < \beta < 1. \tag{3.4.9}$$

The shot noise process S_λ with the shots (3.4.7)–(3.4.8) is called a *power-law shot noise process*.

For slightly different assumptions on the function g, see Exercise 3.8. Note also that the shots X_j^* for the corresponding shot noise process S_λ^* satisfy

$$X_j^*(t) = \int_0^t X_j(y)dy = \Big(\int_0^t g(y)dy\Big)R_j =: g^*(t)R_j, \tag{3.4.10}$$

where g^* is given by

$$g^*(t) = \int_0^t y^{-\beta}L_g(y)dy = t^{1-\beta}L_{g^*}(t), \tag{3.4.11}$$

and where the slowly varying function L_{g^*} satisfies

$$L_{g^*}(t) \sim \frac{L_g(t)}{1-\beta}, \quad \text{as } t \to \infty \tag{3.4.12}$$

(Proposition 2.2.2).

Example 3.4.3 If $g(t) = t^{-\beta}(1_{\{t<2\}} + 1_{\{t\geq 2\}}\log t)$, then

$$\begin{aligned} g^*(t) &= \int_0^t y^{-\beta}(1_{\{y<2\}} + 1_{\{y\geq 2\}}\log y)dy \\ &= \frac{2^{1-\beta}}{(1-\beta)^2}(2-\beta-\log 2) + \log t\frac{t^{1-\beta}}{1-\beta} - \frac{t^{1-\beta}}{(1-\beta)^2}. \end{aligned}$$

The following result shows that the power-law shot noise process S_λ is long-range dependent with parameter

$$d = 1-\beta \in (0,1/2).$$

Proposition 3.4.4 *Consider the power-law shot noise process S_λ in (3.4.4) with shots given by (3.4.7)–(3.4.9). Then, S_λ is a stationary, zero mean process with autocovariance function satisfying*

$$\mathrm{Cov}(S_\lambda(y), S_\lambda(0)) \sim \lambda(\mathbb{E}R^2)B(1-\beta, 2\beta-1)(L_g(y))^2 y^{1-2\beta}, \tag{3.4.13}$$

as $y \to \infty$.

Proof Represent the shot noise process as

$$S_\lambda(y) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} g(y-s)rN(ds,dr),$$

where N is a Poisson random measure with the intensity $\lambda ds F_R(dr)$. Now observe that

$$\begin{aligned} \mathrm{Cov}(S_\lambda(y), S_\lambda(0)) &= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} g(y-s)g(-s)r^2\lambda ds F_R(dr) \\ &= \lambda\mathbb{E}(R^2)\int_0^\infty g(z)g(y+z)dz \\ &= \lambda\mathbb{E}(R^2)\int_0^\infty z^{-\beta}L_g(z)(y+z)^{-\beta}L_g(y+z)dz \end{aligned}$$

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$$
\begin{aligned}
&= \lambda \mathbb{E}(R^2) y^{1-2\beta} \int_0^\infty w^{-\beta}(1+w)^{-\beta} L_g(wy) L_g((1+w)y) dw \\
&\sim \lambda \mathbb{E}(R^2) y^{1-2\beta} (L_g(y))^2 \int_0^\infty w^{-\beta}(1+w)^{-\beta} dw \\
&= \lambda \mathbb{E}(R^2) y^{1-2\beta} (L_g(y))^2 B(1-\beta, 2\beta-1).
\end{aligned}
$$

The equivalence relation above follows from the dominated convergence theorem by using Potter's bound (2.2.2) to bound the integrand in the prelimit by $Cw^{-\beta\pm\delta}(1+w)^{-\beta+\delta}(L_g(y)^2)$, which is integrable on $(0,\infty)$ for sufficiently small $\delta > 0$. The last equality above follows from (2.2.12). □

The next result shows that a scaled power-law shot noise process S_λ^* converges to fractional Brownian motion.

Theorem 3.4.5 *Consider shot noise processes S_λ, S_λ^* in (3.4.4) and (3.4.5) with the shots given by (3.4.7)–(3.4.9). Then, as $a \to \infty$, a scaled process*

$$
\frac{S_\lambda^*(at)}{a^{3/2-\beta} L_{g^*}(a)} \tag{3.4.14}
$$

converges in the sense of finite-dimensional distributions to

$$
S^*(t) = (\lambda \mathbb{E} R^2)^{1/2} \int_{-\infty}^{\infty} \Big((t-s)_+^{1-\beta} - (-s)_+^{1-\beta} \Big) B(du), \tag{3.4.15}
$$

where $B(du)$ is a Gaussian random measure with the control measure du. The limit process S^* is FBM with the self-similarity parameter

$$
H = \frac{3}{2} - \beta \in \Big(\frac{1}{2}, 1\Big). \tag{3.4.16}
$$

Proof Let $b = a^{3/2-\beta} L_{g^*}(a)$. Using the representation

$$
S_\lambda^*(at) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \Big(g^*(at-s) - g^*(-s) \Big) r N(ds, dr)
$$

and (3.3.43), we have[3]

$$
\begin{aligned}
I_a &:= \log \mathbb{E} \exp \Big\{ i \sum_{j=1}^J \theta_j \frac{S_\lambda^*(at_j)}{b} \Big\} \\
&= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \Big(\exp \Big\{ i \sum_{j=1}^J \theta_j \frac{g^*(at_j - s) - g^*(-s)}{b} r \Big\} \\
&\quad - 1 - i \sum_{j=1}^J \theta_j \frac{g^*(at_j - s) - g^*(-s)}{b} r \Big) \lambda ds F_R(dr)
\end{aligned}
$$

[3] Do not confuse I_a with the integrand $I_a(s,r)$ in (3.4.17) below.

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$$= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\Big(\exp\Big\{i\sum_{j=1}^{J}\theta_j\frac{g^*(a(t_j+s))-g^*(as)}{b}r\Big\}$$

$$-1-i\sum_{j=1}^{J}\theta_j\frac{g^*(a(t_j+s))-g^*(as)}{b}r\Big)\lambda a ds F_R(dr)$$

$$=: \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} I_a(s,r)dsF_R(dr). \tag{3.4.17}$$

Note that, since

$$\frac{g^*(at)}{b} = \frac{(at)^{1-\beta}L_g(at)}{a^{3/2-\beta}L_{g^*}(a)} \to 0,$$

we can use the relation $e^{ix}-1-ix \sim -x^2/2$ as $x \to 0$ to get

$$I_a(s,r) \sim -\frac{1}{2}\Big|\sum_{j=1}^{J}\theta_j\frac{g^*(a(t_j+s))-g^*(as)}{b}r\Big|^2\lambda a$$

$$= -\frac{\lambda}{2}\Big|\sum_{j=1}^{J}\theta_j\frac{g^*(a(t_j+s))-g^*(as)}{a^{1-\beta}L_{g^*}(a)}r\Big|^2$$

$$\to -\frac{\lambda}{2}\Big|\sum_{j=1}^{J}\theta_j\Big((t_j+s)_+^{1-\beta}-s_+^{1-\beta}\Big)\Big|^2 r^2 =: I(s,r)$$

for any $s, r \in \mathbb{R}$. We want to conclude that

$$I_a = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} I_a(s,r)dsF_R(dr) \to \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} I(s,r)dsF_R(dr) =: I$$

and it is enough to prove that $|I_a(s,r)|$ is dominated by an integrable function.

Since $|e^{ix}-1-ix| \le |x|^2$ and without loss of generality, it is enough to show that, for each $t>0$, the function $|g^*(a(t+s))-g^*(as)|/(a^{1-\beta}L_g(a))$ is dominated by a square integrable function on $\mathbb{R}$. (We have replaced L_{g^*} by L_g in the denominator for convenience by using (3.4.12).) Observe that

$$\frac{|g^*(a(t+s))-g^*(as)|}{a^{1-\beta}L_g(a)} = \frac{\int_{as}^{a(t+s)}g(u)du}{a^{1-\beta}L_g(a)} = \frac{\int_{s}^{t+s}g(au)du}{a^{-\beta}L_g(a)} = \int_{s_+}^{(t+s)_+} u^{-\beta}\frac{L_g(au)}{L_g(a)}du$$

$$\le C\int_{s_+}^{(t+s)_+} u^{-\beta\pm\delta}du$$

$$= \frac{C}{1-\beta\pm\delta}\Big((t+s)_+^{1-\beta\pm\delta}-s_+^{1-\beta\pm\delta}\Big),$$

where we used Potter's bound (2.2.2). The bound above is a square integrable function on $\mathbb{R}$ for small enough $\delta>0$. □

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3.5 Hierarchical Model

We present here another physical model for long-range dependence. The form of the long-range dependence will be somewhat less conventional. The model nevertheless is interesting, instructive and related to the wavelet expansions discussed in Section 8.2.

Let $U^{(n)} = \{U^{(n)}(t)\}_{t\in\mathbb{R}}$, $n \geq 0$, be a collection of independent, second-order stationary processes, having the same autocovariance function $\gamma_U(h)$, $h \in \mathbb{R}$. For $0 < a < 1$ and $k > 1$, consider a sequence of processes $X^{(n)} = \{X^{(n)}(t)\}_{t\in\mathbb{R}}$, $n \geq 0$, defined as

$$\begin{aligned}
X^{(0)}(t) &= U^{(0)}(t),\\
X^{(1)}(t) &= k^{-1}X^{(0)}(k^{-1}t) + k^{-a}U^{(1)}(t)\\
&= k^{-1}U^{(0)}(k^{-1}t) + k^{-a}U^{(1)}(t),\\
X^{(2)}(t) &= k^{-1}X^{(1)}(k^{-1}t) + k^{-2a}U^{(2)}(t)\\
&= k^{-2}U^{(0)}(k^{-2}t) + k^{-1-a}U^{(1)}(k^{-1}t) + k^{-2a}U^{(2)}(t)
\end{aligned}$$

and, in general,

$$\begin{aligned}
X^{(n)}(t) &= k^{-1}X^{(n-1)}(k^{-1}t) + k^{-na}U^{(n)}(t)\\
&= \sum_{j=0}^{n} k^{-(n-j)-ja}U^{(j)}(k^{-(n-j)}t).
\end{aligned} \tag{3.5.1}$$

Note that for $k \geq 1$, time flows more slowly for $X(k^{-1}t)$ than for the $X(t)$ series since, if $k = 2$ for example, then $X(10)$ occurs at $t = 10$ for $X(t)$ but it occurs at $t = 20$ for $X(k^{-1}t)$. There is a slow-down of the tempo.

The physical interpretation of the model (3.5.1) is that n represents the stage of a process and that at each stage n, the available variation $X^{(n-1)}(t)$ has its time scale extended and a new short-term variation $U^{(n)}(t)$ is introduced. An example of such process is the production of textile yarns, which are produced by repetition of thinning and pulling out. See the bibliographical notes in Section 3.15.

Observe that the process $X^{(n)}$ has the autocovariance function

$$\gamma^{(n)}(h) = \sum_{j=0}^{n}(k^{-(n-j)-ja})^2\gamma_U(k^{-(n-j)}h) = k^{-2na}\sum_{j=0}^{n}k^{-2(1-a)j}\gamma_U(k^{-j}h), \tag{3.5.2}$$

where the last equality follows by the change of summation parameter j to $n-j$. The relation (3.5.2) now implies that, for each h, as $n \to \infty$,

$$k^{2na}\gamma^{(n)}(h) \to \sum_{j=0}^{\infty}k^{-2(1-a)j}\gamma_U(k^{-j}h) =: \gamma_X(h), \tag{3.5.3}$$

where the series in (3.5.3) converges absolutely since $a < 1$ and $|\gamma_U(k^{-j}h)| \leq \gamma_U(0) = \mathrm{Var}(U^{(0)}(0))$.

What can be said about the behavior of $\gamma_X(h)$ as $h \to \infty$? The following result provides a partial answer.

Proposition 3.5.1 *Fix* $0 < a < 1$ *and* $k > 1$ *and let* $\gamma_X(h)$ *be the autocovariance function defined in (3.5.3). Suppose that, for some* $\epsilon > 0$,

$$|\gamma_U(h)| = O(h^{-2(1-a)-\epsilon}), \tag{3.5.4}$$

as $h \to \infty$. *Then, for any* $b > 0$, *there is a constant* $c(b)$ *such that*

$$\gamma_X(bk^m) \sim c(b)(bk^m)^{-2(1-a)}, \tag{3.5.5}$$

as $m \to \infty$, *or equivalently as* $k^m \to \infty$. *Moreover,*

$$c(b) = \sum_{j=-\infty}^{\infty} (k^{-j}b)^{2(1-a)}\gamma_U(k^{-j}b). \tag{3.5.6}$$

Proof Note that

$$\begin{aligned}
\gamma_X(bk^m) &= \sum_{j=0}^{\infty} k^{-2(1-a)j}\gamma_U(k^{-j}k^m b) \\
&= (bk^m)^{-2(1-a)} \sum_{j=0}^{\infty}(k^{-j}k^m b)^{2(1-a)}\gamma_U(k^{-j}k^m b) \\
&= (bk^m)^{-2(1-a)} \sum_{j=-m}^{\infty} (k^{-j}b)^{2(1-a)}\gamma_U(k^{-j}b)
\end{aligned}$$

and that

$$\sum_{j=-m}^{\infty} (k^{-j}b)^{2(1-a)}\gamma_U(k^{-j}b) \to \sum_{j=-\infty}^{\infty} (k^{-j}b)^{2(1-a)}\gamma_U(k^{-j}b) = c(b),$$

as $m \to \infty$. We have convergence by assumption (3.5.4) since for $j \to \infty$, $(k^{-j}b)^{2(1-a)}\gamma_U(k^{-j}b) = O((k^{-j}b)^{-\epsilon})$. □

The relation (3.5.5) resembles the LRD relation (2.1.5) in condition II with $h = bk^m$,

$$d = a - 1/2 \in (-1/2, 1/2)$$

and $L_2(h) \sim c(b)$. This is not exactly an LRD condition in that c depends on h through the factor b. Note also that $d \in (0, 1/2)$ only when $1/2 < a < 1$, and that there is a priori no reason for $c(b)$ to be necessarily positive.

On the other hand, $c(b)$ in (3.5.6) is almost a constant function in the following sense.

Proposition 3.5.2 *Let* $1/2 < a < 1$, $k > 1$ *and* $c(b)$ *be defined in (3.5.6). Suppose that* $\gamma_U(h)$, $h \geq 0$, *is a monotone decreasing function and satisfies the condition (3.5.4). Then,*

$$\frac{1}{k-1}\int_0^\infty h^{2(1-a)-1}\gamma_U(h)dh \leq c(b) \leq \frac{k}{k-1}\int_0^\infty h^{2(1-a)-1}\gamma_U(h)dh. \tag{3.5.7}$$

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Proof We shall use the identities $k^j(k^{-j}-k^{-(j+1)}) = \frac{k-1}{k}$ and $k^j(k^{-(j-1)}-k^{-j}) = k-1$. The upper bound in (3.5.7) follows by expressing $c(b)$ in (3.5.6) as

$$c(b) = \frac{k}{k-1}\sum_{j=-\infty}^{\infty}(k^{-j}b)^{2(1-a)-1}\gamma_U(k^{-j}b)(k^{-j}b - k^{-(j+1)}b)$$

and noting that $h^{2(1-a)-1}\gamma_U(h)$ is monotone decreasing on $h > 0$ since $2(1-a)-1 < 0$ when $1/2 < a$, and $\gamma_U(h)$ is monotone decreasing by assumption. The lower bound in (3.5.7) follows by writing

$$c(b) = \frac{1}{k-1}\sum_{j=-\infty}^{\infty}(k^{-j}b)^{2(1-a)-1}\gamma_U(k^{-j}b)(k^{-(j-1)}b - k^{-j}b). \qquad \square$$

For example, when $k = 2$, the two factors $1/(k-1)$ and $k/(k-1)$ in the bound (3.5.7) are $1/(k-1) = 1$ and $k/(k-1) = 2$, respectively. The ratio of these factors is

$$\frac{k}{k-1}\Big/\frac{1}{k-1} = k$$

and approaches 1, as $k \downarrow 1$. In particular, in the limit $k \downarrow 1$, one has

$$c(b) \sim \frac{1}{k-1}\int_0^{\infty} h^{2(1-a)-1}\gamma_U(h)dh =: \frac{c}{k-1}, \tag{3.5.8}$$

where the constant c no longer depends on b. In this case, however, $c(b)$ diverges.

3.6 Regime Switching

It is part of the folklore, especially in Economics and Finance, that long-range dependence is easily confused with changes in regime (regime switching, structural change). We present below one such model, and a related convergence result. Some other models of this type are moved to exercises, and a related discussion can be found in Section 3.15.

Focus on the so-called *Markov switching model* defined as follows. Let $s^N = \{s_n^N\}_{n=1,\dots,N}$ be a stationary Markov chain, taking the value 0 or 1, with the probability transition matrix

$$P = \begin{pmatrix} p_{00} & 1-p_{00} \\ 1-p_{11} & p_{11} \end{pmatrix}. \tag{3.6.1}$$

We shall take below $p_{00} = p_{00}(N)$ and $p_{11} = p_{11}(N)$ as functions of N, which explains the use of the index N in the notation s^N. Let $\mu_0, \mu_1 \in \mathbb{R}$. Consider the series

$$X_n^N = \mu_{s_n^N} + \epsilon_n, \quad n = 1, \dots, N, \tag{3.6.2}$$

where $\{\epsilon_n\}$ is a sequence of i.i.d. random variables with mean 0 and variance σ_ϵ^2, and its partial sum process

$$S_N(t) = \sum_{n=1}^{[Nt]}(X_n^N - \mathbb{E}X_n^N), \quad t \in [0,1]. \tag{3.6.3}$$

The mean $\mu_{s_n^N} = \mu_0 1_{\{s_n^N=0\}} + \mu_1 1_{\{s_n^N=1\}}$ of X_n^N is not constant. It varies randomly between two values μ_0 and μ_1. The switching mechanism is dictated by the Markov chain s_n^N, $n = 1, \ldots, N$. If one is in regime k, $k = 1, 2$, one stays there with probability p_{kk} and switches to the other regime with probability $1 - p_{kk}$.

The following auxiliary lemma will be used in analyzing the asymptotics of the partial sum process $S_N(t)$ as $N \to \infty$.

Lemma 3.6.1 *The term $\mu_{s_n^N}$ in the Markov switching model (3.6.2) satisfies: for $1 \le m, n \le N$,*

$$\mathrm{Cov}\left(\mu_{s_n^N}, \mu_{s_m^N}\right) = \mu' \Gamma_{|n-m|} \mu = \frac{(1-p_{00})(1-p_{11})}{(2-p_{00}-p_{11})^2} (\mu_0 - \mu_1)^2 \lambda^{|n-m|}, \tag{3.6.4}$$

where

$$\mu = \begin{pmatrix} \mu_0 \\ \mu_1 \end{pmatrix}, \quad \Gamma_j = \frac{(1-p_{00})(1-p_{11})}{(2-p_{00}-p_{11})^2} \begin{pmatrix} \lambda^j & -\lambda^j \\ -\lambda^j & \lambda^j \end{pmatrix} \tag{3.6.5}$$

and

$$\lambda = p_{00} + p_{11} - 1. \tag{3.6.6}$$

Proof Let

$$\xi_n = \begin{pmatrix} 1_{\{s_n^N=0\}} \\ 1_{\{s_n^N=1\}} \end{pmatrix},$$

so that $\mu_{s_n^N} = \mu' \xi_n$, where prime denotes the transpose. Observe further that

$$\mathrm{Cov}\left(\mu_{s_n^N}, \mu_{s_m^N}\right) = \mathrm{Cov}\left(\mu' \xi_n, \mu' \xi_m\right) = \mu' \,\mathrm{Cov}(\xi_n, \xi_m) \mu = \mu' \Gamma_{|n-m|} \mu, \tag{3.6.7}$$

where

$$\Gamma_j = \mathrm{Cov}(\xi_j, \xi_0) = \begin{pmatrix} \mathrm{Cov}(1_{\{s_j^N=0\}}, 1_{\{s_0^N=0\}}) & \mathrm{Cov}(1_{\{s_j^N=1\}}, 1_{\{s_0^N=0\}}) \\ \mathrm{Cov}(1_{\{s_j^N=0\}}, 1_{\{s_0^N=1\}}) & \mathrm{Cov}(1_{\{s_j^N=1\}}, 1_{\{s_0^N=1\}}) \end{pmatrix}$$
$$= \begin{pmatrix} \mathbb{P}(s_j^N=0, s_0^N=0) - \mathbb{P}(s_j^N=0)\mathbb{P}(s_0^N=0) & \mathbb{P}(s_j^N=1, s_0^N=0) - \mathbb{P}(s_j^N=1)\mathbb{P}(s_0^N=0) \\ \mathbb{P}(s_j^N=0, s_0^N=1) - \mathbb{P}(s_j^N=0)\mathbb{P}(s_0^N=1) & \mathbb{P}(s_j^N=1, s_0^N=1) - \mathbb{P}(s_j^N=1)\mathbb{P}(s_0^N=1) \end{pmatrix}.$$

Denoting by $\pi = (\pi_0, \pi_1)'$ the invariant (stationary) probabilities of s^N, and by p_{kl}^j the transition probability from state k to state l in j steps, we have

$$\Gamma_j = \begin{pmatrix} \pi_0(p_{00}^j - \pi_0) & \pi_0(p_{01}^j - \pi_1) \\ \pi_1(p_{10}^j - \pi_0) & \pi_1(p_{11}^j - \pi_1) \end{pmatrix}. \tag{3.6.8}$$

To evaluate Γ_j, we need expressions for π_0, π_1 and p_{kl}^j. One can verify that

$$\pi_0 = \frac{1-p_{11}}{2-p_{00}-p_{11}}, \quad \pi_1 = \frac{1-p_{00}}{2-p_{00}-p_{11}} \tag{3.6.9}$$

and that the probability transition matrix P in (3.6.1) is diagonalizable as

$$P = \begin{pmatrix} 1 & \pi_1 \\ 1 & -\pi_0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & \lambda \end{pmatrix} \begin{pmatrix} \pi_0 & \pi_1 \\ 1 & -1 \end{pmatrix} =: E \begin{pmatrix} 1 & 0 \\ 0 & \lambda \end{pmatrix} F', \tag{3.6.10}$$

where λ is given by (3.6.6) (Exercise 3.10). Note that 1 and λ in (3.6.10) are the eigenvalues of the matrix P and the matrices E and F are made of the right- and left-eigenvectors of P, satisfying $F' = E^{-1}$ (cf. Example 2.3.4). The latter fact implies that

$$\begin{aligned} P^j &= \begin{pmatrix} 1 & \pi_1 \\ 1 & -\pi_0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & \lambda^j \end{pmatrix} \begin{pmatrix} \pi_0 & \pi_1 \\ 1 & -1 \end{pmatrix} = \begin{pmatrix} \pi_0 + \pi_1\lambda^j & \pi_1(1-\lambda^j) \\ \pi_0(1-\lambda^j) & \pi_1 + \pi_0\lambda^j \end{pmatrix} \\ &= \frac{1}{2 - p_{00} - p_{11}} \begin{pmatrix} (1-p_{11}) + (1-p_{00})\lambda^j & (1-p_{00})(1-\lambda^j) \\ (1-p_{11})(1-\lambda^j) & (1-p_{00}) + (1-p_{11})\lambda^j \end{pmatrix} \\ &= \begin{pmatrix} p^j_{00} & p^j_{01} \\ p^j_{10} & p^j_{11} \end{pmatrix} \end{aligned} \tag{3.6.11}$$

(Exercise 3.10). By using (3.6.9) and (3.6.11), we obtain from (3.6.8) that Γ_j is given by (3.6.5) and (3.6.4) is finally established by using (3.6.7). □

The following result provides conditions on the Markov chain s^N for the partial sum process $S_N(t)$ to converge to Brownian motion. Connections to long-range dependence are discussed following the proof.

Proposition 3.6.2 *Assume that*

$$\mu_0 \neq \mu_1 \tag{3.6.12}$$

and that, for $k = 0, 1$,

$$p_{kk} = p_{kk}(N) = 1 - \frac{c_k}{N^{\delta_k}}, \tag{3.6.13}$$

with $\delta_k > 0$, $0 < c_k < 1$. *Suppose that the exponents*

$$0 < \delta_0, \delta_1 < 1 \tag{3.6.14}$$

are such that

$$d := \frac{1}{2}\Big(\min\{\delta_0, \delta_1\} - |\delta_0 - \delta_1|\Big) > 0. \tag{3.6.15}$$

Then,

$$\frac{S_N(t)}{N^{d+1/2}} \xrightarrow{fdd} \sigma B(t), \qquad t \in [0, 1], \tag{3.6.16}$$

where B *is a standard Brownian motion with*

$$\sigma^2 = \frac{2c_0c_1(\mu_0 - \mu_1)^2}{(\widetilde{c})^3} \tag{3.6.17}$$

and

$$\widetilde{c} = \begin{cases} c_0, & \text{if } \delta_0 < \delta_1, \\ c_1, & \text{if } \delta_1 < \delta_0, \\ c_0 + c_1, & \text{if } \delta_0 = \delta_1. \end{cases} \tag{3.6.18}$$

Proof It is enough to show that, for $t_1, \dots, t_J \in [0,1]$ and $\theta_1, \dots, \theta_J \in \mathbb{R}$,

$$\sum_{j=1}^{J} \theta_j \frac{S_N(t_j)}{N^{d+1/2}} \xrightarrow{d} \mathcal{N}(0, \sigma_J^2) \tag{3.6.19}$$

and that

$$\sigma_J^2 = \lim_{N\to\infty} \mathbb{E}\Big(\sum_{j=1}^{J} \theta_j \frac{S_N(t_j)}{N^{d+1/2}}\Big)^2 = \sigma^2 \mathbb{E}\Big(\sum_{j=1}^{J} \theta_j B(t_j)\Big)^2. \tag{3.6.20}$$

Since X_n^N is a stationary series, the relation (3.6.20) follows from

$$\mathbb{E}\Big(\frac{S_N(t)}{N^{d+1/2}}\Big)^2 \to \sigma^2 t, \tag{3.6.21}$$

as $N \to \infty$.

Write

$$S_N(t) = \sum_{n=1}^{[Nt]} (\mu_{s_n^N} - \mathbb{E}\mu_{s_n^N}) + \sum_{n=1}^{[Nt]} \epsilon_n =: S_{1,N}(t) + S_{2,N}(t). \tag{3.6.22}$$

Then, by the independence of s^N and $\{\epsilon_n\}$,

$$\mathbb{E}\Big(\frac{S_N(t)}{N^{d+1/2}}\Big)^2 = \mathbb{E}\Big(\frac{S_{1,N}(t)}{N^{d+1/2}}\Big)^2 + \mathbb{E}\Big(\frac{S_{2,N}(t)}{N^{d+1/2}}\Big)^2$$

and

$$\mathbb{E}\Big(\frac{S_{2,N}(t)}{N^{d+1/2}}\Big)^2 = \frac{\sigma_\epsilon^2 [Nt]}{N^{2d+1}} \to 0.$$

Hence, it is enough to show that

$$\mathbb{E}\Big(\frac{S_{1,N}(t)}{N^{d+1/2}}\Big)^2 \to \sigma^2 t. \tag{3.6.23}$$

Observe that

$$\begin{aligned}\mathbb{E}(S_{1,N}(t))^2 &= \mathrm{Var}\Big(\sum_{n=1}^{[Nt]} \mu_{s_n^N}\Big) = \mathrm{Var}\Big(\sum_{n=1}^{[Nt]} \mu' \xi_n\Big) = \mu' \sum_{n_1,n_2=1}^{[Nt]} \mathrm{Cov}(\xi_{n_1}, \xi_{n_2})\mu \\ &= \mu'\Big([Nt]\Gamma_0 + \sum_{j=1}^{[Nt]-1} ([Nt]-j)(\Gamma_j + \Gamma_j')\Big)\mu.\end{aligned}$$

From Lemma 3.6.1 and by using

$$\sum_{j=1}^{[Nt]-1} \lambda^j = \frac{\lambda^{[Nt]} - \lambda}{\lambda - 1}, \quad \sum_{j=1}^{[Nt]-1} j\lambda^j = \frac{\lambda}{(\lambda-1)^2}(([Nt]-1)\lambda^{[Nt]} - [Nt]\lambda^{[Nt]-1} + 1), \tag{3.6.24}$$

we get

$$\mathbb{E}(S_{1,N}(t))^2 = V_{00}(\mu_0 - \mu_1)^2, \tag{3.6.25}$$

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where

$$V_{00} = \frac{(1-p_{00})(1-p_{11})}{(2-p_{00}-p_{11})^2}\Big(\frac{2[Nt]\lambda}{1-\lambda} + [Nt] + \frac{2(\lambda^{[Nt]+1}-\lambda)}{(1-\lambda)^2}\Big).$$

But by (3.6.6) and (3.6.13),

$$\lambda = p_{00} + p_{11} - 1 = 1 - \frac{c_0}{N^{\delta_0}} - \frac{c_1}{N^{\delta_1}}.$$

Hence, as $N \to \infty$,

$$\frac{1}{1-\lambda} = \frac{1}{c_0 N^{-\delta_0} + c_1 N^{-\delta_1}} \sim \frac{N^{\min\{\delta_0,\delta_1\}}}{\widetilde{c}},$$

where $\widetilde{c}$ is defined in (3.6.18), and

$$\lambda^{[Nt]} = \Big(1 - \frac{c_0}{N^{\delta_0}} - \frac{c_1}{N^{\delta_1}}\Big)^{[Nt]} \sim \Big(1 - \frac{\widetilde{c}}{N^{\min\{\delta_0,\delta_1\}}}\Big)^{[Nt]} \sim \exp\Big\{-\widetilde{c}[Nt]N^{-\min\{\delta_0,\delta_1\}}\Big\} = o(1),$$

since $\delta_0, \delta_1 < 1$. This shows that

$$V_{00} \sim \frac{(1-p_{00})(1-p_{11})}{(2-p_{00}-p_{11})^2}\frac{2[Nt]\lambda}{1-\lambda} \sim \frac{(1-p_{00})(1-p_{11})}{(2-p_{00}-p_{11})^2}\frac{2[Nt]}{\widetilde{c}}N^{\min\{\delta_0,\delta_1\}}.$$

Now, by (3.6.13),

$$\frac{(1-p_{00})(1-p_{11})}{(2-p_{00}-p_{11})^2} \sim \frac{c_0 N^{-\delta_0} c_1 N^{-\delta_1}}{\widetilde{c}^2 N^{-2\min\{\delta_0,\delta_1\}}} \sim \frac{c_0 c_1}{\widetilde{c}^2} N^{-|\delta_0-\delta_1|}, \tag{3.6.26}$$

since $-\delta_0 - \delta_1 + 2\min\{\delta_0,\delta_1\} = -|\delta_0 - \delta_1|$. Thus, by (3.6.15),

$$V_{00} \sim \frac{2c_0c_1[Nt]}{\widetilde{c}^3} N^{\min\{\delta_0,\delta_1\}-|\delta_0-\delta_1|} = \frac{2c_0c_1[Nt]}{\widetilde{c}^3}N^{2d}. \tag{3.6.27}$$

Relations (3.6.25) and (3.6.27) now yield

$$\frac{\mathbb{E}(S_{1,N}(t))^2}{N^{2d+1}} \sim \frac{2c_0c_1(\mu_0-\mu_1)^2 t}{\widetilde{c}^3} = \sigma^2 t,$$

which establishes (3.6.23).

We now turn to the convergence (3.6.19). In view of the decomposition (3.6.22) and since, by the central limit theorem, as $N \to \infty$,

$$\sum_{j=1}^J \theta_j \frac{S_{2,N}(t_j)}{N^{1/2}} \xrightarrow{d} \mathcal{N}(0, \sigma^2_{2,J}),$$

where $0 < \sigma^2_{2,J} < \infty$, it is enough to show that

$$\sum_{j=1}^J \theta_j \frac{S_{1,N}(t_j)}{N^{d+1/2}} \xrightarrow{d} \mathcal{N}(0, \sigma^2_J), \tag{3.6.28}$$

where σ_J^2 is defined in (3.6.20). Write

$$\sum_{j=1}^J \theta_j \frac{S_{1,N}(t_j)}{N^{d+1/2}} = \sum_{n=1}^N f_n^{(N)}(s_n^N),$$

where

$$f_n^{(N)}(x) = \sum_{j=1}^{J} \theta_j \frac{1}{N^{d+1/2}} 1_{\{n \le [Nt_j]\}} (\mu_x - \mathbb{E}\mu_{s_n^N}).$$

Recall that s_n^N may be equal to 0 or 1.

We shall apply next a central limit theorem for functionals of Markov chains. Let

$$\rho_{N,1} = \max_{2\le n\le N} \sup_g \left\{ \frac{\|\mathbb{E}(g(s_n^N)|s_{n-1}^N)\|_2}{\|g(s_n^N)\|_2} : \|g(s_n^N)\|_2 < \infty \quad \text{and} \quad \mathbb{E}g(s_n^N) = 0 \right\}$$

be the so-called maximal coefficient of correlation for the Markov chain $\{s_n^N\}$, where $\|X\|_2 = (\mathbb{E}X^2)^{1/2}$. Denoting $g(S_n^N) = g_0 1_{\{s_n^N=0\}} + g_1 1_{\{s_n^N=1\}}$ with $g_0, g_1 \in \mathbb{R}$, it can be seen from (3.6.9) that $\mathbb{E}g(s_n^N) = 0$ is equivalent to $g_0(1-p_{11}) + g_1(1-p_{00}) = 0$. A simple algebra then yields $\mathbb{E}(g(s_n^N)|s_{n-1}^N) = (p_{00}+p_{11}-1)g(s_{n-1}^N)$ and hence

$$\rho_{N,1} = |p_{00}+p_{11}-1| = 1 - \frac{c_0}{N^{\delta_0}} - \frac{c_1}{N^{\delta_1}}.$$

In particular, by (3.6.13),

$$\eta_N := 1 - \rho_{N,1} = \frac{c_0}{N^{\delta_0}} + \frac{c_1}{N^{\delta_1}} \sim \frac{\widetilde{c}}{N^{\min\{\delta_0,\delta_1\}}}, \tag{3.6.29}$$

as $N \to \infty$. Observe that

$$\sup_{1\le n\le N} \sup_x |f_n^{(N)}(x)| \le C_N := \frac{C}{N^{d+1/2}}$$

and also that

$$\sigma_N^2 := \mathrm{Var}\left(\sum_{n=1}^{N} f_n^{(N)}(s_n^N)\right) \sim C',$$

by using (3.6.20). By Theorem 1 in Peligrad [800], the convergence (3.6.28) follows if

$$\frac{C_N(1+\log \eta_N)}{\eta_N \sigma_N} \to 0. \tag{3.6.30}$$

The condition (3.6.30) holds since C_N/η_N behaves asymptotically up to a constant as

$$N^{-\frac{1}{2}-d+\min\{\delta_0,\delta_1\}} = N^{-\frac{1}{2}+\frac{1}{2}(\min\{\delta_0,\delta_1\}+|\delta_0-\delta_1|)} = N^{-\frac{1}{2}(1-\max\{\delta_0,\delta_1\})} \to 0,$$

by using (3.6.29) and (3.6.15). □

How is the Markov switching model related to long-range dependence? By the convergence (3.6.21) established in the proof above, note that

$$\mathrm{Var}(S_N(1)) = \mathrm{Var}(X_1^N + \cdots + X_N^N) \sim CN^{2d+1}, \tag{3.6.31}$$

as $N \to \infty$. When $d > 0$, this behavior is consistent with that of LRD series given in (2.1.8) in condition V. On the other hand, note that the convergence to Brownian motion in (3.6.16) suggests that the series $\{X_n^N\}$ is SRD. How can these two seemingly contradictory facts be reconciled? There is no contradiction between these two facts because the probabilistic structure of the Markov switching series $\{X_n^N\}$ depends on N. In this sense, the

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Markov switching model exhibits features of both LRD and SRD series. From an estimation perspective, the Markov switching model may indeed suggest long-range dependence but common estimators of the LRD parameter d are very biased and sensitive to the choice of the tuning parameters, such as the parameter m in the GPH estimation in Section 2.10.3 (Diebold and Inoue [312], Baek, Fortuna, and Pipiras [71]).

Several aspects of the Markov switching model presented here are similar to those of other physical models for long-range dependence. The fact that the probabilities p_{00} and p_{11} approach 1, as $N \to \infty$, ensures that the underlying Markov chain s^N will tend to remain a long time in a given state. This is similar to the infinite source Poisson model in Section 3.3 where each arrival stays in the "ON" state for a long, heavy-tailed time. Thus the infinite source Poisson model is also characterized by a kind of regime switching.

The Markov switching model also appears in the spirit of models discussed in Sections 3.1 and 3.2. Its covariance structure described in Lemma 3.6.1 and the proof of Proposition 3.6.2 resembles that of AR(1) series with the autoregressive coefficient approaching 1 at a suitable rate (see also Exercise 3.11). The models of Section 3.1 consider specifically AR(1) series with a random autoregressive coefficient whose distribution has a power-law behavior around 1. Similarly, without the error terms ϵ_n in (3.6.2) and when $p_{00} = p_{11} = p$, the series X_n^N can be thought as the steps of the correlated random walk (CRW) considered in Section 3.2. While the parameter p was random and CRWs were aggregated in Section 3.2, the parameter p depends here on the sample size N and approaches 1 at a suitable rate.

3.7 Elastic Collision of Particles

Consider two particles with equal mass m moving towards each other on the one-dimensional real line $\mathbb{R}$. The first particle has constant velocity v_1, and the second has constant velocity v_2 with $v_1 \neq v_2$. What will be their velocities after an elastic collision? Elastic collisions preserve kinetic energy. Since momentum needs also to be conserved, the speeds u_1 and u_2 after collision must satisfy the following two equations:

$$\frac{1}{2}mv_1^2 + \frac{1}{2}mv_2^2 = \frac{1}{2}mu_1^2 + \frac{1}{2}mu_2^2 \quad \text{(conservation of energy)},$$
$$mv_1 + mv_2 = mu_1 + mu_2 \quad \text{(conservation of momentum)}.$$

Suppose that $v_1 \neq u_1$. Then the only solution to these equations is

$$u_1 = v_2 \quad u_2 = v_1;$$

that is, the two particles merely exchange velocities. Thus, if the second particle is immobile ($v_2 = 0$), then after collision it is the first particle which will be immobile ($u_1 = 0$).

In addition, the order of the particles is preserved since the particles cannot go through each other. Therefore, they exchange trajectories after they collide.

But two particles can collide at most once. Now imagine that there is an infinite countable number of particles with random i.i.d. velocities; that is, suppose that, if there were no collisions, their paths would be $X_i(t) = x_i + \xi_i(t)$, $i \in \mathbb{Z}$, where $X_i(0) = x_i$ and the processes ξ_is are i.i.d. and independent of x_is. The starting points x_i are random and are generated by a stationary Poisson point process on the real line with intensity $\rho > 0$. Focus on the middle

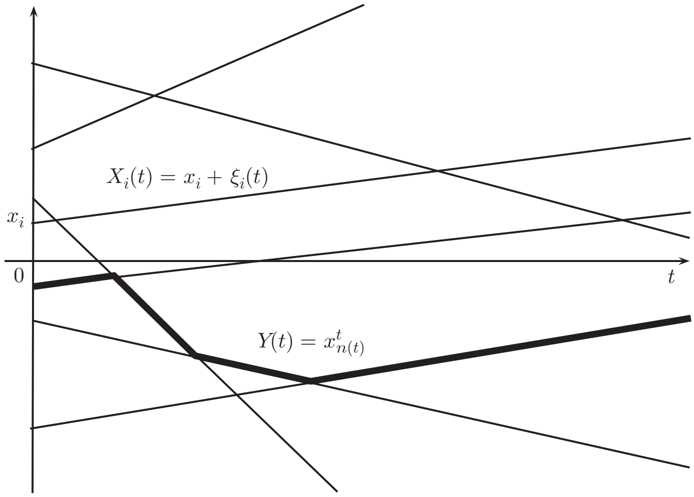

Figure 3.4 Trajectory $Y(t) = x^t_{n(t)}$ of the middle (tagged) particle when the paths $X_i(t) = x_i + \xi_i(t)$ are straight lines.

(tagged) particle, namely the one whose starting position $X_i(0)$ is below but closest to the origin (see Figure 3.4).

Let $Y(t)$, $t \geq 0$, be the trajectory of that middle particle as it undergoes collisions. What happens to it (to its law) after adequate normalization, as $t \to \infty$?

This problem has an interesting history. Harris [455] supposed that $\{\xi_i(t)\}$ is a collection of independent Brownian motions $\{B_i(t)\}$. These are H_0–SSSI with $H_0 = 1/2$. The limit of $Y(Tt)/T^{1/4}$ as $T \to \infty$ turns out to be FBM with self-similarity parameter $H = 1/4$. Spitzer [919] supposed that the paths are straight lines with random slopes; that is, $\xi_i(t) = v_i t$ where the v_is are i.i.d. with $\mathbb{E}|v_i| < \infty$ and mean zero. These are H_0–SSSI with $H_0 = 1$. He showed that the limit of $Y(Tt)/T^{1/2}$ as $T \to \infty$ is Brownian motion; that is, FBM with $H = 1/2$. Gisselquist [407] then chose for the ξ_is another H_0–SSSI process, namely the $S\alpha S$ Lévy motion with $1 < \alpha < 2$ and hence $1/2 < H_0 = 1/\alpha < 1$. He showed that the limit of $Y(Tt)/T^{1/(2\alpha)}$ as $T \to \infty$ is again FBM with $H = 1/(2\alpha)$. In all cases, note that $H = H_0/2$.

Dürr, Goldstein, and Lebowitz [339] provide a more general perspective. They use a Poisson point process framework and show that the limit depends essentially on the function $\mathbb{E}|\xi(t)|$. They do not require that $\{\xi(t)\}$ has stationary increments.

Theorem 3.7.1 *Suppose that the starting points are generated by a Poisson point process on $\mathbb{R}$ with intensity $\rho > 0$. Let $\mathbb{P}_\xi$ denote the probability distribution of $\xi(t)$, $t \geq 0$, and suppose that $\xi(t)$, $t \geq 0$, satisfies the following conditions:*

$$\xi(0) = 0, \quad \mathbb{E}|\xi(t)| = \int |\xi(t)| d\mathbb{P}_\xi < \infty, \quad \mathbb{E}\xi(t) = 0.$$

Moreover, suppose that

$$\mathbb{E}|\xi(t)| = t^\alpha L(t) \tag{3.7.1}$$

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for some $\alpha > 0$*, where* L *is slowly varying at* ∞*, and that there is a function* $c(t, s) < 0$ *such that, for any* $s, t \geq 0$*,*

$$c(t, s) = \lim_{T \to \infty} \frac{\mathbb{E}|\xi(Tt) - \xi(Ts)|}{\mathbb{E}|\xi(T)|} = \lim_{T \to \infty} \frac{\mathbb{E}|\xi(Tt) - \xi(Ts)|}{T^\alpha L(T)}. \tag{3.7.2}$$

Then,

1. $c(t, s)$ *is a homogeneous function of order* α*:* $c(\lambda t, \lambda s) = \lambda^\alpha c(t, s)$*,*
2. *As* $T \to \infty$*,*

$$\frac{Y(Tt)}{\sqrt{T^\alpha L(T)}} \xrightarrow{fdd} \rho^{-1/2} Z(t), \quad t \geq 0, \tag{3.7.3}$$

where $\{Z(t)\}$ *is an* $(\alpha/2)$*–self-similar Gaussian process with mean 0 and covariance:*

$$\mathbb{E}Z(t)Z(s) = \frac{1}{2}\Big(t^\alpha + s^\alpha - c(t, s)\Big), \quad s, t \geq 0. \tag{3.7.4}$$

3. *If, in addition,* $\{\xi(t)\}$ *has stationary increments, then* $0 < \alpha \leq 1$*, and* $c(t, s) = |t - s|^\alpha$*; that is,* $\{Z(t)\}$ *is FBM* $\{B_{\alpha/2}(t)\}$*.*

The basic idea of the proof is to incorporate $\xi(t)$ in the Poisson point process framework, use the Poisson convergence to the normal and exploit the regularly varying assumption $\mathbb{E}|\xi(t)| = t^\alpha L(t)$ to obtain the limiting covariance.

Sketch of proof View $(x_i, \xi_i(\cdot)) \in \mathbb{R} \times \mathbb{R}^{[0,\infty)}$ as a realization of an abstract Poisson point process N with intensity measure

$$d\mu = \rho dx d\mathbb{P}_\xi$$

defined on $\mathbb{R} \times \mathbb{R}^{[0,\infty)}$, where $\mathbb{P}_\xi$ is the law of each process $\xi_i(\cdot)$. The non-collision particle system is $X_i(t) = x_i + \xi_i(t)$, $i \in \mathbb{Z}$, with the starting points ordered as $\ldots, X_{-1}(0) < X_0(0) \leq 0 < X_1(0) \ldots$. Now extend the order-preservation idea to define the collision process.[4] This process describes the motion of the "middle" particle, namely the one which starts at the position closest to and below 0. We use an "order-tracking" process $n(t)$ to keep track of the middle particle. Let

$$n^+(t) = |\{i : i \leq 0, X_i(t) > 0\}|, \quad n^-(t) = |\{i : i > 0, X_i(t) \leq 0\}|. \tag{3.7.5}$$

Here, $n^+(t)$ denotes the (finite) number of particles starting below 0 that would end up above 0 at time t if there were no collision and $n^-(t)$ has the converse interpretation. Define

$$n(t) := n^+(t) - n^-(t). \tag{3.7.6}$$

If $\ldots \leq x^t_{-1} \leq x^t_0 \leq 0 < x^t_1 \leq \ldots$ are the ordered positions at time t of the non-collision particle system $\{X_i(t)\}$, then the collision process is defined as

$$Y(t) = x^t_{n(t)}. \tag{3.7.7}$$

[4] The particle processes $\xi_i(t)$ do not need to have continuous sample paths as in Figure 3.4. The sample paths can belong to the space of right-continuous functions having left limits for the order-preservation processes to be well-defined.

(For later reference and of independent interest, note that (3.7.7) yields

$$\{Y(t) > 0\} = \{n(t) > 0\} = \{n_+(t) > n_-(t)\}.) \tag{3.7.8}$$

Observe that $n(t) = n^+(t) - n^-(t) = N(A^+(t)) - N(A^-(t))$, where N is the Poisson random measure defined above and

$$A^+(t) = \{(x, \xi) : x \le 0, \xi(t) + x > 0\}, \quad A^-(t) = \{(x, \xi) : x > 0, \xi(t) + x \le 0\}$$

are measurable sets in $\mathbb{R} \times \mathbb{R}^{[0,\infty)}$. Since $A^-(t) \cap A^+(t) = \emptyset$, $n^+(t)$ and $n^-(t)$ are i.i.d. Poisson random variables for fixed t. Let us now determine the mean and variance of $n(t)$. Note that

$$\mathbb{E}n^+(t) = \int \int_{-\infty}^{0} 1_{\{\xi(t)>-x\}} \rho dx d\mathbb{P}_\xi = \rho \int \max\{\xi(t), 0\} d\mathbb{P}_\xi = \rho \mathbb{E}\xi(t)_+ = \frac{\rho}{2}\mathbb{E}|\xi(t)|,$$

where the last equality follows from the fact that $\xi(t)$ has 0 mean. Similarly, $\mathbb{E}n^-(t) = \rho\mathbb{E}|\xi(t)|/2$. Hence,

$$\mathbb{E}n(t) = \mathbb{E}n^+(t) - \mathbb{E}n^-(t) = 0$$

and

$$\mathbb{E}n(t)^2 = \mathrm{Var}(n(t)) = \mathrm{Var}(n^+(t)) + \mathrm{Var}(n^-(t)) = \mathbb{E}n^+(t) + \mathbb{E}n^-(t) = 2\mathbb{E}n^+(t) = \rho\mathbb{E}|\xi(t)|,$$

since a Poisson random variable has a variance equal to its mean. Thus, the variance of $n(t)$ equals

$$\phi(t) := \mathrm{Var}(n(t)) = \mathbb{E}n(t)^2 = \rho\mathbb{E}|\xi(t)| = \rho t^\alpha L(t). \tag{3.7.9}$$

One can show that if $\rho^{-1} n(Tt)/\sqrt{\phi(T)}$ converges to some process $Z(t)$ in finite-dimensional distributions, then so does $Y(Tt)/\sqrt{\phi(T)}$ as $T \to \infty$ (see Section 3 of Dürr et al. [339]). Hence to study the scaling limit of the collision process $\{Y(t)\}$, it is equivalent to study the limit of the simpler order-tracking process $n(t)$. But

$$\frac{n(Tt)}{\sqrt{\phi(T)}} \xrightarrow{fdd} Z(t), \tag{3.7.10}$$

where $\{Z(t)\}$ is a Gaussian process with zero mean, unit variance and covariance

$$\lim_{T\to\infty} \mathbb{E}\Big(\frac{n(Tt)}{\sqrt{\phi(T)}} \frac{n(Ts)}{\sqrt{\phi(T)}}\Big).$$

This is because $n(t) = n^+(t) - n^-(t)$ where $n^+(t)$ and $n^-(t)$ at fixed t are i.i.d. Poisson with mean (and variance) $\phi(t)/2$. Since $\phi(t) \to \infty$, by the standard normal approximation to the Poisson (central limit theorem), we get convergence in the sense of marginal distributions. Convergence of the finite-dimensional distributions can be obtained in a similar way. (See Exercise 3.13.)

Note that

$$\mathbb{E}n(t)n(s) = \frac{1}{2}\Big(\mathbb{E}n(t)^2 + \mathbb{E}n(s)^2 - \mathbb{E}(n(t) - n(s))^2\Big).$$

We know by (3.7.9), that $\mathbb{E}n(t)^2 = \phi(t) = \rho t^\alpha L(t)$, so

$$\lim_{T\to\infty} \frac{\mathbb{E}n(Tt)^2}{\phi(T)} = \lim_{T\to\infty} \frac{\rho(Tt)^\alpha L(Tt)}{\rho T^\alpha L(T)} = t^\alpha.$$

We are left to show that

$$\frac{\mathbb{E}(n(Tt)-n(Ts))^2}{\phi(T)} = \lim_{T\to\infty} \frac{\rho\mathbb{E}|\xi(Tt)-\xi(Ts)|}{\rho T^\alpha L(T)} = \lim_{T\to\infty} \frac{\mathbb{E}|\xi(Tt)-\xi(Ts)|}{T^\alpha L(T)} = c(t,s), \tag{3.7.11}$$

which proves (3.7.4). The first equality in (3.7.11) follows as above (see Exercise 3.14).

Since as $T \to \infty$, $Y(Tt)/\sqrt{\phi(T)}$ has the same limit $Z(t)$ as $\rho^{-1}n(Tt)/\sqrt{\phi(T)}$, we get

$$\frac{Y(Tt)}{\sqrt{T^\alpha L(T)}} \sim \frac{\rho^{-1}n(Tt)}{\sqrt{\rho^{-1}\phi(T)}} \xrightarrow{fdd} \rho^{-1/2}Z(t),$$

proving (3.7.3).

When $\{\xi(t)\}$ has stationary increments in the last part of the theorem, the covariance function in (3.7.4) becomes that of FBM with the self-simjilarity parameter $H = \alpha/2$. □

Corollary 3.7.2 *Suppose $\xi(t)$, $t \geq 0$ is H–self-similar and $\mathbb{E}|\xi(1)| < \infty$. Then, as $T \to \infty$,*

$$\frac{Y(Tt)}{T^{H/2}} \xrightarrow{fdd} \rho^{-1/2}\sqrt{\mathbb{E}|\xi(1)|}\, Z(t), \quad t \geq 0,$$

where $\{Z(t)\}$ is a Gaussian process with zero mean and covariance

$$\mathbb{E}Z(t)Z(s) = \frac{1}{2}\Big(t^H + s^H - \frac{\mathbb{E}|\xi(t)-\xi(s)|}{\mathbb{E}|\xi(1)|}\Big), \quad s,t \geq 0. \tag{3.7.12}$$

If, in addition, $\xi(t)$, $t \geq 0$, has stationary increments, then $0 < H \leq 1$ and $\{Z(t)\}$ is the standard FBM $\{B_{H/2}(t)\}$.

Proof The result follows from Theorem 3.7.1 by observing that $\mathbb{E}|\xi(t)| = t^H\mathbb{E}|\xi(1)|$ for H–self-similar process $\{\xi(t)\}$ and hence that we have $L(t) = \mathbb{E}|\xi(1)|$. □

Examples

- Newtonian particles:
  $\xi(t) = vt$: $Y(Tt)/\sqrt{T} \xrightarrow{fdd} \rho^{-1/2}\sqrt{\mathbb{E}|v|}B(t)$.
- Brownian motion:
  $\xi(t) = B(t)$: $Y(Tt)/T^{1/4} \xrightarrow{fdd} \rho^{-1/2}(2/\pi)^{1/4}B_{1/4}(t)$ since $\mathbb{E}|Z| = \sqrt{2/\pi}$ if Z is standard normal.
- Fractional Brownian motion:
  $\xi(t) = B_H(t)$ $(0 < H < 1)$: $Y(Tt)/T^{H/2} \xrightarrow{fdd} \rho^{-1/2}(2/\pi)^{1/4}B_{H/2}(t)$.
- Symmetric α-stable motion:
  $\xi(t) = S_\alpha(t)$ $(1 < \alpha < 2)$: $Y(Tt)/T^{1/\alpha} \xrightarrow{fdd} \rho^{-1/2}\sqrt{\mathbb{E}|S_\alpha(1)|}B_{1/(2\alpha)}(t)$.
- Integrated Brownian motion:

  $$\xi(t) := \int_0^t B(s)ds = \int_0^t \int_0^s dB(r)ds :$$

  Note that $\{\xi(t)\}$ is also a centered Gaussian process, but without stationary increments. It can be seen that $\{\xi(t)\}$ is $3/2$–self-similar. In fact, supposing $0 \leq s \leq t$, its covariance

$\gamma(t, s)$ is

$$\gamma(t,s) = \mathbb{E}\xi(t)\xi(s) = \mathbb{E}\int_0^t B(v)dv\int_0^s B(u)du = \int_0^s\int_0^t \mathbb{E}(B(v)B(u))dvdu$$
$$= \int_0^s\int_0^t \min\{v,u\}dvdu = \int_0^s\Big(\int_0^u vdv + \int_u^t udv\Big)du = \frac{s^2t}{2} - \frac{s^3}{6}$$

and, in particular, $\gamma(t,t) = t^3/3$. Hence,

$$\mathbb{E}(\xi(t)-\xi(s))^2 = \gamma(t,t)+\gamma(s,s)-2\gamma(t,s) = \frac{t^3}{3}+\frac{2s^3}{3}-s^2t.$$

Since $\xi(t)-\xi(s)$ is Gaussian, we have $\mathbb{E}|\xi(t)-\xi(s)| = \sqrt{(2/\pi)|t^3/3+2s^3/3-s^2t|}$, and in particular $\mathbb{E}|\xi(1)| = \sqrt{2/(3\pi)}$.

Therefore, by Corollary 3.7.2, the limit of the collision process $Y(Tt)/T^{3/4}$, properly normalized, is

$$\rho^{-1/2}(2/(3\pi))^{1/4}\, Z(t),$$

where $\{Z(t)\}$ is a 3/4–self-similar Gaussian process with covariance

$$\mathbb{E}Z(t)Z(s) = \frac{1}{2}\Big(t^{3/2}+s^{3/2}-|t^3+2s^3-3s^2t|^{1/2}\Big),\quad 0\le s\le t.$$

3.8 Motion of a Tagged Particle in a Simple Symmetric Exclusion Model

Simple exclusion is one of the canonical interacting particle systems. The corresponding exclusion system η_t, $t \ge 0$, is a Markov process with state space $\{0,1\}^{\mathbb{Z}}$; that is, it consists at each t of an infinite sequence of 0s or 1s. The process describes an evolution of configurations of indistinguishable particles on $\mathbb{Z}$ with at most one particle per site. A particle at $x \in \mathbb{Z}$ waits an exponentially distributed time with mean 1, and then chooses a site $y \in \mathbb{Z}$ with probability $p(x,y)$. If y is vacant at that time, the particle at x moves to y. Otherwise, it stays at x. Whether the particle moves or not, it restarts its exponential clock. All the exponential holding times and choices related to p are independent. Since the exponential holding time is continuous, only one particle moves at a time. A basic quantity of interest is the motion of one of these particles, the so-called *tagged particle*.

In the nearest-neighbor interactions, p satisfies $p(x,y) = 0$ if $|x-y| > 1$; that is, a particle can move only to a neighboring site. The focus here is on symmetric nearest-neighbor interactions; that is, the case $p(x,x+1) = p(x,x-1) = 1/2$. (Asymmetric and other non-nearest-neighbor interactions are discussed in Section 3.15.) The system is assumed to start from the (equilibrium) configuration determined by a product measure ν_ρ on $\{0,1\}^{\mathbb{Z}}$ with marginals ν_ρ(particle at x) $= \rho$, for all $x \in \mathbb{Z}$. The initial measure is conditioned to have a particle at 0, which is tagged and whose motion is denoted $Y(t)$. This is depicted in Figure 3.5.

The basic problem of interest is to understand the behavior of the motion $Y(T)$ of the tagged particle as $T \to \infty$, and more generally the motion viewed as a process $Y(Tt)$, $t \ge 0$, at large scales $T \to \infty$. As stated below, the limiting object is FBM with the self-similarity parameter $H = 1/4$, and the appropriate normalization of $Y(T)$ is $T^H = T^{1/4}$ and that of $Y(Tt)$ is $T^{1/4}$. The simple symmetric exclusion model can be thought as a discrete

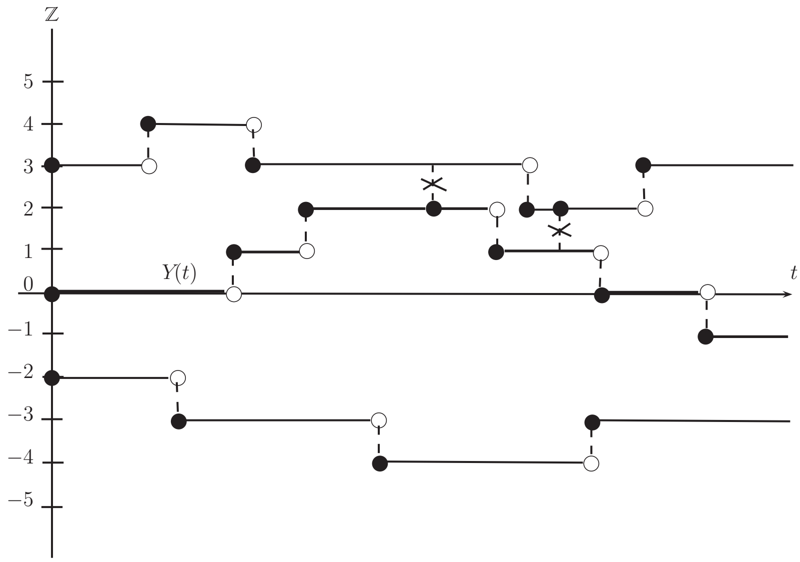

Figure 3.5 Exclusion process with the path $Y(t)$ of the tagged particle. The vertical dashed lines indicate the movement of the particles from one site to another. The vertical dashed lines with crosses indicate instances when the move was not possible because the neighboring site was occupied. A darkened circle indicates the position of the particle at that time.

analogue of the elastic collision model considered in Section 3.7 (with $\xi(t)$ being Brownian motion).

We focus on the behavior of $Y(T)$ only and state that of $Y(Tt)$ at the end of the section. We follow the approach in the original work of Arratia [46]. For other methods, see Rost and Vares [863], De Masi and Ferrari [293], and Liggett [632], Chapter VIII.

The approach of Arratia [46] is based on expressing the symmetric exclusion process in terms of another model, the so-called *stirring motions*. The system $\xi = \{\xi_t^x, x \in \mathbb{Z}, t \geq 0\}$ of random stirrings has exactly one particle per site at all times, with the particles at sites $x < y$ interchanged[5] at rate $p(x, y) = p(y, x)$ at event times determined by independent Poisson processes with intensity 1 for each pair of sites (x, y) with $p(x, y) > 0$. The probability that clocks ring at the same time is zero. Note that in the "stirring motions" model, there is no exclusion and that when the event clock rings, the interchange always occurs. The position at time t of the particle initially at site x is denoted ξ_t^x. For each x, the path ξ_t^x, $t \geq 0$, is a random walk based on p and starting at x.

For a set $A \subset \mathbb{Z}$, let

$$\eta_t^A = \{\xi_t^x : x \in A\}; \tag{3.8.1}$$

that is, η_t^A denotes the set of the positions of random walks ξ_t^x which started at all $x \in A$. The random walks ξ_t^x started at $x \in A$ intersect. To relate these random walks to the simple exclusion process, the next step is to define their "order statistics" representing the non-intersecting motions of the exclusion process. The "order statistics" $Y_t^x (= Y_t^{x,A})$ of the

[5] These interchanges of particles are viewed as "stirrings", hence the term "stirring motions."

stirring paths ξ_t^x for $x \in A$ is defined as the paths satisfying: for all $t \geq 0$,

$$\{Y_t^x : x \in A\} = \eta_t^A,$$
$$Y_t^x < Y_t^y, \quad \text{for all } x < y, \ x, y \in A,$$
$$Y_0^x = x, \quad \text{for all } x \in A,$$

and Y_t^x, $t \geq 0$, is right-continuous with left-limits for all $x \in A$. Thus, the "order statistics" processes Y_t^x maintain their time-zero ordering. As shown in Arratia [46], these processes can also be characterized as follows. Define

$$\mu_{A,x}^{+}(z,t) = \left|\{y \in A : y \leq x, \xi_t^y \geq z\}\right| = \sum_{y \leq x} 1_{\{y \in A, \xi_t^y \geq z\}}, \tag{3.8.2}$$

$$\mu_{A,x}^{-}(z,t) = \left|\{y \in A : y > x, \xi_t^y < z\}\right| = \sum_{y > x} 1_{\{y \in A, \xi_t^y < z\}}, \tag{3.8.3}$$

that is, $\mu_{A,x}^{+}(z,t)$ counts the number of particles y in A initially to the left of x (x included) which moved above z by time t, and $\mu_{A,x}^{-}(z,t)$ is the number of particles y in A initially to the right of x (x excluded) which moved below z by time t. Then, for any $z \in \mathbb{Z}$,

$$\{Y_t^x \geq z\} = \{\mu_{A,x}^{+}(z,t) > \mu_{A,x}^{-}(z,t)\} \tag{3.8.4}$$

(Exercise 3.16). It then follows that

$$Y_t^x = \sup\{z \in \mathbb{Z} : \mu_{A,x}^{+}(z,t) > \mu_{A,x}^{-}(z,t)\}. \tag{3.8.5}$$

The definition (3.8.5) of Y_t^x is valid for symmetric random walk where $p(x,y) = p(y,x)$. In the nearest-neighbor case, at most one of the paths Y_t^x moves at any time, so that Y_t^x is the path of the tagged particle initially at $x \in A$, in the exclusion system determined by (3.8.1). Connecting this observation back to the simple symmetric exclusion process, we also have

$$Y(t) = Y_t^0, \tag{3.8.6}$$

where we take A as the initial configuration η_0 with a particle added at site 0 if necessary, thus $A = \eta_0 \cup \{0\}$. (Strictly speaking, η_0 is an element of $\{0,1\}^{\mathbb{Z}}$ but it can be identified as a subset of $\mathbb{Z}$ in the natural one-to-one manner.)

Theorem 3.8.1 *For the simple symmetric exclusion process on $\mathbb{Z}$, starting from the initial product measure with marginals $\mathbb{P}$(particle at x) $= \rho$, for all $x \in \mathbb{Z} \setminus \{0\}$, $\mathbb{P}$(particle at 0) $= 1$, the position $Y(T)$ of the tagged particle initially at the origin satisfies*

$$\frac{Y(T)}{T^{1/4}} \xrightarrow{d} \mathcal{N}\left(0, \sqrt{\frac{2}{\pi}} \frac{1-\rho}{\rho}\right), \tag{3.8.7}$$

as $T \to \infty$. Furthermore, $\mathrm{Var}(T^{-1/4} Y(T)) \to \sqrt{2/\pi}(1-\rho)/\rho$.

Proof By using the relations (3.8.6) and (3.8.4), we can relate the simple symmetric exclusion process $Y(T)$ to the stirring motions ξ_T^y as follows:

$$\{Y(T) \geq z\} = \{\mu_{A,0}^{+}(z,T) > \mu_{A,0}^{-}(z,T)\} = \{\mu_{A,0}^{+}(z,T) - \mu_{A,0}^{-}(z,T) > 0\}$$

$$= \{\sum_{y\le 0} 1_{\{y\in A, \xi_T^y \ge z\}} - \sum_{y>0} 1_{\{y\in A, \xi_T^y < z\}} > 0\}, \tag{3.8.8}$$

where as before $A = \eta_0 \cup \{0\}$. The idea now is to investigate all those $\xi_T^y \ge z$ for $y \le 0$, and $\xi_T^y < z$ for $y > 0$, and then condition only on the y's belonging to A.

Let

$$N = \mu^+_{\mathbb{Z},0}(z,T) = \sum_{y\le 0} 1_{\{\xi_T^y \ge z\}}, \tag{3.8.9}$$

where A in $\mu^+_{A,0}(z,T)$ is taken as $A = \mathbb{Z}$. It can be shown (Exercise 3.16) that, for $z \ge 1$,

$$\mu^+_{\mathbb{Z},0}(z,T) - \mu^-_{\mathbb{Z},0}(z,T) = -|\mathbb{Z}\cap(0,z)| = -(z-1), \tag{3.8.10}$$

which yields

$$\sum_{y>0} 1_{\{\xi_T^y < z\}} = \mu^-_{\mathbb{Z},0}(z,T) = N + z - 1. \tag{3.8.11}$$

Then, label the sites contributing to (3.8.9) and (3.8.11) as

$$\begin{aligned}\{y \le 0 : \xi_T^y \ge z\} &= \{y_{-N} < y_{-N+1} < \cdots < y_{-1}\},\\ \{y > 0 : \xi_T^y < z\} &= \{y_1 < y_2 < \cdots < y_{N+z-1}\}.\end{aligned} \tag{3.8.12}$$

We put back $A = \eta_0 \cup \{0\}$ into the picture by considering

$$S = \sum_{i=-N}^{-1} 1_{\{y_i\in\eta_0\}} - \sum_{i=1}^{N+z-1} 1_{\{y_i\in\eta_0\}}, \tag{3.8.13}$$

where η_0 is the initial configuration (not conditioned to have particle at 0). It follows from (3.8.8)–(3.8.13) that

$$\{Y(T)\ge z\} = \{S + 1_{\{\xi_T^0\ge z, 0\notin\eta_0\}} > 0\}, \tag{3.8.14}$$

where the last indicator function accounts for the possibility that $0 \notin \eta_0$. In view of the result to prove, we shall take

$$z = \Big\lfloor a\Big(\frac{1-\rho}{\rho}\Big)^{1/2}\Big(\frac{2T}{\pi}\Big)^{1/4}\Big\rfloor, \tag{3.8.15}$$

where $a \in \mathbb{R}$ is fixed.

The right-hand side of (3.8.14) is amenable to straightforward analysis. Indeed, conditionally on N, S is just the difference of the sums of i.i.d. variables for which CLT holds. To be able to condition on N, we shall use a result established next. By Lemma 3.8.2 below, N in (3.8.9) is the sum of negatively correlated variables. Hence,

$$\mathrm{Var}(N) = \mathrm{Var}(\sum_{y\le 0} 1_{\{\xi_T^y\ge z\}}) \le \sum_{y\le 0}\mathrm{Var}(1_{\{\xi_T^y\ge z\}}) \le \sum_{y\le 0}\mathbb{E}1_{\{\xi_T^y\ge z\}} = \mathbb{E}N.$$

Observe next that $\mathbb{E}N = \sum_{y\le 0}\mathbb{P}(\xi_T^y \ge z) = \sum_{y\le 0}\mathbb{P}(\xi_T^0 \ge z-y) = \sum_{v\ge 1}\mathbb{P}(\xi_T^0 \ge z-1+v) = \sum_{v\ge 1}\mathbb{P}(\xi_T^0 - z + 1 \ge v) = \sum_{v\ge 1}\mathbb{P}((\xi_T^0 - z+1)_+ \ge v) = \mathbb{E}(\xi_T^0 - z + 1)_+$ by using the formula $\mathbb{E}X = \sum_{v\ge 1}\mathbb{P}(X\ge v)$ for nonnegative integer-valued random variable X. Since ξ_t^0 is the random walk $\sum_{k=1}^{P(t)} X_k$ with a Poisson process $P(t)$ having intensity 1 and $X_k = \pm 1$ with probability $1/2$, ξ_T^0 follows approximately the distribution $\mathcal{N}(0, T)$, as $T \to \infty$. Then, since $z = z(T) = O(T^{1/2}) = o(T^2)$ as $T \to \infty$, we have $\mathbb{E}N = \mathbb{E}(\xi_T^0 - z + 1)_+ \sim T^{1/2}\mathbb{E}\mathcal{N}(0,1)_+ = T^{1/2}\mathbb{E}|\mathcal{N}(0,1)|/2 = (T/(2\pi))^{1/2}$. By Chebyshev's inequality,

$$\mathbb{P}\Big(\Big|N - \Big(\frac{T}{2\pi}\Big)^{1/2}\Big| > T^{3/8}\Big) \le \mathbb{P}\Big(\Big|N - \mathbb{E}N\Big| > \frac{T^{3/8}}{2}\Big) + \mathbb{P}\Big(\Big|\mathbb{E}N - \Big(\frac{T}{2\pi}\Big)^{1/2}\Big| > T^{3/8}\Big)$$
$$\le \frac{4\mathrm{Var}(N)}{T^{3/4}} + \mathbb{P}\Big(\Big|\frac{(2\pi)^{1/2}\mathbb{E}N}{T^{1/2}} - 1\Big| > (2\pi)^{1/2}T^{-1/8}\Big) \to 0, \tag{3.8.16}$$

by using $\mathrm{Var}(N) \le \mathbb{E}N \sim T^{1/2}/(2\pi)^{1/2}$ from above. The convergence of the last term to 0 above is left as Exercise 3.17.

The result (3.8.16) shows that N can be assumed to satisfy $|N - (T/(2\pi))^{1/2}| < T^{3/8}$ asymptotically. We argue next that, conditionally on any such value of N, the quantity S in (3.8.13) is asymptotically normal. Fixing $N = n$, let

$$S_{n,T} = \sum_{i=-n}^{-1} 1_{\{y_i \in \eta_0\}} - \sum_{i=1}^{n+z-1} 1_{\{y_i \in \eta_0\}},$$

where $z = z(T)$ is given in (3.8.15) and $n = n(T)$ satisfies $|n - (T/(2\pi))^{1/2}| < T^{3/8}$. Since $\mathbb{P}(y_i \in \eta_0) = \rho$, we have $\mathrm{Var}(\sum_{i=-n}^{-1} 1_{\{y_i\in\eta_0\}}) = n\rho(1-\rho)$ and $\mathrm{Var}(S_{n,T}) = (2n + z - 1)\rho(1-\rho)$. Since $z(T) = o(n(T))$, this yields $\mathrm{Var}(\sum_{i=-n}^{-1} 1_{\{y_i\in\eta_0\}})/\mathrm{Var}(S_{n,T}) = n/(2n + z - 1) \to 1/2$ and hence

$$\frac{\sum_{i=-n}^{-1} 1_{\{y_i\in\eta_0\}} - n\rho}{(\mathrm{Var}(S_{n,T}))^{1/2}} \xrightarrow{d} \mathcal{N}(0, 1/2), \qquad \frac{\sum_{i=1}^{n+z-1} 1_{\{y_i\in\eta_0\}} - (n+z-1)\rho}{(\mathrm{Var}(S_{n,T}))^{1/2}} \xrightarrow{d} \mathcal{N}(0, 1/2).$$

For $n(T) \sim (T/(2\pi))^{1/2}$ and $z(T)$ as in (3.8.15), $\mathbb{E}S_{n,T}/(\mathrm{Var}(S_{n,T}))^{1/2} = -(z - 1)\rho/(\rho(1-\rho)(2n + z - 1))^{1/2} \to -a$, so that $S_{n,T}/(\mathrm{Var}(S_{n,T}))^{1/2} \xrightarrow{d} \mathcal{N}(-a, 1)$. We now conclude that, for any fixed $b \in \mathbb{R}$, as $T \to \infty$,

$$\sup_{n:|n-(\frac{T}{2\pi})^{1/2}|<T^{3/8}} \Big|\mathbb{P}(S_{n,T} \ge b) - \int_a^\infty \frac{e^{-u^2/2}}{\sqrt{2\pi}}du\Big| \to 0, \tag{3.8.17}$$

since under the supremum, $n \to \infty$ as $T \to \infty$ (b will be taken as 0 and 1 below).

The combination of (3.8.16) and (3.8.17) now yields the desired result (3.8.7). Indeed, let $E_T = \{|N - (\frac{T}{2\pi})^{1/2}| < T^{3/8}\}$ so that $\mathbb{P}(E_T) \to 1$ as $T \to \infty$ by (3.8.16). Then, by (3.8.14) with z in (3.8.15),

$$\mathbb{P}(Y(T) \ge z) \le \mathbb{P}(S > -1) \le \mathbb{P}(E_T^c) + \mathbb{P}(S > -1|E_T)\mathbb{P}(E_T) \to \mathbb{P}(\mathcal{N}(0,1) > a),$$

by using (3.8.17) with $b = -1$. Similarly, $\mathbb{P}(Y(T) \ge z) \ge \mathbb{P}(S > 0) \ge \mathbb{P}(S > 0|E_T)\mathbb{P}(E_T) \to \mathbb{P}(\mathcal{N}(0,1) > a)$. Therefore, $\mathbb{P}(Y(T) \ge z) \to \mathbb{P}(\mathcal{N}(0,1) > a)$, as $T \to \infty$. Since $z = z(T)$ is given by (3.8.15), we thus have shown that

$$\mathbb{P}\Big(\Big(\frac{\rho}{1-\rho}\Big)^{1/2}\Big(\frac{\pi}{2}\Big)^{1/4}\frac{Y(T)}{T^{1/4}} \ge a\Big) \to \mathbb{P}(\mathcal{N}(0,1) > a)$$

and hence (3.8.7).

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The stated convergence of the second moments can be proved by showing that $\{T^{-1/2}Y(T)^2, t \geq 1\}$ is uniformly integrable. See Arratia [46], p. 368. $\square$

The following auxiliary result was used in the proof above. It is proved in Arratia [46], Lemma 1, by employing the result of Harris [456] on Markov processes preserving the class of measures having positive correlations.

Lemma 3.8.2 *In a system $\{\xi_t^x, x \in \mathbb{Z}, t \geq 0\}$ of random stirrings, based on symmetric, nearest-neighbor random walk $p(x, x+1) = p(x, x-1) = 1/2$, the events $\{\xi_t^v \geq a\}$ and $\{\xi_t^w \geq b\}$ are negatively correlated for all $t \geq 0$, $v, w, a, b \in \mathbb{Z}$ with $v \neq w$.*

We conclude this section by stating the convergence in law of the corresponding scaled path of the tagged particle.

Theorem 3.8.3 *Under the assumptions and notation of Theorem 3.8.1,*

$$\left\{\frac{Y(Tt)}{T^{1/4}}\right\}_{t\geq 0} \xrightarrow{d} \left\{\left(\sqrt{\frac{2}{\pi}}\frac{1-\rho}{\rho}\right)^{1/2} B_H(t)\right\}_{t\geq 0}, \tag{3.8.18}$$

where B_H is a standard FBM with the self-similarity parameter $H = 1/4$, and the convergence in distribution is in the space $D[0,1]$ of right-continuous functions on $[0,1]$ with left limits, endowed with the uniform topology.

The proof of Theorem 3.8.3 can be found in Peligrad and Sethuraman [803].

3.9 Power-Law Pólya's Urn

The correlated random walk presented in this section was introduced by Hammond and Sheffield [449]. Their informal description of the walk is as follows (Hammond and Sheffield [449], p. 697). Let μ be a probability law on the natural numbers $\mathbb{N} = \{1, 2, \ldots\}$. The walk associated with law μ will have dependent increments each of which is either -1 or 1. The series of the increments $\{X_n\}_{n\in\mathbb{Z}}$ of the random walk is such that, given the values $\{X_k\}_{k<n}$, X_n is set equal to the increment X_{n-m_n} obtained by looking back m_n steps in the series. The m_ns are random and are obtained at each vertex $n \in \mathbb{Z}$ by sampling from μ independently. This description is reminiscent of the traditional Pólya's urn process where the value of X_n represents the color of the nth ball added, and the conditional law of X_n given the past is that of a uniform sample from $\{X_1, X_2, \ldots, X_{n-1}\}$.

The look-back involves not the uniform distribution as in the traditional Pólya's urn case, but a power-law distribution μ satisfying: for $\alpha > 0$,

$$\mu\{m, m+1, \ldots\} = m^{-\alpha} L(m), \quad m \geq 1, \tag{3.9.1}$$

where L is a (positive) slowly varying function at infinity. Thus, the tail distribution of μ is regularly varying with the index $(-\alpha)$. Denote the class of distributions μ satisfying (3.9.1) by Γ_α. For the convergence to FBM below, the parameter α will be taken as $\alpha \in (0, 1/2)$.

A more rigorous construction of the random walk is as follows. For a fixed law μ, let G_μ be the random directed graph on $\mathbb{Z}$ in which each vertex z has a unique edge pointing

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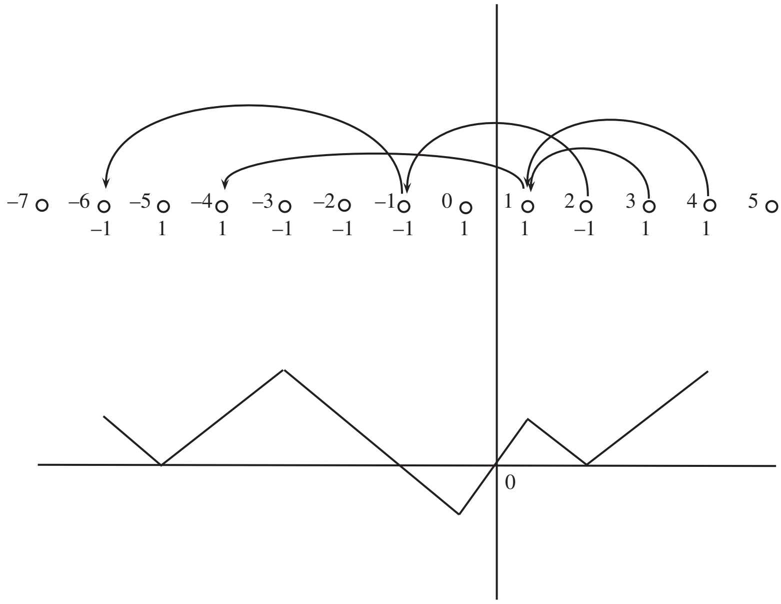

Figure 3.6 In the top plot, the ± 1 values are determined by looking at the ancestral lines. In the bottom plot, they are summed and thus define the random walk.

backwards to $z - m_z$, where m_z is sampled independently according to the measure μ. The point $z - m_z$ is called the *parent* of z. The *ancestral line* of z is the decreasing sequence whose first element is z and each of whose terms is the parent of the previous one. Starting from different zs, the ancestral lines may meet at some point. See the top of Figure 3.6 where the two vertices 3 and 4 have their ancestral lines meet at 1. So each realization of G_μ partitions $\mathbb{Z}$ into connected components consisting of those zs whose ancestral lines meet.

The next result, stated here without proof (see Proposition 2 in Hammond and Sheffield [449]), describes the structure of G_μ when $\mu \in \Gamma_\alpha$.

Proposition 3.9.1 *Let $\mu \in \Gamma_\alpha$, $\alpha > 0$. If $\alpha > 1/2$, then G_μ has one component almost surely. If $\alpha < 1/2$, then G_μ has infinitely many components almost surely.*

In view of Proposition 3.9.1, we shall suppose $\alpha < 1/2$. For $p \in (0, 1)$ and $\mu \in \Gamma_\alpha$, define a law $\lambda_p = \lambda_{p,\mu}$ on functions mapping $\mathbb{Z}$ to $\{-1, 1\}$ in two steps as follows. First, determine the random components of G_μ. Then, independently give each component of G_μ a value of 1 with probability p, or -1 with probability $1 - p$, and assign the corresponding value to all vertices in that component.

To the measure λ_p, we associate a (correlated) random walk $S_p : \mathbb{Z} \to \mathbb{Z}$ by setting $S_p(0) = 0$, and

$$S_p(n) = \begin{cases} \sum_{k=1}^{n} X_p(k), & n \geq 1, \\ -\sum_{k=-n+1}^{0} X_p(k), & n \leq -1, \end{cases}$$

where $X_p : \mathbb{Z} \to \{-1, 1\}$ is a realization of the law λ_p. See Figure 3.6. Note that $S_p(n) - S_p(n-1) = X_p(n)$, $n \in \mathbb{Z}$.

In the next result, we study the asymptotic behavior of the variance of the random walk $S_p(N)$, as $N \to \infty$. To do so, we will need to understand when ancestral lines starting from two different vertices j and k meet, in which case $X_p(j) = X_p(k)$ in the definition of the random walk $S_p(n)$. To be able to trace ancestral lines, we introduce a new random walk $\widetilde{S}_j$ as follows. Using the probability distribution μ on $\mathbb{N}$ defined in (3.9.1), let $\widetilde{S}_0 = 0$ and $\widetilde{S}_j = \sum_{l=1}^{j} Y_l$, $j \geq 1$, where Y_l are i.i.d. copies drawn from the law μ. Let also R_μ be the set of values assumed by the random walk $\widetilde{S}_j$; that is,

$$R_\mu = \Big\{ m \in \mathbb{N} : \ m = \widetilde{S}_j \text{ for some } j \geq 0 \Big\}. \tag{3.9.2}$$

Note that R_μ is a random set. Do not confuse the correlated random walk $S_p(n)$ above whose increments equal ± 1 with $\widetilde{S}_j$ whose increments are i.i.d. random variables with law μ. Set also, for $m \geq 0$,

$$q_m = \mathbb{P}(m \in R_\mu). \tag{3.9.3}$$

Note that $q_0 = 1$ since $0 \in R_\mu$. Thus, q_m denotes the probability that an ancestral line started at a vertex will visit the vertex located m steps back in the past.

Proposition 3.9.2 *Let $\alpha \in (0, 1/2)$, $\mu \in \Gamma_\alpha$ and S_p be the random walk defined above for $p \in (0, 1)$ and associated with the law μ. Then,*

$$\operatorname{Var}\Big(S_p(N) \Big) \sim \frac{4p(1-p)K_\alpha}{\sum_{m=0}^{\infty} q_m^2} N^{2\alpha+1} L(N)^{-2}, \tag{3.9.4}$$

where

$$K_\alpha = \frac{1}{2\alpha(2\alpha+1)} \Big(\Gamma(1-2\alpha)^2 \Gamma(2\alpha) \cos(\alpha\pi) \Big)^{-1}.$$

Proof Denote the Fourier transform of the sequence $\{q_m\}$ as

$$Q(t) = \sum_{m=0}^{\infty} q_m e^{-imt}. \tag{3.9.5}$$

Note that $(|Q|^2)_j = \sum_{m=0}^{\infty} q_m q_{j+m}$ is the jth Fourier coefficients of $|Q|^2 = Q\overline{Q}$. We will show first that

$$\operatorname{Var}(S_p(N)) = \frac{4p(1-p)}{(|Q|^2)_0} \Big(2 \sum_{n=1}^{N-1} \sum_{j=1}^{n} (|Q|^2)_j + N(|Q|^2)_0 \Big). \tag{3.9.6}$$

Observe that

$$\operatorname{Var}(S_p(N)) = \sum_{j=1}^{N} \sum_{k=1}^{N} \operatorname{Cov}(X_p(j), X_p(k)). \tag{3.9.7}$$

Two cases need to be considered for $\operatorname{Cov}(X_p(j), X_p(k))$. In one case, j and k are in different components of the random graph G_μ. Hence, conditioned on this, the covariance of $X_p(j)$ and $X_p(k)$ is 0. This case can be expressed as $A_j \cap A_k = \emptyset$, where A_j and A_k denote the ancestral lines of j and k, respectively. (Note that A_j and A_k are random sets.) In the other case, j and k are in the same component of the random graph G_μ, corresponding to $A_j \cap A_k \neq \emptyset$. Note that, in this case, $X_p(j) = X_p(k) = 1$ with probability p, and $X_p(j) = X_p(k) = -1$ with probability $1-p$. In particular, conditioned on this case, the covariance of $X_p(j)$ and $X_p(k)(= X_p(j))$ is the variance of $X_p(j)$, which is $4p(1-p)$. This shows that $\mathrm{Cov}(X_p(j), X_p(k)) = 4p(1-p)\mathbb{P}(A_j \cap A_k \neq \emptyset)$ and hence, from (3.9.7),

$$\mathrm{Var}(S_p(N)) = 4p(1-p)\sum_{j=1}^{N}\sum_{k=1}^{N}\mathbb{P}(A_j \cap A_k \neq \emptyset). \tag{3.9.8}$$

To compute $\mathbb{P}(A_j \cap A_k \neq \emptyset)$, that is, the probability that the ancestral lines of j and k meet, note first that $\mathbb{P}(A_j \cap A_k \neq \emptyset) = \mathbb{P}(A_j \cap A'_k \neq \emptyset)$, where A'_k is the ancestral line of k in an independent copy of G_μ. Indeed, this is easier to understand thinking of the equality $\mathbb{P}(A_j \cap A_k = \emptyset) = \mathbb{P}(A_j \cap A'_k = \emptyset)$ of the complements, since in this case, the non-intersecting ancestral lines A_j and A_k can be thought independent. To compute $\mathbb{P}(A_j \cap A'_k \neq \emptyset)$, let $|A_j \cap A'_k|$ be the number of points $z \in \mathbb{Z}$ in $A_j \cap A'_k$. Note that, for $j \leq k$, $|A_j \cap A'_k| = \sum_{h=-\infty}^{j} 1_{\{h \in A_j \cap A'_k\}}$; that is, $|A_j \cap A'_k|$ is the number of $h \leq j$ such that $h \in A_j \cap A'_k$. Hence, by using independence of A_j and A'_k,

$$\mathbb{E}|A_j \cap A'_k| = \sum_{h=-\infty}^{j} \mathbb{P}(h \in A_j)\mathbb{P}(h \in A'_k).$$

Note that, for fixed j,

$$A_j \stackrel{d}{=} j - R_\mu := \{j - m : m \in R_\mu\}, \tag{3.9.9}$$

where R_μ is defined in (3.9.2). Hence,

$$\mathbb{E}|A_j \cap A'_k| = \sum_{h=-\infty}^{j} \mathbb{P}(j-h \in R_\mu)\mathbb{P}(k-h \in R_\mu)$$

and, by using (3.9.3),

$$\mathbb{E}|A_j \cap A'_k| = \sum_{h=-\infty}^{j} q_{j-h}q_{k-h} = \sum_{m=0}^{\infty} q_m q_{k-j+m} = (|Q|^2)_{k-j}. \tag{3.9.10}$$

On the other hand, note also that

$$\begin{aligned}\mathbb{E}|A_j \cap A'_k| &= \mathbb{E}\Big(|A_j \cap A'_k|\Big|A_j \cap A'_k \neq \emptyset\Big)\mathbb{P}(A_j \cap A'_k \neq \emptyset)\\ &= \Big(1 + \sum_{m=1}^{\infty} q_m^2\Big)\mathbb{P}(A_j \cap A_k \neq \emptyset) = \Big(\sum_{m=0}^{\infty} q_m^2\Big)\mathbb{P}(A_j \cap A_k \neq \emptyset)\\ &= (|Q|^2)_0\,\mathbb{P}(A_j \cap A_k \neq \emptyset), \end{aligned} \tag{3.9.11}$$

since $\mathbb{P}(A_j \cap A'_k \neq \emptyset) = \mathbb{P}(A_j \cap A_k \neq \emptyset)$ as discussed above, and given $A_j \cap A'_k \neq \emptyset$, the two ancestral lines will first meet at some point (contributing 1 in (3.9.11)) and then develop as two independent related partial sum process $\widetilde{S}$ of i.i.d. steps with distribution μ (contributing $\sum_{m=1}^{\infty} q_m^2$ for the conditional expected number).

Combining (3.9.10) and (3.9.11) yields $\mathbb{P}(A_j \cap A_k \neq \emptyset) = (|Q|^2)_{k-j}/(|Q|^2)_0$, $j \leq k$, and hence, from (3.9.8),

$$\mathrm{Var}(S_p(N)) = \frac{4p(1-p)}{(|Q|^2)_0} \sum_{j=1}^{N} \sum_{k=1}^{N} (|Q|^2)_{|k-j|}, \tag{3.9.12}$$

which can be written as (3.9.6) (see also the proof of Proposition 2.2.5). In view of (3.9.12) and by using Proposition 2.2.1, it is then enough to show that

$$\sum_{j=1}^{n} (|Q|^2)_j \sim n^{2\alpha} L(n)^{-2} \frac{K_\alpha(2\alpha+1)}{2} = n^{2\alpha} L(n)^{-2} \frac{1}{4\alpha} \Big(\Gamma(1-\alpha)^2 \Gamma(2\alpha) \cos(\alpha\pi) \Big)^{-1}, \tag{3.9.13}$$

as $n \to \infty$. To do so, introduce $p_m = \mu\{m\}$, that is, the probability that a μ–distributed random variable takes the value m, and let $P(t) = \sum_{m=1}^{\infty} p_m e^{-imt}$ be the Fourier transform of the sequence $\{p_m\}$. By Lemma 3.9.3, $Q(t) = (1 - P(t))^{-1}$.

By using (A.2.12) in Appendix A.2 with $\alpha \in (0, 1/2)$, we have, as $t \to 0$,

$$|P(t) - 1| \sim C_\alpha t^\alpha L(t^{-1}) \Big| 1 - i \tan\Big(\frac{\alpha\pi}{2}\Big) \Big| = \Gamma(1-\alpha) t^\alpha L(t^{-1}),$$

since $C_\alpha = \Gamma(1-\alpha)\cos(\alpha\pi/2)$ (see (A.2.13)) and

$$\Big| 1 - i \tan\Big(\frac{\alpha\pi}{2}\Big) \Big| = \frac{|\cos(\alpha\pi/2) - i \sin(\alpha\pi/2)|}{\cos(\alpha\pi/2)} = \frac{|e^{-i\alpha\pi/2}|}{\cos(\alpha\pi/2)} = \frac{1}{\cos(\alpha\pi/2)}.$$

Hence, since $|Q(t)| = |P(t) - 1|^{-1}$,

$$|Q(t)| \sim \Gamma(1-\alpha)^{-1} t^{-\alpha} L(t^{-1})^{-1},$$

so that

$$|Q(t)|^2 \sim \Gamma(1-\alpha)^{-2} t^{-2\alpha} L(t^{-1})^{-2},$$

as $t \to 0$. Note by symmetry that $|Q(t)|^2 = \sum_{m=0}^{\infty} q_m^2 + 2 \sum_{j=1}^{\infty} (|Q|^2)_j \cos(jt)$ and hence, as $t \to 0$,

$$\begin{aligned} \sum_{j=1}^{\infty} (|Q|^2)_j \cos(jt) &\sim \frac{\Gamma(1-\alpha)^{-2}}{2} t^{-2\alpha} L(t^{-1})^{-2} \\ &= \frac{\ell(t^{-1})}{t^{2\alpha}} \Gamma(2\alpha)\cos(\alpha\pi) = \frac{\ell(t^{-1})}{t^{1-\alpha'}} \Gamma(1-\alpha')\sin(\alpha'\pi/2), \end{aligned}$$

where

$$\alpha' = 1 - 2\alpha, \quad \ell(t^{-1}) = \frac{\Gamma(1-\alpha)^{-2} L(t^{-1})^{-2}}{2\Gamma(2\alpha)\cos(\alpha\pi)} = \frac{\Gamma(1-\alpha)^{-2} L(t^{-1})^{-2}}{2\Gamma(1-\alpha')\sin(\alpha'\pi/2)}.$$

By applying Theorem 4.10.1, (a), in Bingham et al. [157] with α replaced by α', we conclude that, as $n \to \infty$,

$$\sum_{j=1}^{n} (|Q|^2)_j \sim n^{1-\alpha'} \frac{\ell(n)}{1-\alpha'},$$

which is the desired relation (3.9.13) after substituting α' and the slowly varying function $\ell(n)$. □

The following auxiliary lemma was used in the proof above.

Lemma 3.9.3 *Let μ be a probability measure on $\mathbb{N}$, $p_m = \mu\{m\}$, $m \in \mathbb{N}$, and $P(t) = \sum_{m=1}^{\infty} p_m e^{-imt}$ be the Fourier transform of the sequence $\{p_m\}$. Let q_m be the probabilities (3.9.3) of a random walk $\widetilde{S}_j$ with i.i.d. steps drawn from the distribution μ visiting the sites $m \geq 0$ and $Q(t)$ be the Fourier transform (3.9.5) of the sequence $\{q_m\}$. Then,*

$$Q(t) = \sum_{j=0}^{\infty} (P(t))^j = \frac{1}{1 - P(t)}. \tag{3.9.14}$$

Proof Observe that, for $m \geq 0$,

$$q_m = \mathbb{P}(m \in R_\mu) = 1_{\{m=0\}} + \sum_{j=1}^{\infty} \mathbb{P}(\widetilde{S}_j = m) = 1_{\{m=0\}} + \sum_{j=1}^{\infty} (p^{*j})_m,$$

where p_m^{*j} denotes the mth element of the jth convolution p^{*j} of the sequence $p = \{p_m\}$. By taking the Fourier transform of the relation above, we obtain the first relation in (3.9.14). □

The scaling of the order $N^{2\alpha+1}$ derived in Proposition 3.9.2 shows that the stationary series of the increments of the correlated random walk is long-range dependent with parameter $d = \alpha$ in the sense (2.1.8) in condition V. Furthermore, as the next result shows, the scaled correlated random walk in fact converges to fractional Brownian motion. See Hammond and Sheffield [449], Proposition 3, for a proof based on a martingale central limit theorem.

Theorem 3.9.4 *Let $\alpha \in (0, 1/2)$, $\mu \in \Gamma_\alpha$ and S_p be the random walk defined above for $p \in (0, 1)$ and associated with the law μ. Let also*

$$\widetilde{c} = \Big(\frac{4p(1-p)K_\alpha}{(|Q|^2)_0} \Big)^{-1/2}$$

be the reciprocal square root of the constant appearing on the right-hand side of (3.9.4). Then,

$$\widetilde{c} N^{-\frac{1}{2}-\alpha} L(N) \Big(S_p([Nt]) - Nt(2p-1) \Big) \xrightarrow{fdd} B_H(t), \quad t \geq 0, \tag{3.9.15}$$

where B_H is a standard FBM with the self-similarity parameter

$$H = \frac{1}{2} + \alpha \in \Big(\frac{1}{2}, 1 \Big). \tag{3.9.16}$$

3.10 Random Walk in Random Scenery

Kesten and Spitzer [553] considered sums of random variables with summands $Y_n = \xi(S_n)$, where

- $\{S_n, n \geq 1\}$ is a zero mean (integer-valued) random walk,
- $\xi(k)$, $k \in \mathbb{Z}$, are independent and identically distributed random variables independent of the S_ns.

The sequence $Y_n = \xi(S_n)$, $n \geq 1$, describes the motion of the random walker S_n viewed through the perspective of the random scenery $\xi(\cdot)$.[6] If one views S_n as describing the "position" of the "state" of a user at time n, and $\xi(k)$ the "reward" earned by, "cost" incurred by or "amount of work" produced by the user in state k, then $\xi(S_n)$ can also be thought as the reward earned at time n. The $\xi(S_n)$s are dependent because the random variable $\xi(S_n)$ takes the same value whenever the random walk visits the same state $\xi(k)$. By choosing the distributions as indicated in the following theorem, one obtains a self-similar process.

Theorem 3.10.1 *Suppose that, as $N \to \infty$, $N^{-1/\alpha} S_N$ converges to a $S\alpha S$ random variable with $1 < \alpha \leq 2$ and $N^{-1/\beta} \sum_{k=1}^{N} \xi(k)$ converges to a $S\beta S$ random variable with $0 < \beta \leq 2$. Then,*

$$\frac{1}{N^{1-1/\alpha+1/(\alpha\beta)}} \sum_{n=1}^{[Nt]} \xi(S_n) \xrightarrow{fdd} \Delta_t = \int_{\mathbb{R}} \ell_t(x) X(dx), \quad t \geq 0, \tag{3.10.1}$$

where $\ell_t(x)$ is the local time of a $S\alpha S$ Lévy motion and $X(dx)$ is a $S\beta S$ random measure on $\mathbb{R}$ with the Lebesgue control measure dx, which is independent of $\ell_t(\cdot)$. The limit process is H–SSSI with

$$H = 1 - \frac{1}{\alpha} + \frac{1}{\alpha\beta} \in \left(\frac{1}{2}, \infty\right). \tag{3.10.2}$$

The local time $\ell_t(x)$, $x \in \mathbb{R}$, of a process represents roughly the density of time in $[0, t]$ that the process spends at x. For more information about local times, see Berman [134], Section 5.12, as well as Section 7.2.5 below in the case of FBM. The local time $\ell_t(x)$ of a process $Y(\cdot)$ satisfies the following occupation time formula:

$$\int_0^t 1_A(Y(s))ds = \int_A \ell_t(x)dx, \tag{3.10.3}$$

which expresses in two different ways the total amount of time in $[0, t]$ that the process $Y(\cdot)$ spends in the set A, doing so in the left-hand side of (3.10.3) by integrating over time $s \in [0, t]$ and in the right-hand side by integrating over space $x \in \mathbb{R}$. More generally, one has

$$\int_0^t f(Y(s))ds = \int_{\mathbb{R}} f(x)\ell_t(x)dx, \tag{3.10.4}$$

for suitable functions f.

Observe now that the right-hand side of (3.10.1) has the same meaning than its left-hand side, but this meaning is expressed in terms of the continuous limit processes. Indeed, the convergence (3.10.1) can informally be derived as follows. Let $\{X(t)\}$ and $\{Y(t)\}$ be $S\beta S$

[6] Random walk in random scenery should not be confused with the so-called random walk in random environment, where the motion of the walk is dictated by the environment. See, for example, Kesten, Kozlov, and Spitzer [554], Sznitman [940], Bogachev [166].

and $S\alpha S$ Lévy motions approximating the partial sums S_N and $\sum_{k=1}^{N}\xi(k)$, respectively; that is,

$$\frac{1}{N^{1/\alpha}}S_{[Nt]} \approx Y(t), \qquad \frac{1}{N^{1/\beta}}\sum_{k=1}^{[Nt]}\xi(k) \approx X(t). \tag{3.10.5}$$

The left-hand side of (3.10.3) is then approximated as

$$\int_0^t 1_A(Y(s))ds \approx \int_0^t 1_A\Big(\frac{1}{N^{1/\alpha}}S_{[Ns]}\Big)ds. \tag{3.10.6}$$

On the other hand, note that

$$\begin{aligned}
\int_0^t 1_A\Big(\frac{1}{N^{1/\alpha}}S_{[Ns]}\Big)ds &\approx \frac{1}{N}\sum_{n=1}^{[Nt]} 1_{\{\frac{1}{N^{1/\alpha}}S_n\in A\}} = \frac{1}{N}\sum_{N^{-1/\alpha}y\in A,\, y\in\mathbb{Z}}\ \sum_{n=1}^{[Nt]} 1_{\{S_n=y\}}\\
&= \sum_{N^{-1/\alpha}y\in A,\, y\in\mathbb{Z}}\Big(\frac{1}{N^{1-1/\alpha}}\sum_{n=1}^{[Nt]}1_{\{S_n=y\}}\Big)\frac{1}{N^{1/\alpha}}\\
&= \sum_{N^{-1/\alpha}y\in A,\, y\in\mathbb{Z}}\ell_{N,t}\Big(\frac{y}{N^{1/\alpha}}\Big)\frac{1}{N^{1/\alpha}} \approx \int_A \ell_{N,t}(x)dx,
\end{aligned} \tag{3.10.7}$$

where

$$\ell_{N,t}(x) = \frac{1}{N^{1-1/\alpha}}\sum_{n=1}^{[Nt]}1_{\{S_n=[N^{1/\alpha}x]\}}.$$

The relations (3.10.3), (3.10.6) and (3.10.7) suggest that $\ell_{N,t}(x)$ serves as an approximation to $\ell_t(x)$; that is,

$$\ell_{N,t}(x) \approx \ell_t(x). \tag{3.10.8}$$

Given this approximation and using the $(1/\beta)$–self-similarity of the $S\beta S$ Lévy motion $\{X(t)\}$ (Section 2.6.4), it remains to observe that

$$\begin{aligned}
&\frac{1}{N^{1-1/\alpha+1/(\alpha\beta)}}\sum_{n=1}^{[Nt]}\xi(S_n) = \frac{1}{N^{1-1/\alpha+1/(\alpha\beta)}}\sum_{n=1}^{[Nt]}\sum_{y\in\mathbb{Z}}\xi(y)1_{\{S_n=y\}} = \sum_{y\in\mathbb{Z}}\ell_{N,t}\Big(\frac{y}{N^{1/\alpha}}\Big)\frac{\xi(y)}{N^{1/(\alpha\beta)}}\\
&= \sum_{N^{1/\alpha}x\in\mathbb{Z}}\ell_{N,t}(x)\frac{\xi(N^{1/\alpha}x)}{N^{1/(\alpha\beta)}} \approx \sum_{N^{1/\alpha}x\in\mathbb{Z}}\ell_t(x)\frac{N^{1/\beta}}{N^{1/(\alpha\beta)}}\Big(X\Big(\frac{N^{1/\alpha}x}{N}\Big) - X\Big(\frac{N^{1/\alpha}(x-1)}{N}\Big)\Big)\\
&\stackrel{d}{=} \sum_{N^{1/\alpha}x\in\mathbb{Z}}\ell_t(x)\frac{1}{N^{1/(\alpha\beta)}}\Big(X(N^{1/\alpha}x) - X(N^{1/\alpha}(x-1))\Big)\\
&\stackrel{d}{=} \sum_{N^{1/\alpha}x\in\mathbb{Z}}\ell_t(x)\Big(X(x) - X(x-N^{-1/\alpha})\Big) \approx \int_{\mathbb{R}}\ell_t(x)dX(x).
\end{aligned}$$

The informal approach outlined above is formalized by Dombry and Guillotin-Plantard [322], and applies in a more general setting than that considered in Theorem 3.10.1.

The process $\{\Delta_t,\ t \geq 0\}$ in (3.10.1) is H–SSSI because the law of $\ell_{t+h}(x) - \ell_t(x)$ does not depend on $h \geq 0$ and because if $\ell_t(x)$ is the local time of an H'–SS process, then the self-similarity and the relation (3.10.3) imply

$$\{\ell_{ct}(c^{H'}x),\ x \in \mathbb{R},\ t \geq 0\} \stackrel{d}{=} \{c^{1-H'}\ell_t(x),\ x \in \mathbb{R},\ t \geq 0\} \tag{3.10.9}$$

(Exercise 3.18). Consequently, for any $c > 0$,

$$\begin{aligned}\Delta_{ct} &= \int_{\mathbb{R}} \ell_{ct}(x)dX(x) = \int_{\mathbb{R}} \ell_{ct}(c^{H'}x)dX(c^{H'}x)\\ &\stackrel{d}{=} \int_{\mathbb{R}} c^{1-H'}\ell_t(x)c^{H'/\beta}dX(x) = c^{1-H'+H'/\beta}\int_{\mathbb{R}} \ell_t(x)dX(x) = c^{1-1/\alpha+1/\alpha\beta}\Delta_t,\end{aligned}$$

since the $S\alpha S$ (resp. $S\beta S$) Lévy stable motion is self-similar with parameter $H' = 1/\alpha$ (resp. $1/\beta$).

Example 3.10.2 (*Simple random walk in Bernoulli-like scenery*) If S_N is a simple random walk and if the $\xi(k)$s with $k \in \mathbb{Z}$ take values ± 1 with probability $1/2$, then they are both in the domain of attraction of Brownian motion with $\alpha = \beta = 2$ and thus $H = 3/4$. The corresponding limit in (3.10.1) is $\int_{\mathbb{R}} \ell_t(x)B(dx)$; that is, intuitively $\int_0^t W(\widetilde{B}(s))ds$, where $W = B(dx)/dx$ is a Gaussian white noise independent of the Brownian motion $\widetilde{B}$ (see (3.10.4)).

By conditioning on ℓ_t, the finite-dimensional distributions of the process $\Delta_t = \int_{-\infty}^{\infty} \ell_t(z)X(dx)$, $t \geq 0$, in (3.10.1) are given by

$$\mathbb{E}\exp\Big\{i\sum_{j=1}^{J}\theta_j\Delta_{t_j}\Big\} = \mathbb{E}\exp\Big\{-C\int_{-\infty}^{\infty}\Big|\sum_{j=1}^{J}\theta_j\ell_{t_j}(x)\Big|^{\beta}dx\Big\}, \tag{3.10.10}$$

where C is a positive constant.

3.11 Two-Dimensional Ising Model

Another very interesting model associated with long-range dependence is the celebrated Ising[7] model, which originated in Statistical Mechanics. This is one of the few models of Statistical Mechanics which is "exactly solvable." A number of ingenious methods have been proposed to study the Ising model. We shall present here one such method, and show that the Ising model exhibits long-range dependence at the so-called critical temperature. We should add that our presentation will barely scratch the surface of what is known for the Ising model – for further notes see Section 3.15. Our goal is to provide the reader (perhaps one little familiar with Statistical Mechanics) with an idea of another interesting situation which leads to long-range dependence.

[7] Pronounced as [EEH sing].

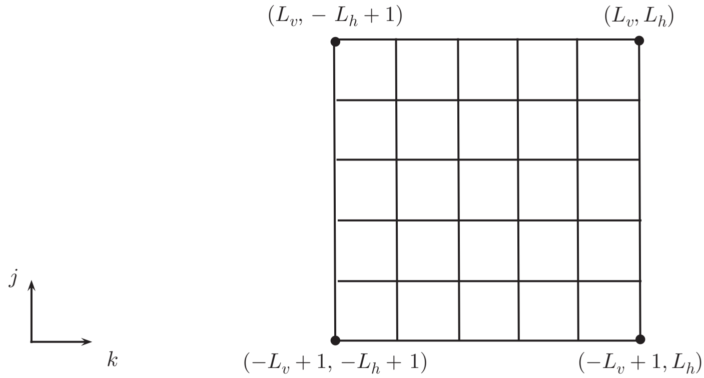

Figure 3.7 The Ising lattice.

3.11.1 Model Formulation and Result

We consider the Ising model in two dimensions. The reason is that, in one dimension, the Ising model does not exhibit long-range dependence (Exercise 3.19) and, in three dimensions, the Ising model has not been solved exactly yet (in fact, this is still one of the famous open problems in Statistical Mechanics and, more generally, Physics, though which may remain unsolved; see Cipra [245]). Thus, consider a two-dimensional square lattice (Figure 3.7), referred to as the *Ising lattice*, with

$$2L_v : \text{the number of rows,}$$
$$2L_h : \text{the number of columns}$$

("v" stands for "vertical", and "h" stands for "horizontal"). Denote a general point on the lattice as (j,k), $-L_v+1 \le j \le L_v$, $-L_h+1 \le k \le L_h$. Let

$$\sigma_{j,k} = \pm 1$$

be a variable associated with a lattice point (j,k).[8] We will consider below correlations between $\sigma_{0,0}$ and $\sigma_{0,N}$, and $\sigma_{0,0}$ and $\sigma_{N,N}$. In fact, we will first let $L_v, L_h \to \infty$ (the so-called thermodynamic limit), and then investigate these correlations as $N \to \infty$.

In the Ising model, the collection of variables $\{\sigma_{j,k}, -L_v+1 \le j \le L_v, -L_h+1 \le k \le L_h\}$ is assumed to follow the joint distribution

$$\frac{1}{Z_{L_v,L_h}} e^{-\beta\mathcal{E}}, \tag{3.11.1}$$

where

$$\mathcal{E} = -\sum_{j=-L_v+1}^{L_v}\sum_{k=-L_h+1}^{L_h} (E^h\sigma_{j,k}\sigma_{j,k+1} + E^v\sigma_{j,k}\sigma_{j+1,k}) \tag{3.11.2}$$

[8] Note that j and k in (j,k) refer to row (vertical) and column (horizontal) directions, respectively. This is in accord with the matrix notation where (j,k) is used for a matrix element $a_{j,k}$. But this is the opposite of the (x,y) convention, where for example x refers to the horizontal direction.

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with some positive constants E^h, $E^v > 0$, and where

$$Z_{L_v,L_h} = \sum_{\{\sigma_{j,k}\}} e^{-\beta\mathcal{E}} \tag{3.11.3}$$

is the normalization constant, where the sum is over all possible *configurations* (outcomes) $\{\sigma_{j,k}\}$. Since each $\sigma_{j,k}$ can take 2 values, there are $2^{(2L_v)(2L_h)}$ configurations. The parameter $\beta > 0$ is expressed as

$$\beta = \frac{1}{k_B T}, \tag{3.11.4}$$

where $k_B = 1.38 \times 10^{-23}$ is Boltzmann's constant and $T > 0$.

The term $\mathcal{E}$ in (3.11.2) is referred to as *energy* (*Hamiltonian*) of a configuration $\{\sigma_{j,k}\}$. Note that the energy is determined only by interactions between neighboring sites, with the contributions E^h for horizontal interactions and E^v for vertical interactions. Note also the following effect of individual terms on the overall energy of the system. The energy is lower when neighboring sites align as $+1$ and $+1$, or as -1 and -1. Lower (negative) energy $\mathcal{E}$ translates into higher probability (3.11.1); that is, as expected in physical terms, the system favors configurations with lower energy.

The parameter $T > 0$ in (3.11.4) is referred to as *temperature* (and β as *inverse temperature*). When $T \to \infty$ (high temperature), note that $\beta \to 0$ and hence (3.11.1) approaches the uniform probability distribution on the lattice. In this case, the variables $\sigma_{j,k}$ tend to be independent and the system is in the disordered state. On the other hand, when $T \to 0$ (low temperature), note that $\beta \to \infty$ and the aligned configurations (all $+1$ or -1) are more likely. This is because the exponent in (3.11.1) will be large and positive. In this case, the system is in the ordered state.

Moving from the disordered to ordered state (large T to small T), one could expect that the system undergoes a *phase transition* at some *critical temperature* $T = T_c > 0$. We shall not define T_c in rigorous terms. Informally, the system exhibits very different characteristics for $T < T_c$ and $T > T_c$. The two-dimensional Ising model turns out to have such a phase transition at $T_c > 0$.

Here are few other notes regarding the Ising model. The distribution (3.11.1) is also known as a *Boltzmann distribution* or a *Gibbs distribution*. The normalization factor

$$Z_{L_v,L_h} = Z_{L_v,L_h}(\beta) = \sum_{\{\sigma_{j,k}\}} e^{-\beta\mathcal{E}} \tag{3.11.5}$$

in (3.11.3) is called the *partition function*. In (3.11.5), the sum is over every value ± 1 of the variables $\sigma_{j,k}$ in the lattice. The energy $\mathcal{E}$ in (3.11.2) is often written more generally as

$$\mathcal{E} = -\sum_{j=-L_v+1}^{L_v} \sum_{k=-L_h+1}^{L_h} (E^h\sigma_{j,k}\sigma_{j,k+1} + E^v\sigma_{j,k}\sigma_{j+1,k} + H\sigma_{j,k}), \tag{3.11.6}$$

where the terms $H\sigma_{j,k}$ in (3.11.6) account for the so-called external magnetic field[9] (see the discussion below for the reason of using "magnetic"). With (3.11.2), we thus focus only on the situation of zero magnetic field $H = 0$. (In fact, the presence of the external magnetic

[9] Do not confuse the magnetic field H in (3.11.6) with the Hurst parameter.

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field complicates the matters considerably and the Ising model with a magnetic field $H > 0$ has not been solved explicitly yet. See, for example, McCoy [702], pp. 277–280.)

Note also that the energy $\mathcal{E}$ in (3.11.2) involves the variables σ_{j,L_h+1} and $\sigma_{L_v+1,k}$ "outside" the lattice. What exactly are these variables? This is related to the so-called *boundary conditions*. The case

$$\sigma_{j,L_h+1} = \sigma_{j,-L_h+1}, \quad \sigma_{L_v+1,k} = \sigma_{-L_v+1,k} \tag{3.11.7}$$

is referred to as that of *periodic boundary* (toroidal or doughnut-shaped) conditions. The case

$$\sigma_{j,L_h+1} = 0, \quad \sigma_{L_v+1,k} = 0 \tag{3.11.8}$$

is known as that of *free boundary* conditions. It effectively corresponds to the situation where the terms involving σ_{j,L_h+1} and $\sigma_{L_v+1,k}$ are not present in (3.11.2).

Finally, one of the original applications of the Ising model is to the phenomenon of *ferromagnetism*. The variables $\sigma_{j,k}$ model a magnetic dipole of atoms of ferromagnetic material (e.g., iron). The variables $\sigma_{j,k}$ are referred to as *spins*. The critical temperature T_c is known as the Curie temperature. Below this temperature, ferromagnetic materials exhibit spontaneous magnetization.

We are interested here in the behavior of the correlation function of the spin at the origin and the spin at position (m,n), namely

$$C(m,n) = \mathrm{Corr}(\sigma_{0,0},\sigma_{m,n}) = \frac{\mathbb{E}\sigma_{0,0}\sigma_{m,n} - \mathbb{E}\sigma_{0,0}\mathbb{E}\sigma_{m,n}}{\sqrt{\mathbb{E}\sigma_{0,0}^2 - (\mathbb{E}\sigma_{0,0})^2}\sqrt{\mathbb{E}\sigma_{0,0}^2 - (\mathbb{E}\sigma_{0,0})^2}} = \mathbb{E}\sigma_{0,0}\sigma_{m,n},$$

since $\mathbb{E}\sigma_{j,k} = 0$ and $\mathbb{E}\sigma_{j,k}^2 = 1 = \sigma_{j,k}^2$. Thus correlations reduce to covariances. For the sake of simplicity and illustration, we shall only consider row and diagonal covariances

$$C(0,n) = \mathbb{E}\sigma_{0,0}\sigma_{0,n}, \quad C(n,n) = \mathbb{E}\sigma_{0,0}\sigma_{n,n}. \tag{3.11.9}$$

We shall also work in the case of free boundary conditions (3.11.8). Our goal is to show the following:

Theorem 3.11.1 *At the critical temperature $T = T_c$ characterized by*

$$1 = \sinh(2\beta E_h)\sinh(2\beta E_v) \tag{3.11.10}$$

and as $n \to \infty$,

$$\lim_{L_h,L_v\to\infty} \mathbb{E}\sigma_{0,0}\sigma_{0,n} \sim \left(\frac{1+\alpha_1}{1-\alpha_1}\right)^{1/4} A n^{-1/4}, \tag{3.11.11}$$

$$\lim_{L_h,L_v\to\infty} \mathbb{E}\sigma_{0,0}\sigma_{n,n} \sim A n^{-1/4}, \tag{3.11.12}$$

where $A = 0.6450024\ldots$ and

$$\alpha_1 = \frac{z_h(1-z_v)}{1+z_v}, \quad z_h = \tanh(\beta E_h), \quad z_v = \tanh(\beta E_v).$$

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Sketch of proof The idea is to introduce the notion of bonds in the Ising lattice, then replace this lattice by a larger one, called the *counting lattice*, to which a number of matrices are associated. The covariances of interest involve these matrices, their inverse and related determinants. Szegö's limit theorem describes the asymptotic behavior of these determinants.

The asymptotic behavior of $\mathbb{E}\sigma_{0,0}\sigma_{n,n}$ in (3.11.12) is stated in McCoy and Wu [703], (4.43), p. 265, and proved. But the asymptotic behavior of $\mathbb{E}\sigma_{0,0}\sigma_{0,n}$ in (3.11.11), stated in McCoy and Wu [703], p. 267, is only conjectured (McCoy and Wu [703], p. 266).

Whereas, as expected, the asymptotic behavior of $\mathbb{E}\sigma_{0,0}\sigma_{n,n}$ is invariant under permutation of h and v, this is not the case for $\mathbb{E}\sigma_{0,0}\sigma_{0,n}$, for which

$$\frac{1+\alpha_1}{1-\alpha_1} = \frac{1+z_v+z_h-z_hz_v}{1+z_v-z_h+z_hz_v}. \tag{3.11.13}$$

Observe that in the special case $E_h = E_v = E$, this ratio simplifies and one has

Corollary 3.11.2 *If $E_h = E_v$, then at the critical temperature $T = T_c$, one has as $n \to \infty$,*

$$\lim_{L_h, L_v \to \infty} \mathbb{E}\sigma_{0,0}\sigma_{0,n} \sim 2^{1/8} A n^{-1/4}. \tag{3.11.14}$$

Since, at the critical temperature T_c, the covariances behave like $n^{-1/4}$ for large n, the Ising model exhibits LRD in the sense (2.1.5) in condition II. By identifying the exponent $2d-1$ with $-1/4$ (see (2.1.5)) and using $H = d + 1/2$ (see (2.8.4)), one gets that the Ising model exhibits LRD with

$$d = \frac{3}{8} \quad \text{and} \quad H = \frac{7}{8}. \tag{3.11.15}$$

Theorem 3.11.1 and Corollary 3.11.2 are proved in Section 3.11.5.

3.11.2 Correlations, Dimers and Pfaffians

We shall express the correlations $C(0,n)$ and $C(n,n)$ in terms of the determinants of $n \times n$ matrices in the so-called thermodynamic limit $L_v, L_h \to \infty$ first, and then analyze them as $n \to \infty$.

Observe that

$$\mathbb{E}\sigma_{0,0}\sigma_{0,n} = \frac{1}{Z_{L_v,L_h}} \sum_{\{\sigma_{j,k}\}} \sigma_{0,0}\sigma_{0,n} e^{-\beta\mathcal{E}}. \tag{3.11.16}$$

We shall analyze the two terms Z_{L_v,L_h} and $\sum_{\{\sigma_{j,k}\}} \sigma_{0,0}\sigma_{0,n} e^{-\beta\mathcal{E}}$ separately. The focus will be on the partition function Z_{L_v,L_h}, and similar arguments will be outlined for the second term. As for the diagonal correlation $\mathbb{E}\sigma_{0,0}\sigma_{n,n}$, we shall give its corresponding expression without proof.

Partition function Z_{L_v,L_h}

Observe that

$$
\begin{aligned}
Z_{L_v,L_h} &= \sum_{\{\sigma_{j,k}\}} e^{-\beta\mathcal{E}} = \sum_{\{\sigma_{j,k}\}} e^{\beta\sum_{j=-L_v+1}^{L_v}\sum_{k=-L_h+1}^{L_h}(E_h\sigma_{j,k}\sigma_{j,k+1}+E_v\sigma_{j,k}\sigma_{j+1,k})} \\
&= \sum_{\{\sigma_{j,k}\}} \prod_{j=-L_v+1}^{L_v}\prod_{k=-L_h+1}^{L_h} e^{\beta E_h\sigma_{j,k}\sigma_{j,k+1}} \prod_{j=-L_v+1}^{L_v}\prod_{k=-L_h+1}^{L_h} e^{\beta E_v\sigma_{j,k}\sigma_{j+1,k}} \\
&= \sum_{\{\sigma_{j,k}\}} \prod_{j=-L_v+1}^{L_v}\prod_{k=-L_h+1}^{L_h} \Big(\cosh\beta E_h + \sigma_{j,k}\sigma_{j,k+1}\sinh\beta E_h\Big) \\
&\quad\times \prod_{j=-L_v+1}^{L_v}\prod_{k=-L_h+1}^{L_h} \Big(\cosh\beta E_v + \sigma_{j,k}\sigma_{j+1,k}\sinh\beta E_v\Big),
\end{aligned}
\tag{3.11.17}
$$

since $\cosh a = \frac{1}{2}(e^a + e^{-a})$, $\sinh a = \frac{1}{2}(e^a - e^{-a})$ and thus

$$e^{ax} = \cosh a + x\sinh a, \quad x = \pm 1.$$

Under the free boundary conditions, we get further that

$$Z_{L_v,L_h} = (\cosh\beta E_h)^{2L_v(2L_h-1)}(\cosh\beta E_v)^{(2L_v-1)2L_h}\widetilde{Z}_{L_v,L_h}, \tag{3.11.18}$$

where

$$
\begin{aligned}
\widetilde{Z}_{L_v,L_h} &= \sum_{\{\sigma_{j,k}\}} \prod_{j=-L_v+1}^{L_v}\prod_{k=-L_h+1}^{L_h} \Big(1 + \sigma_{j,k}\sigma_{j,k+1}z_h\Big) \\
&\quad\times \prod_{j=-L_v+1}^{L_v}\prod_{k=-L_h+1}^{L_h} \Big(1 + \sigma_{j,k}\sigma_{j+1,k}z_v\Big) =: \sum_{\{\sigma_{j,k}\}} \widetilde{Z}_{L_v,L_h}(\{\sigma_{j,k}\})
\end{aligned}
\tag{3.11.19}
$$

and

$$z_h = \frac{\sinh\beta E_h}{\cosh\beta E_h} = \tanh\beta E_h, \quad z_v = \tanh\beta E_v. \tag{3.11.20}$$

Consider now $\widetilde{Z}_{L_v,L_h}(\{\sigma_{j,k}\})$ given by (3.11.19). Expand all the products to express $\widetilde{Z}_{L_v,L_h}(\{\sigma_{j,k}\})$ as the sum over a number of terms as

$$
\begin{aligned}
\widetilde{Z}_{L_v,L_h}(\{\sigma_{j,k}\}) &= 1 + \sigma_{-L_v+1,-L_h+1}\sigma_{-L_v+1,-L_h+2}z_h + \cdots + \sigma_{L_v-1,L_h}\sigma_{L_v,L_h}z_v \\
&+ \cdots + \prod_{j=-L_v+1}^{L_v}\prod_{k=-L_h+1}^{L_h} \sigma_{j,k}\sigma_{j,k+1}z_h \prod_{j=-L_v+1}^{L_v}\prod_{k=-L_h+1}^{L_h} \sigma_{j,k}\sigma_{j+1,k}z_v.
\end{aligned}
\tag{3.11.21}
$$

Each term in the sum (3.11.21) can be thought as containing a factor for every pair of nearest-neighbor sites (j,k) and (j',k') on the lattice. This factor is either 1, when 1 from $1 + \sigma_{j,k}\sigma_{j',k'}z$ is selected in the product, or $\sigma_{j,k}\sigma_{j',k'}z$, when $\sigma_{j,k}\sigma_{j',k'}z$ is selected from the product (z denotes z_h or z_v). For example, the first term in (3.11.21) occurs when each pair of nearest-neighbor sites contributes 1, and the last term in (3.11.21) occurs when each contributes $\sigma_{j,k}\sigma_{j',k'}z$. Each term in the sum (3.11.21) will therefore include either 1, $\sigma_{j,k}$,

$\sigma_{j,k}^2$, $\sigma_{j,k}^3$ or $\sigma_{j,k}^4$ and hence one of the five quantities can be associated with each site (j,k). Note now that the terms in the sum $\widetilde{Z}_{L_v,L_h}(\{\sigma_{j,k}\})$ with sites having $\sigma_{j,k}$ or $\sigma_{j,k}^3$ will vanish in the total sum $\sum_{\{\sigma_{j,k}\}}$ in (3.11.19) because

$$\sum_{\sigma_{j,k}=\pm 1} \sigma_{j,k} = \sum_{\sigma_{j,k}=\pm 1} \sigma_{j,k}^3 = 0.$$

Thus only

$$\text{the terms having } 1,\ \sigma_{j,k}^2 \equiv 1 \text{ or } \sigma_{j,k}^4 \equiv 1 \tag{3.11.22}$$

will not vanish in the total sum $\sum_{\{\sigma_{j,k}\}}$.

The presence of a term $\sigma_{j,k}\sigma_{j,k+1}z_h$ in (3.11.21) can be interpreted as indicating the presence of a horizontal bond z_h connecting the sites (j,k) and $(j,k+1)$. A corresponding interpretation holds for $\sigma_{j,k}\sigma_{j+1,k}z_v$. From a different perspective, when 1, $\sigma_{j,k}^2$ or $\sigma_{j,k}^4$ is associated with the site (j,k), then zero, two or four bonds traverse the site (j,k). Therefore, the terms in the sum (3.11.21) where the conditions (3.11.22) hold can be regarded as depicting a figure on a lattice with the following properties:

(i) each bond between nearest neighbors may be used, at most, once;
(ii) an even number of bonds terminate at each site.

Figure 3.8 depicts one such figure. Observe that such figure would contribute a term

$$z_h^p z_v^q$$

to the total sum $\sum_{\{\sigma_{j,k}\}}$, where p is the number of horizontal bonds in the figure ($p = 18$ in Figure 3.8) and q is the number of vertical bonds in the figure ($q = 16$ in Figure 3.8). Statement *(ii)* above implies that p and q are even. Since there are

$$\sum_{\{\sigma_{j,k}\}} 1 = 2^{2L_v 2L_h}$$

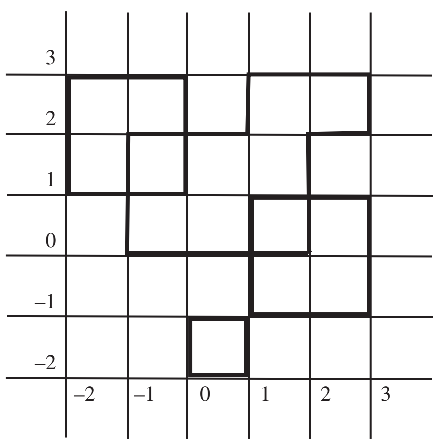

Figure 3.8 A figure (lines in bold) on the Ising lattice.

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different configurations, one gets that

$$\widetilde{Z}_{L_v,L_h} = 2^{2L_v 2L_h} \sum_{p,q} N_{p,q} z_h^p z_v^q =: 2^{2L_v 2L_h} g(z_h, z_v), \tag{3.11.23}$$

where $N_{p,q}$ is the number of figures on the lattice with the properties (i) and (ii) above, and where p, q are the numbers of horizontal and vertical bonds of the figure. By convention, $N_{0,0} = 1$. The next step is to compute the "generating function"

$$g(z_1, z_2) = \sum_{p,q} N_{p,q} z_1^p z_2^q \tag{3.11.24}$$

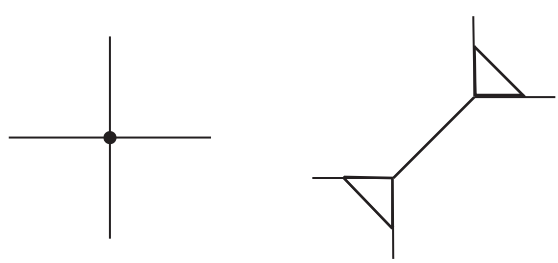
associated with counting of the figures as in Figure 3.8. Observe that $g(z_1, z_2) \geq 0$ since p and q are even.

Dimers

There is an ingenious way to turn the problem of computing (3.11.24) into the so-called "problem of counting closest-packed dimer coverings" on a suitable lattice, as explained in McCoy and Wu [703]. We will see later that the latter problem has a solution involving a matrix determinant which can then be studied more easily.

The first step is to replace the Ising lattice by a larger lattice, which we will call the *counting lattice*. More specifically, replace each site of the Ising lattice, as on the left side of Figure 3.9, by a six-site cluster depicted on the right side of Figure 3.9. Note that, as in the Ising lattice, the six-site cluster has four connecting lines. The difference is that one site is now replaced by six sites. The new, counting lattice has then $2K = 6(2L_v)(2L_h)$ sites and is formed by connecting six-site clusters, as depicted in Figure 3.10.

In the next step, the idea is to replace each of eight possible Ising site configuration bonds with a suitable configuration of bonds on the counting lattice. The replacement is done according to Figure 3.11. With this replacement, each figure counted in (3.11.24) is replaced by another figure in the counting lattice. In the Ising lattice, two adjacent sites may be either unconnected or connected by a bond. In the counting lattice, each site has exactly one bond with one of its neighbors indicated by a double line in Figure 3.11. Each



(a) (b)

Figure 3.9 (a) A site of the Ising lattice. (b) A corresponding six-site cluster of the counting lattice.

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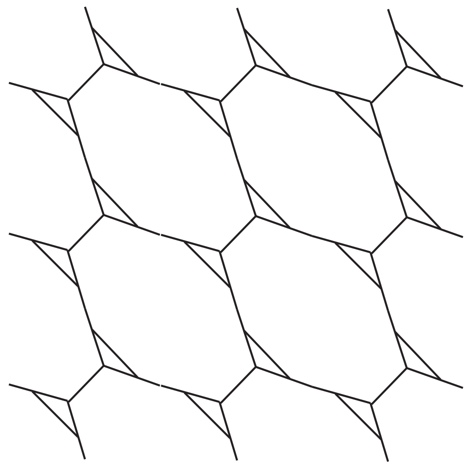

Figure 3.10 The counting lattice.

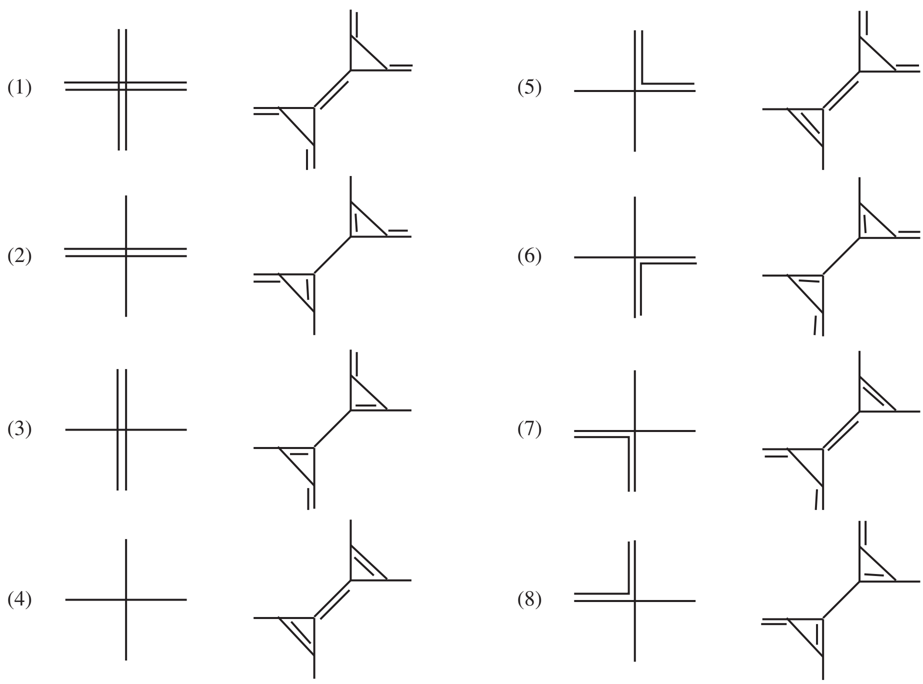

Figure 3.11 Replacement of eight possible Ising site configuration bonds by suitable configuration bonds on the counting lattice. The connecting bonds are indicated by *double* lines. Observe that a horizontal (resp. vertical) bond at an Ising site is associated with a horizontal (resp. vertical) bond connecting clusters.

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such bond is called a *dimer*.[10] The figure one gets in the counting lattice is called a *closest-packed dimer configuration*.[11] Then, the problem of counting in (3.11.24) can be replaced by counting dimer configurations in the counting lattice.

To make this connection more precise, three classes of bonds need to be distinguished in the counting lattice: (1) horizontal bonds between clusters, (2) vertical bonds between clusters, and (3) bonds within a cluster. Let

$$G(z_1, z_2, z_3) = \sum_{p,q,r} N_{p,q,r} z_1^p z_2^q z_3^r \tag{3.11.25}$$

be the corresponding generating function, where $N_{p,q,r}$ is the number of dimer configurations on the counting lattice with p bonds of type (1), q bonds of type (2) and r bonds of type (3). In particular, one can think that bonds of types (1), (2) and (3) carry weights of sizes z_1, z_2 and z_3, respectively. The generating functions (3.11.24) and (3.11.25) are then related as follows:

$$g(z_1, z_2) = G(z_1, z_2, 1), \tag{3.11.26}$$

by assigning the weight $z_3 = 1$ to bonds within clusters. This is because a horizontal (resp. vertical) bond in the Ising lattice is associated with a horizontal (resp. vertical) bond between clusters (see Figure 3.11).

The advantage of the formulation (3.11.25) or (3.11.26) is that it can be related to the determinant of a matrix. We shall do this in two steps. In the *first step*, we shall express (3.11.26) as

$$g(z_1, z_2) = G(z_1, z_2, 1) = {\sum_p}' \overline{b}_{p_1,p_2} \overline{b}_{p_3,p_4} \dots \overline{b}_{p_{2K-1},p_{2K}}, \tag{3.11.27}$$

where $\overline{B} = (\overline{b}_{pq})_{1\le p,q\le 2K}$ is a suitable $2K \times 2K$ matrix (the bar does not refer here to the complex conjugate) and $\sum_p'$ is the sum over all permutations $p_1, p_2 \dots, p_{2K}$ of $1, 2, \dots, 2K$ satisfying

$$p_{2m-1} < p_{2m}, \ 1 \le m < K, \quad p_{2m-1} < p_{2m+1}, \ 1 \le m < K - 1. \tag{3.11.28}$$

For example, if $2K = 4$, the sum is over the permutations 1234, 1324 and 1423 of 1234. Relation (3.11.28) is satisfied since these permutations are such that $p_1 < p_2$, $p_1 < p_3$ and $p_3 < p_4$, where, in the permutation 1324, for example, $p_1 = 1$, $p_2 = 3$, $p_3 = 2$, $p_4 = 4$. In the *second step*, the expression (3.11.27) will be written as the so-called *Pfaffian* of a matrix. As stated below, the Pfaffian of a matrix is the square root of its determinant. We shall now detail these two steps.

In the *first step*, the expression (3.11.27) and the permutations (3.11.28) above are naturally related to closest-packed dimer configurations on *any* lattice. Suppose a lattice (that is, *any* lattice) consists of $2K$ sites and enumerate these sites by $1, 2, \dots, 2K$. Observe that there is a one-to-one correspondence between closest-packed dimer configurations on the lattice and permutations satisfying (3.11.28): the permutation

$$p_1 p_2 \mid p_3 p_4 \mid \dots \mid p_{2K-1} p_{2K}$$

[10] In Chemistry, a dimer is a structure formed by two sub-units.

[11] One uses this term whenever each site has exactly one bond with one of its neighbors.

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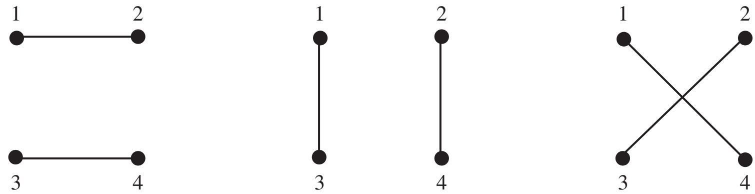

Figure 3.12 All possible closest-packed dimer configurations in a lattice of four sites.

is associated with the closest-packed dimer configuration where dimers connect the sites p_1 and p_2, p_3 and p_4, $\dots$, p_{2K-1} and p_{2K}. Here, note that a dimer can connect any two sites. For example, with a lattice of $2K = 4$ sites, there are three closest-packed dimer configurations associated with

$$12\,|\,34, \quad 13\,|\,24 \quad \text{and} \quad 14\,|\,23, \tag{3.11.29}$$

as illustrated in Figure 3.12. Suppose, in addition, that a dimer connecting sites p and q carries a weight $\overline{b}_{p,q}$. Then, we define the weight of the closest-packed dimer configuration associated with the permutation $p_1 p_2 \,|\, p_3 p_4 \,|\, \dots \,|\, p_{2K-1} p_{2K}$ as

$$\overline{b}_{p_1,p_2}\overline{b}_{p_3,p_4}\dots\overline{b}_{p_{2K-1},p_{2K}}.$$

Therefore,

$$\sum_p{}' \overline{b}_{p_1,p_2}\overline{b}_{p_3,p_4}\dots\overline{b}_{p_{2K-1},p_{2K}}$$

is exactly the sum of the weights of all closest-packed dimer configurations.

With the latter fact in mind, we now turn to the relation (3.11.27). The function $G(z_1, z_2, 1)$ can then be written as

$$G(z_1, z_2, 1) = \sum_p{}' \overline{b}_{p_1,p_2}\overline{b}_{p_3,p_4}\dots\overline{b}_{p_{2K-1},p_{2K}},$$

where

$$2K = 6(2L_v)(2L_h)$$

is the number of sites in the counting lattice, the sum $\sum_p'$ can be viewed as over all closest-packed dimer configurations, and the weights $\overline{b}_{p,q}$ are such that

$$\overline{b}_{pq} = \begin{cases} 1, & \text{if } p \text{ and } q \text{ are neighboring sites within the same six-site cluster,} \\ z_1, & \text{if sites } p \text{ and } q \text{ connect two six-site clusters in the vertical direction,} \\ z_2, & \text{if sites } p \text{ and } q \text{ connect two six-site clusters in the horizontal direction,} \\ 0, & \text{otherwise.} \end{cases} \tag{3.11.30}$$

Figure 3.13 illustrates this. The weight 0 in $\overline{b}_{pq}$ ensures that only closest-packed dimer configurations with dimers connecting neighboring sites count for the function $G(z_1, z_2, 1)$. This means, in particular, that most of the elements $\overline{b}_{pq}$ are actually zero.

A matrix $\overline{B} = (\overline{b}_{pq})$ satisfying (3.11.30) can be constructed as follows. Consider first 6×6 matrices $\overline{B}(j, k;\, j', k')$ labeled by the Ising lattice sites (j, k) and (j', k'), where $-L_v + 1 \leq$

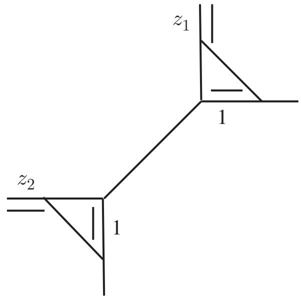

Figure 3.13 The connecting bonds between sites are indicated by double lines. The weight b_{pq} connecting two sites p and q is indiacted. It is 1, z_1 or z_2 according to whether p and q are within the same six-site cluster, connect two six-site clusters in the vertical direction or connect two six-site clusters in the horizontal direction. All other bonds have weight 0.

$j, j' \leq L_v$, $-L_h + 1 \leq k, k' \leq L_h$. They are zero matrices except in the following cases, where they are defined as:

$$\overline{B}(j,k;\,j,k) = \begin{array}{c c} & \begin{array}{cccccc} R & L & U & D & 1 & 2 \end{array} \\ \begin{array}{c} R \\ L \\ U \\ D \\ 1 \\ 2 \end{array} & \begin{pmatrix} 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 \end{pmatrix} \end{array},$$

for $-L_v + 1 \leq j \leq L_v,\ -L_h + 1 \leq k \leq L_h$,

$$\overline{B}(j,k;\,j,k+1) = \overline{B}(j,k+1;\,j,k)' = \begin{array}{c c} & \begin{array}{cccccc} R & L & U & D & 1 & 2 \end{array} \\ \begin{array}{c} R \\ L \\ U \\ D \\ 1 \\ 2 \end{array} & \begin{pmatrix} 0 & z_1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \end{array},$$

for $-L_v + 1 \leq j \leq L_v,\ -L_h + 1 \leq k \leq L_h - 1$,

$$\overline{B}(j,k;\,j+1,k) = \overline{B}(j+1,k;\,j,k)' = \begin{array}{c c} & \begin{array}{cccccc} R & L & U & D & 1 & 2 \end{array} \\ \begin{array}{c} R \\ L \\ U \\ D \\ 1 \\ 2 \end{array} & \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & z_2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \end{array},$$

for $-L_v + 1 \leq j \leq L_v - 1,\ -L_h + 1 \leq k \leq L_h$. Each site (j, k) of the Ising lattice corresponds to a six-site cluster in the counting lattice. Think now of $\overline{B}(j, k;\ j', k')$ as containing weights $\overline{b}_{pq}$ between sites in six-site clusters denoted (j, k) and (j', k') (within the same

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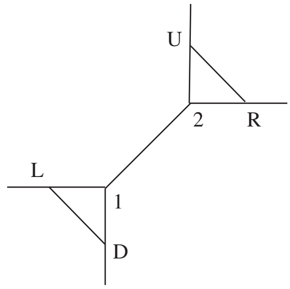

Figure 3.14 Labels for the six sites of a cluster on the counting lattice.

cluster if $(j,k) = (j',k')$). The labels R, L, U, D, 1 and 2 correspond to the six sites in a cluster, depicted in Figure 3.14. (R stands for "Right", L for "Left", U for "Up" and D for "Down".) Note that these matrices are exactly such that their elements $\overline{b}_{pq}$ satisfy (3.11.30). For example, within a cluster, a site R can only connect to U or 2 with the weight of 1.

Finally, the matrix $\overline{B} = (\overline{b}_{pq})_{1\le p,q\le 2K}$ is made of the 6×6 blocks or submatrices $\overline{B}(j,k;\, j',k')$, $-L_v+1 \le j, j' \le L_v$, $-L_h+1 \le k,k' \le L_h$, we just defined. The exact placement of the blocks inside of the matrix $\overline{B}$ is as follows. Order the $d = (2L_v)(2L_h)$ sites (j,k) along the rows for increasing columns k; that is,

$$\begin{gathered}(-L_v+1, -L_h+1), \ldots, (-L_v+1, L_h),\\ (-L_v+2, -L_h+1), \ldots, (-L_v+2, L_h),\\ \ldots\\ (L_v, -L_h+1), \ldots, (L_v, L_h)\end{gathered}$$

and renumber the sites $1 \le m \le d$. This renumbering associates then to each pair (j,k), (j',k') a pair (m_1, m_2), $1 \le m_1, m_2 \le d$. In the matrix $\overline{B}$, its (m_1,m_2) block of size 6 is then defined as $\overline{B}(j,k;\, j',k')$.

Pfaffians

We now turn to the *second step* described in the discussion around (3.11.27), namely, to express (3.11.27) as the Pfaffian of a matrix. We first define the Pfaffian.[12]

Definition 3.11.3 Consider a $2K \times 2K$ antisymmetric real-valued matrix $A = (a_{jk})_{1\le j,k\le 2K}$, that is, with elements $a_{jk} = -a_{kj}$ and $a_{jj} = 0$. Its Pfaffian is defined as

$$\mathrm{Pf}A = \sum_p{}' \delta_p a_{p_1p_2} a_{p_3p_4} \ldots a_{p_{2K-1}p_{2K}}, \tag{3.11.31}$$

where, as in (3.11.28), $\sum_p'$ is the sum over all permutations $p_1, p_2 \ldots, p_{2K}$ of $1, 2, \ldots, 2K$ satisfying

$$p_{2m-1} < p_{2m},\ 1 \le m < K, \quad p_{2m-1} < p_{2m+1},\ 1 \le m < K-1 \tag{3.11.32}$$

and where δ_p, the parity of the permutation p, is 1 if the permutation p is made up of an even number of transpositions and -1 if p is made of an odd number of transpositions.

[12] After a German mathematician J. F. Pfaff (1765–1825), who was a formal research supervisor of C. F. Gauss.

For example, if $2K = 4$, and

$$A = \begin{pmatrix} 0 & a_{12} & a_{13} & a_{14} \\ -a_{12} & 0 & a_{23} & a_{24} \\ -a_{13} & -a_{23} & 0 & a_{34} \\ -a_{14} & -a_{24} & -a_{34} & 0 \end{pmatrix},$$

then by (3.11.29),

$$\mathrm{Pf}A = a_{12}a_{34} - a_{13}a_{24} + a_{14}a_{23}.$$

This is because 1234, 1324, 1423 involve respectively 0, 1 and 2 transpositions.

The usefulness of the Pfaffian comes from the following formula: for an antisymmetric matrix A,

$$(\mathrm{Pf}A)^2 = \det(A) \tag{3.11.33}$$

(see, for example, pp. 47–51 in McCoy and Wu [703] for a proof). For example, if $2K = 2$ and

$$A = \begin{pmatrix} 0 & a_{12} \\ -a_{12} & 0 \end{pmatrix},$$

then one sees immediately that $(\mathrm{Pf}A)^2 = \det(A)$ because $\mathrm{Pf}A = a_{12}$ and $\det(A) = a_{12}^2$.

The relation (3.11.27) is not exactly the Pfaffian of the matrix $\overline{B}$ in that it does not include the parity factor δ_p. In fact, the parity factor can be introduced by suitably altering the signs of the elements of the matrix $\overline{B}$. We shall describe but not prove the assignment for doing so. See, for example, pp. 51–67 in McCoy and Wu [703] for more details.

For the sum (3.11.27) to have the parity factor δ_p, the blocks of the matrix $\overline{B}$ have to be replaced by the blocks:

$$\overline{A}(j,k;j,k) = \begin{array}{c} \\ R \\ L \\ U \\ D \\ 1 \\ 2 \end{array} \begin{array}{c} \begin{array}{cccccc} R & L & U & D & 1 & 2 \end{array} \\ \begin{pmatrix} 0 & 0 & -1 & 0 & 0 & 1 \\ 0 & 0 & 0 & -1 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 & -1 \\ 0 & 1 & 0 & 0 & -1 & 0 \\ 0 & -1 & 0 & 1 & 0 & 1 \\ -1 & 0 & 1 & 0 & -1 & 0 \end{pmatrix} \end{array}, \tag{3.11.34}$$

for $-L_v + 1 \le j \le L_v,\ -L_h + 1 \le k \le L_h$,

$$\overline{A}(j,k;j,k+1) = -\overline{A}(j,k+1;j,k)' = \begin{array}{c} \\ R \\ L \\ U \\ D \\ 1 \\ 2 \end{array} \begin{array}{c} \begin{array}{cccccc} R & L & U & D & 1 & 2 \end{array} \\ \begin{pmatrix} 0 & z_1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \end{array}, \tag{3.11.35}$$

for $-L_v + 1 \le j \le L_v,\ -L_h + 1 \le k \le L_h - 1$,

$$\overline{A}(j,k;\,j+1,k) = -\overline{A}(j+1,k;\,j,k)' = \begin{array}{c c} & \begin{array}{cccccc} R & L & U & D & 1 & 2 \end{array} \\ \begin{array}{c} R \\ L \\ U \\ D \\ 1 \\ 2 \end{array} & \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & z_2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \end{array}, \quad (3.11.36)$$

for $-L_v + 1 \le j \le L_v - 1,\ -L_h + 1 \le k \le L_h$. Denote the corresponding matrix by $\overline{A}$ and its elements by $\overline{a}_{p,q}$. (The $2K \times 2K$ matrix $\overline{A}$ is defined in the same way as $\overline{B}$ but using blocks $\overline{A}(j,k;\,j',k')$ instead of $\overline{B}(j,k;\,j',k')$.) Note that the only difference between the elements of $\overline{A}$ and $\overline{B}$ is that some of the entries have different signs. It is convenient to think of the sign assignment graphically as depicted in Figure 3.15. Each bond between sites of the counting lattice now not only carries a weight (z_1, z_2, 1 or 0) but also a direction. The sign of the elements of the matrix blocks $\overline{A}(j,k;\,j',k')$ now corresponds to this direction. For example, the sign is positive if the connecting bond is in the forward direction, as indicated by the arrows in Figure 3.16. Thus, in the $\overline{A}(j,k;\,j,k)$ block, the (U,R) entry is 1 because it corresponds to the arrow $U \to R$ in Figure 3.16, but the $(U,2)$ entry is -1 because $U \to 2$ is not in the direction of the arrow. Thus, one can show (McCoy and Wu [703], pp. 81, 51–58) that

$$g(z_1,z_2) = \sum_p{}' \delta_p \overline{a}_{p_1p_2}\overline{a}_{p_3p_4}\ldots\overline{a}_{p_{6d-1}p_{6d}} = \mathrm{Pf}\,\overline{A} = (\det(\overline{A}))^{1/2}, \quad (3.11.37)$$

where the sum is as in (3.11.27), δ_p is the parity factor and $\mathrm{Pf}\,\overline{A}$ is the Pfaffian in Definition 3.11.3. The sign at the square root in (3.11.37) is positive since $g(z_1,z_2)$ is defined in (3.11.24) with even exponents.

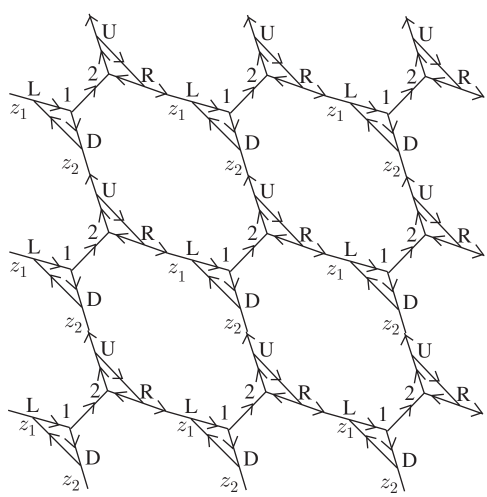

Figure 3.15 The counting lattice with directions between sites.

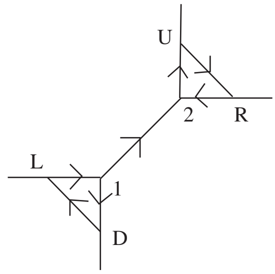

Figure 3.16 Directions in the six-site cluster of the counting lattice.

Let $c_R, c_L, c_U, c_D, c_1, c_2$ denote the columns of any 6×6 block $\overline{A}(j, k; j', k')$, and let us perform the following operations:

$$c_R - c_1, \; c_U + c_1, \; c_L + c_2, \; c_D - c_2.$$

After these operations, the block $\overline{A}(j, k; j, k)$ becomes

$$\begin{array}{c} \\ R \\ L \\ U \\ D \\ 1 \\ 2 \end{array} \begin{array}{c} \begin{array}{cccccc} R & L & U & D & 1 & 2 \end{array} \\ \begin{pmatrix} 0 & 1 & -1 & -1 & 0 & 1 \\ -1 & 0 & 1 & -1 & 1 & 0 \\ 1 & -1 & 0 & 1 & 0 & -1 \\ 1 & 1 & -1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & -1 & 0 \end{pmatrix} \end{array} \tag{3.11.38}$$

and the other blocks $\overline{A}(j, k; j', k')$ remain the same. Note that the last two rows of the block (3.11.38) are zero except the 2×2 submatrix

$$\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \tag{3.11.39}$$

in the bottom-right corner. Since the determinant of the submatrix (3.11.39) equals 1, the rows and columns of the matrix $\overline{A}$ corresponding to the submatrix can be eliminated without affecting the determinant of $\overline{A}$. We thus have

$$\det(\overline{A}) = \det(A), \tag{3.11.40}$$

where

$$A(j, k; j, k) = \begin{array}{c} \\ R \\ L \\ U \\ D \end{array} \begin{array}{c} \begin{array}{cccc} R & L & U & D \end{array} \\ \begin{pmatrix} 0 & 1 & -1 & -1 \\ -1 & 0 & 1 & -1 \\ 1 & -1 & 0 & 1 \\ 1 & 1 & -1 & 0 \end{pmatrix} \end{array} \tag{3.11.41}$$

and all other $A(j, k; j', k')$ are identical to $\overline{A}(j, k; j', k')$ with the rows and columns labeled 1 and 2 removed. Thus, we also have

$$g(z_1, z_2) = (\det(A))^{1/2}, \tag{3.11.42}$$

where A is now a $2K \times 2K$ matrix with

$$2K = 4(2L_v)(2L_h). \tag{3.11.43}$$

Combining (3.11.18), (3.11.23) and (3.11.42), we obtain that

$$Z_{L_v,L_h} = (\cosh \beta E_h)^{2L_v(2L_h-1)}(\cosh \beta E_v)^{(2L_v-1)2L_h} 2^{2L_v 2L_h}(\det(A))^{1/2}, \tag{3.11.44}$$

where z_1, z_2 in A are replaced by z_h and z_v.

The right-hand side of (3.11.44) provides an expression for the denominator (partition function) of (3.11.16). One can derive similarly an expression for the numerator of (3.11.16). Proceeding as in (3.11.17) and (3.11.18), we have

$$\begin{aligned}\sum_{\{\sigma_{j,k}\}} \sigma_{0,0}\sigma_{0,n} e^{-\beta \mathcal{E}} &= (\cosh \beta E_h)^{2L_v(2L_h-1)}(\cosh \beta E_v)^{(2L_v-1)2L_h} \\ &\quad \times \sum_{\{\sigma_{j,k}\}} \sigma_{0,0}\sigma_{0,n} \prod_{j=-L_v+1}^{L_v} \prod_{k=-L_h+1}^{L_h} \Big(1 + \sigma_{j,k}\sigma_{j,k+1} z_h\Big) \\ &\quad \times \prod_{j=-L_v+1}^{L_v} \prod_{k=-L_h+1}^{L_h} \Big(1 + \sigma_{j,k}\sigma_{j+1,k} z_v\Big).\end{aligned} \tag{3.11.45}$$

Since $\sigma_{i,j}^2 = 1$, we can write

$$\sigma_{0,0}\sigma_{0,n} = (\sigma_{0,0}\sigma_{0,1})(\sigma_{0,1}\sigma_{0,2})\dots(\sigma_{0,n-1}\sigma_{0,n})$$

and

$$\sigma_{0,k}\sigma_{0,k+1}(1 + \sigma_{0,k}\sigma_{0,k+1} z_h) = z_h(1 + \sigma_{0,k}\sigma_{0,k+1} z_h^{-1}),$$

so that

$$\sigma_{0,0}\sigma_{0,n} \prod_{k=0}^{n-1}(1 + \sigma_{0,k}\sigma_{0,k+1} z_h) = z_h^n \prod_{k=0}^{n-1}(1 + \sigma_{0,k}\sigma_{0,k+1} z_h^{-1}).$$

Then the relation (3.11.45) can be expressed as

$$\begin{aligned}\sum_{\{\sigma_{j,k}\}} \sigma_{0,0}\sigma_{0,n} e^{-\beta \mathcal{E}} &= (\cosh \beta E_h)^{2L_v(2L_h-1)}(\cosh \beta E_v)^{(2L_v-1)2L_h}\ z_h^n \\ &\quad \times \sum_{\{\sigma_{j,k}\}} \prod_{k=0}^{n-1}(1 + \sigma_{0,k}\sigma_{0,k+1} z_h^{-1}) \prod_{j=-L_v+1}^{L_v} {\prod_{k=-L_h+1}^{L_h}}' \Big(1 + \sigma_{j,k}\sigma_{j,k+1} z_h\Big) \\ &\quad \times \prod_{j=-L_v+1}^{L_v} \prod_{k=-L_h+1}^{L_h} \Big(1 + \sigma_{j,k}\sigma_{j+1,k} z_v\Big),\end{aligned} \tag{3.11.46}$$

where ${\prod_{k=-L_h+1}^{L_h}}'$ means that the terms with $j = 0$, $k = 0, \dots, n-1$, are omitted. The last expression is similar to (3.11.18) and (3.11.19) except for the factor z_h^n and for the fact that the weights z_h of the bonds on the horizontal line connecting $(0,0)$ and $(0,n)$ are now replaced by z_h^{-1}. This is also reflected in the fact that the term $\prod_{k=0}^{n-1}(1+\sigma_{0,k}\sigma_{0,k+1} z_h^{-1})$ now replaces the corresponding term $\prod_{k=0}^{n-1}(1 + \sigma_{0,k}\sigma_{0,k+1} z_h)$ in (3.11.19). One can then argue

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in the same way as for Z_{L_v,L_h}. Since Z_{L_v,L_h} is expressible as (3.11.44), by combining the expressions for the numerator and denominator of (3.11.16), we obtain that

$$\mathbb{E}\sigma_{0,0}\sigma_{0,n} = \frac{1}{Z_{L_v,L_h}} \sum_{\{\sigma_{j,k}\}} \sigma_{0,0}\sigma_{0,n} e^{-\beta\mathcal{E}} = z_h^n \Big(\frac{\det(A^0)}{\det(A)}\Big)^{1/2}, \tag{3.11.47}$$

where

$$A^0 = A + \delta$$

or equivalently $\delta = A^0 - A$ with

$$\delta(0,k;0,k+1) = -\delta(0,k+1;0,k)' = \begin{array}{c c} & \begin{array}{cccc} R & L & U & D \end{array} \\ \begin{array}{c} R \\ L \\ U \\ D \end{array} & \begin{pmatrix} 0 & z_h^{-1} - z_h & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \end{array} \tag{3.11.48}$$

if $0 \le k \le n-1$ and zero otherwise. Observe that the presence of δ affects only the "horizontal" edges $(0,k) \to (0,k+1)$ and $(0,k+1) \to (0,k)$ of the Ising lattice.

The matrix δ, which has dimensions $2K \times 2K$ with $2K = 4(2L_v)(2L_h)$, is zero everywhere except on its $2n$ columns and $2n$ rows. Let y be that $2n \times 2n$ submatrix of δ where it does not vanish. We have from (3.11.48) that

$$y = \begin{pmatrix} y_{RR} & y_{RL} \\ y_{LR} & y_{LL} \end{pmatrix}, \tag{3.11.49}$$

where

$$y_{RR} = \begin{array}{c c c} & & \begin{array}{cccc} 00 & 01 & \dots & 0n-1 \\ R & R & \dots & R \end{array} \\ \begin{array}{c} 00 \\ 01 \\ \vdots \\ 0n-1 \end{array} & \begin{array}{c} R \\ R \\ \vdots \\ R \end{array} & \begin{pmatrix} 0 & 0 & \dots & 0 \\ 0 & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 0 \end{pmatrix} \end{array},$$

$$y_{LR} = \begin{array}{c c c} & & \begin{array}{cccc} 00 & 01 & \dots & 0n-1 \\ R & R & \dots & R \end{array} \\ \begin{array}{c} 01 \\ 02 \\ \vdots \\ 0n \end{array} & \begin{array}{c} L \\ L \\ \vdots \\ L \end{array} & \begin{pmatrix} -(z_h^{-1} - z_h) & 0 & \dots & 0 \\ 0 & -(z_h^{-1} - z_h) & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & -(z_h^{-1} - z_h) \end{pmatrix} \end{array},$$

$$y_{RL} = \begin{array}{c c c} & & \begin{array}{cccc} 01 & 02 & \dots & 0n \\ L & L & \dots & L \end{array} \\ \begin{array}{c} 00 \\ 01 \\ \vdots \\ 0n-1 \end{array} & \begin{array}{c} R \\ R \\ \vdots \\ R \end{array} & \begin{pmatrix} z_h^{-1} - z_h & 0 & \dots & 0 \\ 0 & z_h^{-1} - z_h & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & z_h^{-1} - z_h \end{pmatrix} \end{array},$$

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$$
y_{LL} = \begin{array}{cc} & \\ & \\ 01 & L \\ 02 & L \\ \vdots & \vdots \\ 0n & L \end{array}
\begin{array}{c} \begin{array}{cccc} 01 & 02 & \dots & 0n \\ L & L & \dots & L \end{array} \\ \begin{pmatrix} 0 & 0 & \dots & 0 \\ 0 & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 0 \end{pmatrix} \end{array}.
$$

In the expressions above, following δ in (3.11.48), $00, 01, \dots, 0n-1, 01, \dots, 0n$ refer to the Ising lattice sites with nonzero weights, and R, L refer to the site in the cluster of the counting lattice.

Let also Q be the $2n \times 2n$ submatrix of A^{-1} in this same subspace as y. Then, in view of (3.11.47),

$$
\begin{aligned}
(\mathbb{E}\sigma_{0,0}\sigma_{0,n})^2 &= z_h^{2n}\frac{\det(A+\delta)}{\det(A)} = z_h^{2n}\det(A^{-1})\det(A+\delta) \\
&= z_h^{2n}\det(I + A^{-1}\delta) = z_h^{2n}\det(I+Qy) = z_h^{2n}\det(y)\det(y^{-1}+Q) \\
&= z_h^{2n}(z_h^{-1} - z_h)^{2n}\det(y^{-1}+Q) = (1-z_h^2)^{2n}\det(y^{-1}+Q), \qquad (3.11.50)
\end{aligned}
$$

so that

$$
(\mathbb{E}\sigma_{0,0}\sigma_{0,n})^2 = \det\Big((1-z_h^2)(y^{-1}+Q)\Big). \qquad (3.11.51)
$$

3.11.3 Computation of the Inverse

To evaluate $\det(y^{-1}+Q)$ in (3.11.51), we need an expression for the inverse A^{-1} of the $2K \times 2K$ matrix A and, more specifically, for the elements of the $2n \times 2n$ submatrix Q of A^{-1} in the same subspace as y. Then, we have

$$
Q = \begin{pmatrix} Q_{RR} & Q_{RL} \\ Q_{LR} & Q_{LL} \end{pmatrix}, \qquad (3.11.52)
$$

where

$$
Q_{RR} = \begin{pmatrix} 0 & \dots & A^{-1}(0,0;0,n-1)_{RR} \\ A^{-1}(0,1;0,0)_{RR} & \dots & A^{-1}(0,1;0,n-1)_{RR} \\ \vdots & \ddots & \vdots \\ A^{-1}(0,n-1;0,0)_{RR} & \dots & 0 \end{pmatrix},
$$

$$
Q_{LR} = \begin{pmatrix} A^{-1}(0,1;0,0)_{LR} & \dots & A^{-1}(0,1;0,n-1)_{LR} \\ A^{-1}(0,2;0,0)_{LR} & \dots & A^{-1}(0,2;0,n-1)_{LR} \\ \vdots & \ddots & \vdots \\ A^{-1}(0,n;0,0)_{LR} & \dots & A^{-1}(0,n;0,n-1)_{LR} \end{pmatrix},
$$

$$
Q_{RL} = \begin{pmatrix} A^{-1}(0,0;0,1)_{RL} & \dots & A^{-1}(0,0;0,n)_{RL} \\ A^{-1}(0,1;0,1)_{RL} & \dots & A^{-1}(0,1;0,n)_{RL} \\ \vdots & \ddots & \vdots \\ A^{-1}(0,n-1;0,1)_{RL} & \dots & A^{-1}(0,n-1;0,n)_{RL} \end{pmatrix},
$$

$$Q_{LL} = \begin{pmatrix} 0 & \dots & A^{-1}(0,1;0,n)_{LL} \\ A^{-1}(0,2;0,1)_{LL} & \dots & A^{-1}(0,2;0,n)_{LL} \\ \vdots & \ddots & \vdots \\ A^{-1}(0,n;0,1)_{LL} & \dots & 0 \end{pmatrix},$$

and where zeroes are on the diagonal since A^{-1} and hence Q is antisymmetric. The entries $A^{-1}(\cdot,\cdot,\cdot,\cdot)_{..}$ are place holders for the corresponding entries of the matrix A^{-1} and will need to be evaluated.

Recall that the matrix A has dimension $2K \times 2K$ where $2K = 4(2L_h)(2L_v)$ by (3.11.43). It is made up of blocks $A(j,k;j',k')$ given by (3.11.41) if $j' = j, k' = k$, and otherwise, by (3.11.35) and (3.11.36) with the rows and columns labeled 1 and 2 removed. Note that A can thus be written as

$$\begin{aligned} A = {} & I_{2L_v} \otimes I_{2L_h} \otimes \begin{pmatrix} 0 & 1 & -1 & -1 \\ -1 & 0 & 1 & -1 \\ 1 & -1 & 0 & 1 \\ 1 & 1 & -1 & 0 \end{pmatrix} \\ & + I_{2L_v} \otimes K_{2L_h} \otimes \begin{pmatrix} 0 & z_h & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} + I_{2L_v} \otimes K'_{2L_h} \otimes \begin{pmatrix} 0 & 0 & 0 & 0 \\ -z_h & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \\ & + K_{2L_v} \otimes I_{2L_h} \otimes \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & z_v \\ 0 & 0 & 0 & 0 \end{pmatrix} + K'_{2L_v} \otimes I_{2L_h} \otimes \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & -z_v & 0 \end{pmatrix}, \end{aligned} \tag{3.11.53}$$

where $\otimes$ indicates the Kronecker product and

$$K_p = \begin{pmatrix} 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 1 \\ 0 & 0 & 0 & \dots & 0 \end{pmatrix} \tag{3.11.54}$$

is an $p \times p$ matrix.[13] The nonzero blocks $A(j,k;j,k)$, $A(j,k;j,k+1)$, $A(j,k+1;j,k)$, $A(j,k;j+1,k)$ and $A(j+1,k;j,k)$ in the matrix A are captured by the respective 5 terms in the sum (3.11.53).

[13] Recall that the Kronecker product of an $p_1 \times q_1$ matrix U by a $p_2 \times q_2$ matrix V is the $p_1 p_2 \times q_1 q_2$ matrix

$$U \otimes V = \begin{pmatrix} u_{11} V & \dots & u_{1q_1} V \\ \vdots & \ddots & \vdots \\ u_{p_1 1} V & \dots & u_{p_1 q_1} V \end{pmatrix}.$$

It is, in general, not commutative.

The matrices K_p are given by (3.11.54). For computational purpose, it is convenient to replace them in (3.11.53) by the $p \times p$ matrices

$$H_p = \begin{pmatrix} 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 1 \\ 1 & 0 & 0 & \dots & 0 \end{pmatrix}, \tag{3.11.55}$$

which are circulant (Section 2.11). Thus, consider the following matrix

$$\begin{aligned} \widetilde{A} = I_{2L_v} \otimes I_{2L_h} \otimes & \begin{pmatrix} 0 & 1 & -1 & -1 \\ -1 & 0 & 1 & -1 \\ 1 & -1 & 0 & 1 \\ 1 & 1 & -1 & 0 \end{pmatrix} \\ + I_{2L_v} \otimes H_{2L_h} \otimes & \begin{pmatrix} 0 & z_h & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} + I_{2L_v} \otimes H'_{2L_h} \otimes \begin{pmatrix} 0 & 0 & 0 & 0 \\ -z_h & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \\ + H_{2L_v} \otimes I_{2L_h} \otimes & \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & z_v \\ 0 & 0 & 0 & 0 \end{pmatrix} + H'_{2L_v} \otimes I_{2L_h} \otimes \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & -z_v & 0 \end{pmatrix}. \end{aligned} \tag{3.11.56}$$

The matrix $\widetilde{A}$ differs from A only at the rows and columns corresponding to the boundary sites (j, k); that is, when $j = -L_h + 1, L_h$ or $k = -L_v + 1, L_v$.

We will consider below the so-called thermodynamic limit of $L_h, L_v \to \infty$. Since the difference between A and $\widetilde{A}$ is only at the columns and rows corresponding to the (j, k) which are at boundary sites, in the thermodynamic limit, the elements of A^{-1} away from these columns and rows will be those of $\widetilde{A}^{-1}$. Since the elements in (3.11.52) are away from the boundary, we will thus suppose without loss of generality that A is actually given by (3.11.56).

The advantage of working with A given by (3.11.56) is that it now has a circulant structure which makes it easier to compute its inverse.

Lemma 3.11.4 *Let*

$$\alpha_1 = \frac{z_h(1 - z_v)}{1 + z_v}, \quad \alpha_2 = \frac{z_h^{-1}(1 - z_v)}{1 + z_v}, \tag{3.11.57}$$

$$a_m = \frac{1}{2\pi} \int_0^{2\pi} d\theta \, e^{-im\theta} \phi(\theta), \tag{3.11.58}$$

$$\phi(\theta) = \Big(\frac{(1 - \alpha_1 e^{i\theta})(1 - \alpha_2 e^{-i\theta})}{(1 - \alpha_1 e^{-i\theta})(1 - \alpha_2 e^{i\theta})} \Big)^{1/2} \tag{3.11.59}$$

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and δ_{ij} denote the Kronecker delta. Then,

$$\lim_{L_h, L_v \to \infty} A^{-1}(0, k; 0, k')_{RL} = \frac{1}{1 - z_h^2} (z_h \delta_{k'-k-1,0} - a_{k'-k-1}), \tag{3.11.60}$$

$$\lim_{L_h, L_v \to \infty} A^{-1}(0, k; 0, k')_{LR} = \frac{1}{1 - z_h^2} (-z_h \delta_{k-k'-1,0} + a_{k-k'-1}), \tag{3.11.61}$$

$$\lim_{L_h, L_v \to \infty} A^{-1}(0, k; 0, k')_{RR} = \lim_{L_h, L_v \to \infty} A^{-1}(0, k; 0, k')_{LL} = 0. \tag{3.11.62}$$

Remark 3.11.5 For $a > 0$, the factor

$$g_a(e^{i\theta}) = \left(\frac{1 - ae^{\pm i\theta}}{1 - ae^{\mp i\theta}}\right)^{1/2} \tag{3.11.63}$$

appearing twice in (3.11.59) is interpreted as

$$g_a(e^{i\theta}) = \left(\frac{1 - ae^{\pm i\theta}}{1 - ae^{\mp i\theta}}\right)^{1/2} \left(\frac{1 - ae^{\pm i\theta}}{1 - ae^{\pm i\theta}}\right)^{1/2} = \frac{1 - ae^{\pm i\theta}}{((1 - ae^{i\theta})(1 - ae^{-i\theta}))^{1/2}} = \frac{1 - ae^{\pm i\theta}}{|1 - ae^{i\theta}|}. \tag{3.11.64}$$

If $a \neq 1$, the function $\frac{1-aw^{\pm 1}}{1-aw^{\mp 1}}$ is analytic and has nonzero real part on the unit disk $|w| \leq 1$. In this case, $g(e^{i\theta})$ can also be interpreted as

$$g_a(e^{i\theta}) = e^{\frac{1}{2}\mathrm{Log}\left(\frac{1-ae^{\pm i\theta}}{1-ae^{\mp i\theta}}\right)} = e^{\frac{1}{2}\mathrm{Log}(1-ae^{\pm i\theta})} e^{-\frac{1}{2}\mathrm{Log}(1-ae^{\mp i\theta})}, \tag{3.11.65}$$

where Log is the principal branch of the logarithm. If $a = 1$, in view of the interpretation (3.11.64),

$$\begin{aligned} g_1(e^{i\theta}) &= \frac{1 - e^{\pm i\theta}}{|1 - e^{i\theta}|} = \frac{1 - \cos\theta \pm i \sin\theta}{2 \sin\frac{\theta}{2}} = \sin\frac{\theta}{2} \pm i \cos\frac{\theta}{2} \\ &= \pm i\left(\cos\frac{\theta}{2} \mp i \sin\frac{\theta}{2}\right) = \pm i e^{\mp\theta/2}. \end{aligned} \tag{3.11.66}$$

We now turn to the proof of Lemma 3.11.4.

Proof The matrix H_p in (3.11.55) is circulant and by Proposition 2.11.1, it can be factorized using a discrete Fourier basis,

$$f_q = \frac{1}{\sqrt{p}}\left(1,\ e^{-i\frac{2\pi q}{p}},\ e^{-i\frac{2\pi 2q}{p}}, \dots, e^{-i\frac{2\pi(p-1)q}{p}}\right)',$$

$q = 0, \dots, p-1$. In this basis, by using (2.11.5) in Proposition 2.11.1, the eigenvalues of H_p are $e^{-i\frac{2\pi q}{p}}$, $q = 0, \dots, p-1$. We can thus write

$$H_p = F_p \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & e^{-i\frac{2\pi}{p}} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & e^{-i\frac{2\pi(p-1)}{p}} \end{pmatrix} F_p^*, \tag{3.11.67}$$

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where $F_p = (f_0 \dots f_{p-1})$ and F_p^* denotes the Hermitian transpose. Similarly, since H_p is real-valued,[14]

$$H_p' = F_p \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & e^{i\frac{2\pi}{p}} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & e^{i\frac{2\pi(p-1)}{p}} \end{pmatrix} F_p^*. \tag{3.11.68}$$

The matrix F_p, moreover, is unitary; that is,

$$F_p F_p^* = I_p. \tag{3.11.69}$$

Consequently, if we denote by D_p the diagonal matrix in (3.11.67), we have $F_p^* H_p F_p = F_p^* F_p D_p F_p^* F_p = D_p$ and similarly for (3.11.68). Now, in view of (3.11.67), (3.11.68) and (3.11.69), by multiplying the left sides of the equality (3.11.56)[15] by $F_{2L_v}^* \otimes F_{2L_h}^* \otimes I_4$ and the right sides of (3.11.56) by $F_{2L_v} \otimes F_{2L_h} \otimes I_4$, we obtain[16] that

$$(F_{2L_v}^* \otimes F_{2L_h}^* \otimes I_4) A (F_{2L_v} \otimes F_{2L_h} \otimes I_4) = I_{2L_v} \otimes I_{2L_h} \otimes \begin{pmatrix} 0 & 1 & -1 & -1 \\ -1 & 0 & 1 & -1 \\ 1 & -1 & 0 & 1 \\ 1 & 1 & -1 & 0 \end{pmatrix}$$

$$+ I_{2L_v} \otimes \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & e^{-i\frac{2\pi}{2L_h}} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & e^{-i\frac{2\pi(2L_h-1)}{2L_h}} \end{pmatrix} \otimes \begin{pmatrix} 0 & z_h & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$+ I_{2L_v} \otimes \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & e^{i\frac{2\pi}{2L_h}} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & e^{i\frac{2\pi(2L_h-1)}{2L_h}} \end{pmatrix} \otimes \begin{pmatrix} 0 & 0 & 0 & 0 \\ -z_h & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$+ \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & e^{-i\frac{2\pi}{2L_v}} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & e^{-i\frac{2\pi(2L_v-1)}{2L_v}} \end{pmatrix} \otimes I_{2L_h} \otimes \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & z_v \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

[14] Recall that $F^* = \overline{F}'$ where bar denotes "conjugate". Since H is real-valued, one has $H' = H^* = (FDF^*)^* = F^{**} D^* F^* = F\overline{D}F^*$.

[15] Recall that $\widetilde{A}$ in (3.11.56) is our A as noted before the statement of the lemma.

[16] If U_1 and U_2 are $m \times m$ and V_1 and V_2 are $p \times p$ matrices, then $(U_1 \otimes V_1)(U_2 \otimes V_2) = (U_1 U_2) \otimes (V_1 V_2)$. In particular, $(F_{2L_h}^* \otimes F_{2L_v}^*)(I_{2L_h} \otimes I_{2L_v})(F_{2L_h} \otimes F_{2L_v}) = I_{2L_h} \otimes I_{2L_v} = I_{(2L_h)(2L_v)}$ by (3.11.69).

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$$+\begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & e^{i\frac{2\pi}{2L_v}} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & e^{i\frac{2\pi(2L_v-1)}{2L_v}} \end{pmatrix} \otimes I_{2L_h} \otimes \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & -z_v & 0 \end{pmatrix}.$$

This can be written as

$$(F^*_{2L_v} \otimes F^*_{2L_h} \otimes I_4) A (F_{2L_v} \otimes F_{2L_h} \otimes I_4)$$
$$= \mathrm{diag}\Big(A(\theta_{h,t}, \theta_{v,s}),\ \theta_{h,t} = \frac{2\pi t}{2L_h},\ t = 0, \dots, 2L_h-1,\ \theta_{v,s} = \frac{2\pi s}{2L_v},\ s = 0, \dots, 2L_v-1\Big), \tag{3.11.70}$$

where[17]

$$A(\theta_h, \theta_v) = \begin{array}{c} \\ R \\ L \\ U \\ D \end{array} \begin{array}{c} \begin{array}{cccc} R & L & U & D \end{array} \\ \begin{pmatrix} 0 & 1 + z_h e^{-i\theta_h} & -1 & -1 \\ -1 - z_h e^{i\theta_h} & 0 & 1 & -1 \\ 1 & -1 & 0 & 1 + z_v e^{-i\theta_v} \\ 1 & 1 & -1 - z_v e^{i\theta_v} & 0 \end{pmatrix} \end{array} \tag{3.11.71}$$

and diag in (3.11.70) refers to placing the blocks $A(\theta_{h,t}, \theta_{v,s})$ on the diagonal. (The exact placement is $A(\theta_{h,0}, \theta_{v,0})$, $A(\theta_{h,1}, \theta_{v,0})$, ..., $A(\theta_{h,2L_h-1}, \theta_{v,0})$, $A(\theta_{h,0}, \theta_{v,1})$, $A(\theta_{h,1}, \theta_{v,1})$, ..., $A(\theta_{h,2L_h-1}, \theta_{v,1})$, and so on.)

It follows from (3.11.70) that

$$(F^*_{2L_v} \otimes F^*_{2L_h} \otimes I_4) A^{-1} (F_{2L_v} \otimes F_{2L_h} \otimes I_4) = \mathrm{diag}\Big(A^{-1}(\theta_{h,t}, \theta_{v,s})\Big)$$

and hence that

$$A^{-1} = (F_{2L_v} \otimes F_{2L_h} \otimes I_4)\mathrm{diag}\Big(A^{-1}(\theta_{h,t}, \theta_{v,s})\Big)(F^*_{2L_v} \otimes F^*_{2L_h} \otimes I_4).$$

For the blocks $A^{-1}(j, k;\ j', k')$ of the matrix A^{-1}, this yields[18]

$$A^{-1}(j, k;\ j', k') = \frac{1}{(2L_h)(2L_v)} \sum_{t=0}^{2L_h-1} \sum_{s=0}^{2L_v-1} e^{i\theta_{h,t}(k'-k)+i\theta_{v,s}(j'-j)} A^{-1}(\theta_{h,t}, \theta_{v,s}). \tag{3.11.72}$$

[17] As illustration, if $V = \begin{pmatrix} 0 & z \\ 0 & 0 \end{pmatrix}$, then

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & e^{i\theta_1} & 0 \\ 0 & 0 & e^{i\theta_2} \end{pmatrix} \otimes V = \begin{pmatrix} V & 0 & 0 \\ 0 & e^{i\theta_1} V & 0 \\ 0 & 0 & e^{i\theta_2} V \end{pmatrix} = \begin{pmatrix} 0 & z & & & & \\ 0 & 0 & & & & \\ & & 0 & e^{i\theta_1} z & & \\ & & 0 & 0 & & \\ & & & & 0 & e^{i\theta_2} z \\ & & & & 0 & 0 \end{pmatrix}.$$

[18] To show (3.11.72), one may suppose without loss of generality that $A^{-1}(j, k;\ j', k')$, $A^{-1}(\theta_{h,t}, \theta_{v,s})$ are 1×1 rather than 4×4 blocks and show that $A^{-1} = (F_{2L_v} \otimes F_{2L_h})\mathrm{diag}\Big(A^{-1}(\theta_{h,t}, \theta_{v,s})\Big)(F^*_{2L_v} \otimes F^*_{2L_h})$ implies (3.11.72). The latter fact is left as an exercise.

In the thermodynamic limit $L_h, L_v \to \infty$, the relation (3.11.72) becomes

$$\lim_{L_h, L_v \to \infty} A^{-1}(j,k;j',k') = \frac{1}{(2\pi)^2}\int_0^{2\pi} d\theta_h \int_0^{2\pi} d\theta_v\, e^{i(k'-k)\theta_h + i(j'-j)\theta_v} A^{-1}(\theta_h,\theta_v). \tag{3.11.73}$$

We are interested in the case $j = j' = 0$,

$$\lim_{L_h, L_v \to \infty} A^{-1}(0,k;0,k') = \frac{1}{(2\pi)^2}\int_0^{2\pi} d\theta_h\, e^{i(k'-k)\theta_h} \int_0^{2\pi} d\theta_v\, A^{-1}(\theta_h,\theta_v) \tag{3.11.74}$$

and, more specifically, in the entries at the columns and rows labeled R, L (see (3.11.60), (3.11.61) and (3.11.62)).

The inverse of $A(\theta_h, \theta_v)$ in (3.11.71) can be computed (McCoy [702], p. 358) or verified directly as

$$A^{-1}(\theta_h,\theta_v) = \frac{1}{\Delta(\theta_h,\theta_v)}$$

$$\times \begin{array}{c} \\ R \\ L \\ U \\ D \end{array} \begin{array}{c} \begin{array}{cccc} \quad R \quad & \quad L \quad & \quad U \quad & \quad D \quad \end{array} \\ \begin{pmatrix} b - b^* & b + b^* - abb^* & 2 - ab^* & 2 - ab \\ -b^* - b + a^*b^*b & b^* - b & -2 + a^*b^* & 2 - a^*b \\ -2 + a^*b & 2 - ab & a^* - a & a + a^* - aa^*b \\ -2 + a^*b^* & -2 + ab^* & -a^* - a + a^*ab^* & a - a^* \end{pmatrix} \end{array}, \tag{3.11.75}$$

where

$$a = 1 + z_h e^{-i\theta_h}, \quad b = 1 + z_v e^{-i\theta_v} \tag{3.11.76}$$

and

$$\begin{aligned}\Delta(\theta_h,\theta_v) &= 4 + aa^*bb^* - (a+a^*)(b+b^*) \\ &= (1+z_h^2)(1+z_v^2) - 2z_h(1-z_v^2)\cos\theta_h - 2z_v(1-z_h^2)\cos\theta_v.\end{aligned} \tag{3.11.77}$$

We now want to compute $A^{-1}(0,k;0,k')_{ll'}$ at $l, l' = R, L$ in the thermodynamic limit (see (3.11.60), (3.11.61) and (3.11.62)). In view of (3.11.74), we need first to consider

$$B^{-1}(\theta_h)_{ll'} := \frac{1}{2\pi}\int_0^{2\pi} d\theta_v\, A^{-1}(\theta_h,\theta_v)_{ll'}. \tag{3.11.78}$$

The entries $A^{-1}(\theta_h,\theta_v)_{RR}$ and $A^{-1}(\theta_h,\theta_v)_{LL}$ involve $b - b^*$. Note that

$$\int_0^{2\pi} d\theta_v \frac{b-b^*}{\Delta(\theta_h,\theta_v)} = \int_0^{2\pi} d\theta_v \frac{z_h e^{-i\theta_v}}{(1+z_h^2)(1+z_v^2) - 2z_h(1-z_v^2)\cos\theta_h - 2z_v(1-z_h^2)\cos\theta_v}$$

$$- \int_0^{2\pi} d\theta_v \frac{z_h e^{i\theta_v}}{(1+z_h^2)(1+z_v^2) - 2z_h(1-z_v^2)\cos\theta_h - 2z_v(1-z_h^2)\cos\theta_v} = 0,$$

since the last two integrals are equal, by making the change of variables $\theta_v \to 2\pi - \theta_v$ in the second integral. It follows that

$$\lim_{L_h,L_v\to\infty} A^{-1}(0,k;0,k')_{RR} = \lim_{L_h,L_v\to\infty} A^{-1}(0,k;0,k')_{LL} = 0,$$

that is, the relations (3.11.62) hold.

We now turn to $A^{-1}(0,k;0,k')_{RL}$ and $A^{-1}(0,k;0,k')_{LR}$ in the thermodynamic limit. We first want to factor the quantity $\Delta(\theta_h,\theta_v)$ in (3.11.77) as a function of $e^{i\theta_v}$. Write

$$\Delta(\theta_h,\theta_v) = (1+z_h^2)(1+z_v^2) - 2z_h(1-z_v^2)\cos\theta_h - z_v(1-z_h^2)(e^{i\theta_v}+e^{-i\theta_v})$$

$$= -z_v(1-z_h^2)e^{-i\theta_v}\Big(e^{2i\theta_v} - \frac{(1+z_h^2)(1+z_v^2)-2z_h(1-z_v^2)\cos\theta_h}{z_v(1-z_h^2)}e^{i\theta_v}+1\Big) \quad (3.11.79)$$

and consider the polynomial

$$f(x) = x^2 - \frac{(1+z_h^2)(1+z_v^2)-2z_h(1-z_v^2)\cos\theta_h}{z_v(1-z_h^2)}x + 1. \quad (3.11.80)$$

The polynomial (3.11.80) has roots x_1 and x_2 satisfying

$$x_1x_2 = 1, \quad x_1+x_2 = \frac{(1+z_h^2)(1+z_v^2)-2z_h(1-z_v^2)\cos\theta_h}{z_v(1-z_h^2)}. \quad (3.11.81)$$

One can verify that

$$(1+z_h^2)(1+z_v^2)-2z_h(1-z_v^2)\cos\theta_h - 2z_v(1-z_h^2) = (1-z_v)^2(1-\alpha_2^{-1}e^{i\theta_h})(1-\alpha_2^{-1}e^{-i\theta_h}), \quad (3.11.82)$$

where α_2 is defined in (3.11.57). This shows that $x_1+x_2 \geq 2$ in (3.11.81), and hence that the roots x_1 and x_2 are both positive and real. Since $x_1x_2 = 1$ in (3.11.81), we can set $x_1 = \alpha$ and $x_2 = \alpha^{-1}$ with

$$x_1 = \alpha > 1 > x_2 = \alpha^{-1} > 0. \quad (3.11.83)$$

Thus (3.11.80) becomes $f(x) = x^2 - (\alpha+\alpha^{-1})x + 1$ so that $f(e^{i\theta_v}) = e^{2i\theta_v} - (\alpha + \alpha^{-1})e^{i\theta_v} + 1$. Hence, in view of (3.11.79), we have

$$\Delta(\theta_h,\theta_v) = (e^{i\theta_v}-\alpha)(\alpha^{-1}e^{-i\theta_v}-1)z_v(1-z_h^2). \quad (3.11.84)$$

Solving for the roots $x_1 = \alpha$ and $x_2 = \alpha^{-1}$, we can write

$$\alpha^{\pm 1} = \frac{1}{2z_v(1-z_h^2)}\Big((1+z_h^2)(1+z_v^2) - z_h(1-z_v^2)(e^{i\theta_h}+e^{-i\theta_h}) \pm \Big(\Big((1+z_h^2)(1+z_v^2) - z_h(1-z_v^2)(e^{i\theta_h}+e^{-i\theta_h})\Big)^2 - (2z_v(1-z_h^2))^2\Big)^{1/2}\Big). \quad (3.11.85)$$

The term in the square root can be written as

$$\Big((1+z_h^2)(1+z_v^2) - z_h(1-z_v^2)(e^{i\theta_h}+e^{-i\theta_h})\Big)^2 - \Big(2z_v(1-z_h^2)\Big)^2$$
$$= \Big((1+z_h^2)(1+z_v^2) - z_h(1-z_v^2)(e^{i\theta_h}+e^{-i\theta_h}) - 2z_v(1-z_h^2)\Big)$$
$$\times \Big((1+z_h^2)(1+z_v^2) - z_h(1-z_v^2)(e^{i\theta_h}+e^{-i\theta_h}) + 2z_v(1-z_h^2)\Big).$$

The relation (3.11.82) provides a factorization for the first term in the last product. One can similarly verify that the second term in the product can be written as

$$(1+z_h^2)(1+z_v^2)-2z_h(1-z_v^2)\cos\theta_h + 2z_v(1-z_h^2) = (1+z_v)^2(1-\alpha_1e^{i\theta_h})(1-\alpha_1e^{-i\theta_h}), \quad (3.11.86)$$

where α_1 is defined in (3.11.57). Combining (3.11.82) and (3.11.86), we obtain from (3.11.85) that (see also McCoy and Wu [703], Eqs. (3.2)–(3.4) on pp. 86–87)

$$\alpha^{\pm 1} = \frac{1}{2z_v(1-z_h^2)}\Big((1+z_h^2)(1+z_v^2) - z_h(1-z_v^2)(e^{i\theta_h}+e^{-i\theta_h}) \pm (1-z_v^2)\Big((1-\alpha_1 e^{i\theta_h})(1-\alpha_1 e^{-i\theta_h})(1-\alpha_2^{-1}e^{i\theta_h})(1-\alpha_2^{-1}e^{-i\theta_h})\Big)^{1/2}\Big). \tag{3.11.87}$$

Consider now, for example, the entry RL in (3.11.75). It is

$$b + b^* - abb^* = 1 - z_v^2 - z_h e^{-i\theta_h}(1 + z_v^2 + z_v(e^{i\theta_v} + e^{-i\theta_v})). \tag{3.11.88}$$

By using (3.11.84) and (3.11.88), it follows that

$$\begin{aligned} B^{-1}(\theta_h)_{RL} &= \frac{1}{2\pi}\int_0^{2\pi} d\theta_v\, A^{-1}(\theta_h,\theta_v)_{RL} \\ &= \frac{1}{2\pi}\int_0^{2\pi} d\theta_v\, \frac{e^{i\theta_v}(1-z_v^2-z_h e^{-i\theta_h}(1+z_v^2+z_v(e^{i\theta_v}+e^{-i\theta_v})))}{(e^{i\theta_v}-\alpha)(\alpha^{-1}-e^{i\theta_v})z_v(1-z_h^2)} \\ &= -\frac{1}{2\pi i}\int_{|w|=1} dw\, \frac{1-z_v^2-z_h e^{-i\theta_h}(1+z_v^2+z_v(w+w^{-1}))}{(w-\alpha)(w-\alpha^{-1})z_v(1-z_h^2)} \\ &= -\frac{1}{2\pi i}\int_{|w|=1} dw\, \frac{1-z_v^2-z_h e^{-i\theta_h}(1+z_v^2+z_v w)}{(w-\alpha)(w-\alpha^{-1})z_v(1-z_h^2)} \\ &\quad + \frac{1}{2\pi i}\int_{|w|=1} dw\, \frac{z_h z_v e^{-i\theta_h}}{w(w-\alpha)(w-\alpha^{-1})z_v(1-z_h^2)}. \end{aligned}$$

Recall from (3.11.83) that $\alpha > 1 > \alpha^{-1} > 0$. Then, writing the last integral as the sum of two integrals by Cauchy's theorem,

$$\begin{aligned} B^{-1}(\theta_h)_{RL} = &-\frac{1}{2\pi i}\int_{|w|=1} dw\, \frac{1-z_v^2-z_h e^{-i\theta_h}(1+z_v^2+z_v w)}{(w-\alpha)(w-\alpha^{-1})z_v(1-z_h^2)} \\ &+ \frac{1}{2\pi i}\int_{|w|=\epsilon} dw\, \frac{z_h z_v e^{-i\theta_h}}{w(w-\alpha)(w-\alpha^{-1})z_v(1-z_h^2)} \\ &+ \frac{1}{2\pi i}\int_{|w-\alpha^{-1}|=\epsilon} dw\, \frac{z_h z_v e^{-i\theta_h}}{w(w-\alpha)(w-\alpha^{-1})z_v(1-z_h^2)}, \end{aligned}$$

for small enough $\epsilon > 0$ so that $\epsilon < \alpha^{-1} - \epsilon$. By using Cauchy's integral formula,[19] we conclude that

$$B^{-1}(\theta_h)_{RL} = -\frac{1-z_v^2-z_h e^{-i\theta_h}(1+z_v^2+z_v\alpha^{-1})}{(\alpha^{-1}-\alpha)z_v(1-z_h^2)}$$

[19] Cauchy's integral formula states that

$$f(a) = \frac{1}{2\pi i}\int_\gamma \frac{f(z)}{z-a}dz,$$

where $f : U \mapsto \mathbb{C}$ is analytic on an open subset U of $\mathbb{C}$ and γ is the boundary of a circle completely contained in U.

$$
\begin{aligned}
&+\frac{z_h z_v e^{-i\theta_h}}{(-\alpha)(-\alpha^{-1})z_v(1-z_h^2)}+\frac{z_h z_v e^{-i\theta_h}}{\alpha^{-1}(\alpha^{-1}-\alpha)z_v(1-z_h^2)} \\
&=\frac{z_h e^{-i\theta_h}}{1-z_h^2}+e^{-i\theta_h}\frac{(1-z_v^2)e^{i\theta_h}-z_h(1+z_v^2+z_v(\alpha^{-1}+\alpha))}{z_v(1-z_h^2)(\alpha-\alpha^{-1})} \\
&=:\frac{z_h e^{-i\theta_h}}{1-z_h^2}+e^{-i\theta_h}\widetilde{B}(\theta_h)_{RL}. \qquad (3.11.89)
\end{aligned}
$$

Computing $\alpha+\alpha^{-1}$ from (3.11.87), we get that

$$
1+z_v^2+z_v(\alpha^{-1}+\alpha)=\frac{2(1+z_v^2)-z_h(1-z_v^2)(e^{i\theta_h}+e^{-i\theta_h})}{1-z_h^2}
$$

and

$$
z_v(1-z_h^2)(\alpha-\alpha^{-1})=(1-z_v^2)\Big((1-\alpha_1 e^{i\theta_h})(1-\alpha_1 e^{-i\theta_h})(1-\alpha_2^{-1}e^{i\theta_h})(1-\alpha_2^{-1}e^{-i\theta_h})\Big)^{1/2}.
$$

Then, the term $\widetilde{B}(\theta_h)_{RL}$ in (3.11.89) becomes

$$
\begin{aligned}
\widetilde{B}(\theta_h)_{RL}&=\frac{(1-z_v^2)e^{i\theta_h}+z_h^2(1-z_v^2)e^{-i\theta_h}-2z_h(1+z_v^2)}{(1-z_h^2)(1-z_v^2)((1-\alpha_1 e^{i\theta_h})(1-\alpha_1 e^{-i\theta_h})(1-\alpha_2^{-1}e^{i\theta_h})(1-\alpha_2^{-1}e^{i\theta_h}))^{1/2}} \\
&=-\frac{z_h(1+z_v)^2(1-\alpha_1 e^{-i\theta_h})(1-\alpha_2 e^{i\theta_h})}{(1-z_h^2)(1-z_v^2)((1-\alpha_1 e^{i\theta_h})(1-\alpha_1 e^{-i\theta_h})(1-\alpha_2^{-1}e^{i\theta_h})(1-\alpha_2^{-1}e^{-i\theta_h}))^{1/2}},
\end{aligned}
$$

where the last equality can be verified directly using (3.11.57). Using the fact that $z_h(1+z_v)/(1-z_v)=\alpha_2^{-1}$ by (3.11.57), we can write

$$
\begin{aligned}
\widetilde{B}(\theta_h)_{RL}&=-\frac{(1-\alpha_1 e^{-i\theta_h})(1-\alpha_2 e^{i\theta_h})}{(1-z_h^2)\alpha_2((1-\alpha_1 e^{i\theta_h})(1-\alpha_1 e^{-i\theta_h})(1-\alpha_2^{-1}e^{i\theta_h})(1-\alpha_2^{-1}e^{-i\theta_h}))^{1/2}} \\
&=-\frac{(1-\alpha_1 e^{-i\theta_h})(1-\alpha_2 e^{i\theta_h})}{(1-z_h^2)((1-\alpha_1 e^{i\theta_h})(1-\alpha_1 e^{-i\theta_h})(1-\alpha_2 e^{i\theta_h})(1-\alpha_2 e^{-i\theta_h}))^{1/2}},
\end{aligned}
$$

where we used the identity $\alpha_2^2(1-\alpha_2^{-1}e^{i\theta_h})(1-\alpha_2^{-1}e^{-i\theta_h})=(1-\alpha_2 e^{i\theta_h})(1-\alpha_2 e^{-i\theta_h})$. With the interpretation in Remark 3.11.5, we can write the last expression as

$$
\widetilde{B}(\theta_h)_{RL}=-\frac{1}{(1-z_h^2)}\Big(\frac{(1-\alpha_1 e^{-i\theta_h})(1-\alpha_2 e^{i\theta_h})}{(1-\alpha_1 e^{i\theta_h})(1-\alpha_2 e^{-i\theta_h})}\Big)^{1/2}. \qquad (3.11.90)
$$

Combining (3.11.89) and (3.11.90), we obtain that

$$
B^{-1}(\theta_h)_{RL}=\frac{1}{1-z_h^2}\Big(z_h e^{-i\theta_h}-e^{-i\theta_h}\Big(\frac{(1-\alpha_1 e^{-i\theta_h})(1-\alpha_2 e^{i\theta_h})}{(1-\alpha_1 e^{i\theta_h})(1-\alpha_2 e^{-i\theta_h})}\Big)^{1/2}\Big). \qquad (3.11.91)
$$

In view of (3.11.74) and (3.11.78), this yields

$$
\lim_{L_h,L_v\to\infty}A^{-1}(0,k;0,k')_{RL}=\frac{z_h}{1-z_h^2}\frac{1}{2\pi}\int_0^{2\pi}e^{i(k'-k)\theta}e^{-i\theta}d\theta
$$

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$$
\begin{aligned}
&- \frac{1}{(1-z_h^2)2\pi} \int_0^{2\pi} e^{i(k'-k-1)\theta} \Big(\frac{(1-\alpha_1 e^{-i\theta})(1-\alpha_2 e^{i\theta})}{(1-\alpha_1 e^{i\theta})(1-\alpha_2 e^{-i\theta})} \Big)^{1/2} d\theta \\
&= \frac{z_h}{1-z_h^2} \delta_{k'-k-1,0} - \frac{1}{(1-z_h^2)2\pi} \int_0^{2\pi} e^{-i(k'-k-1)\theta} \phi(\theta) d\theta \\
&= \frac{1}{1-z_h^2} (z_h \delta_{k'-k-1,0} - a_{k'-k-1}),
\end{aligned}
$$

after defining a_n and $\phi(\theta)$ as in (3.11.58) and (3.11.59), and using $\int_0^{2\pi} e^{i\theta(k'-k-1)} d\theta = 2\pi \delta_{k'-k-1,0}$. Since the matrix $A^{-1}(\theta_h, \theta_v)$ in (3.11.75) satisfies $A^{-1}(\theta_h, \theta_v)^* = -A^{-1}(\theta_h, \theta_v)$, one has

$$
B^{-1}(\theta_h)_{LR} = -B^{-1}(\theta_h)^*_{RL}
$$

and one gets similarly

$$
\lim_{L_h, L_v \to \infty} A^{-1}(0, k; 0, k')_{LR} = \frac{1}{1-z_h^2} (-z_h \delta_{k-k'-1,0} + a_{k-k'-1}).
$$

This establishes (3.11.60) and (3.11.61), and concludes the proof of the lemma. $\Box$

We now want to substitute (3.11.60), (3.11.61) and (3.11.62) into the expression (3.11.51). Observe from (3.11.52) and (3.11.60)–(3.11.62) that the $2n \times 2n$ matrix Q becomes, in the thermodynamic limit,

$$
\lim_{L_h, L_v \to \infty} Q = \begin{pmatrix}
0 & 0 & \dots & 0 & \frac{z_h - a_0}{1-z_h^2} & \frac{-a_1}{1-z_h^2} & \dots & \frac{-a_{n-1}}{1-z_h^2} \\
0 & 0 & \dots & 0 & \frac{-a_{-1}}{1-z_h^2} & \frac{z_h - a_0}{1-z_h^2} & \dots & \frac{-a_{n-2}}{1-z_h^2} \\
\vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\
0 & 0 & \dots & 0 & \frac{-a_{-n+1}}{1-z_h^2} & \frac{-a_{-n+2}}{1-z_h^2} & \dots & \frac{z_h - a_0}{1-z_h^2} \\
\frac{-z_h + a_0}{1-z_h^2} & \frac{a_{-1}}{1-z_h^2} & \dots & \frac{a_{-n+1}}{1-z_h^2} & 0 & 0 & \dots & 0 \\
\frac{a_1}{1-z_h^2} & \frac{-z_h + a_0}{1-z_h^2} & \dots & \frac{a_{-n+2}}{1-z_h^2} & 0 & 0 & \dots & 0 \\
\vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\
\frac{a_{n-1}}{1-z_h^2} & \frac{a_{n-2}}{1-z_h^2} & \dots & \frac{-z_h + a_0}{1-z_h^2} & 0 & 0 & \dots & 0
\end{pmatrix}.
$$

Note from (3.11.49) that

$$
y^{-1} = \begin{pmatrix}
0 & 0 & \dots & 0 & \frac{-z_h}{1-z_h^2} & 0 & \dots & 0 \\
0 & 0 & \dots & 0 & 0 & \frac{-z_h}{1-z_h^2} & \dots & 0 \\
\vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\
0 & 0 & \dots & 0 & 0 & 0 & \dots & \frac{-z_h}{1-z_h^2} \\
\frac{z_h}{1-z_h^2} & 0 & \dots & 0 & 0 & 0 & \dots & 0 \\
0 & \frac{z_h}{1-z_h^2} & \dots & 0 & 0 & 0 & \dots & 0 \\
\vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\
0 & 0 & \dots & \frac{z_h}{1-z_h^2} & 0 & 0 & \dots & 0
\end{pmatrix}.
$$

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Then,

$$(1-z_h^2)(y^{-1}+Q) = \left(\begin{array}{cccccccc} 0 & 0 & \dots & 0 & -a_0 & -a_1 & \dots & -a_{n-1} \\ 0 & 0 & \dots & 0 & -a_{-1} & -a_0 & \dots & -a_{n-2} \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 0 & -a_{-n+1} & -a_{-n+2} & \dots & -a_0 \\ a_0 & a_{-1} & \dots & a_{-n+1} & 0 & 0 & \dots & 0 \\ a_1 & a_0 & \dots & a_{-n+2} & 0 & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{n-1} & a_{n-2} & \dots & a_0 & 0 & 0 & \dots & 0 \end{array}\right)$$

or

$$(1-z_h^2)(y^{-1}+Q) = \begin{pmatrix} 0 & -B' \\ B & 0 \end{pmatrix},$$

where $B = (a_{j-k})_{1\le j,k\le n}$ and 0 is a $n\times n$ zero matrix. By using the formula $\det((A\ B;\ C\ D)) = \det(AD-CB)$ if $AB=BA$ for $n\times n$ blocks A, B, C and D, we deduce that

$$\det((1-z_h^2)(y^{-1}+Q)) = (\det(B))^2.$$

Using the expression (3.11.51) for $(\mathbb{E}\sigma_{0,0}\sigma_{0,n})^2$, this leads to the following expression which now involves the determinant of an $n\times n$ matrix:

$$S_n := \lim_{L_h,L_v\to\infty}\mathbb{E}\sigma_{0,0}\sigma_{0,n} = \det\begin{pmatrix} a_0 & a_{-1} & a_{-2} & \dots & a_{-n+1} \\ a_1 & a_0 & a_{-1} & \dots & a_{-n+2} \\ a_2 & a_1 & a_0 & \dots & a_{-n+2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{n-1} & a_{n-2} & a_{n-3} & \dots & a_0 \end{pmatrix}. \tag{3.11.92}$$

One can similarly show (McCoy and Wu [703], p. 199, formula (3.31)) that

$$\widetilde{S}_n := \lim_{L_h,L_v\to\infty}\mathbb{E}\sigma_{0,0}\sigma_{n,n} = \det\begin{pmatrix} \widetilde{a}_0 & \widetilde{a}_{-1} & \widetilde{a}_{-2} & \dots & \widetilde{a}_{-n+1} \\ \widetilde{a}_1 & \widetilde{a}_0 & \widetilde{a}_{-1} & \dots & \widetilde{a}_{-n+2} \\ \widetilde{a}_2 & \widetilde{a}_1 & \widetilde{a}_0 & \dots & \widetilde{a}_{-n+2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \widetilde{a}_{n-1} & \widetilde{a}_{n-2} & \widetilde{a}_{n-3} & \dots & \widetilde{a}_0 \end{pmatrix}, \tag{3.11.93}$$

where $\widetilde{a}_m$ is defined as a_m in (3.11.58) but using $\widetilde{\phi}$ instead of ϕ, defined by

$$\widetilde{\phi}(\theta) = \left(\frac{\sinh 2\beta E_h \sinh 2\beta E_v - e^{-i\theta}}{\sinh 2\beta E_h \sinh 2\beta E_v - e^{i\theta}}\right)^{1/2}. \tag{3.11.94}$$

3.11.4 The Strong Szegö Limit Theorem

We are now interested in the behavior of the determinants S_n and $\widetilde{S}_n$ in (3.11.92) and (3.11.93), resp., as $n\to\infty$ at the critical temperature $T=T_c$, which is such that

$$1 = \sinh 2\beta E_h \sinh 2\beta E_v, \quad \beta = 1/T_c. \tag{3.11.95}$$

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The determinants S_n and $\widetilde{S}_n$ are defined in terms of functions $\phi(\theta)$ and $\widetilde{\phi}(\theta)$ in (3.11.59) and (3.11.94). We will need expressions of these functions at the critical temperature.

Lemma 3.11.6 *At the critical temperature T_c, $\phi(\theta)$ in (3.11.59) becomes*

$$\phi(\theta) = \Big(\frac{1-\alpha_1 e^{i\theta}}{1-\alpha_1 e^{-i\theta}}\Big)^{1/2} e^{i(-1/2)(\theta-\pi)} \tag{3.11.96}$$

and $\widetilde{\phi}(\theta)$ in (3.11.94) becomes

$$\widetilde{\phi}(\theta) = e^{i(-1/2)(\theta-\pi)}. \tag{3.11.97}$$

Proof Consider first $\widetilde{\phi}(\theta)$ in (3.11.94), which becomes by (3.11.95) and (3.11.66) in Remark 3.11.5,

$$\widetilde{\phi}(\theta) = \Big(\frac{1-e^{-i\theta}}{1-e^{i\theta}}\Big)^{1/2} = ie^{-i\theta/2} = e^{i(-1/2)(\theta-\pi)}. \tag{3.11.98}$$

Turning to the function $\phi(\theta)$ in (3.11.59), recall that the parameters α_1 and α_2 were defined in (3.11.57), and z_h and z_v in (3.11.20). We first show that if (3.11.95) holds, then $\alpha_2 = 1$. To do so, use the relations $(\cosh x)^2 - (\sinh x)^2 = 1$ and $2(\sinh x)(\cosh x) = \sinh 2x$. Then, by using $(1-z_k^2)/(2z_k) = (\sinh 2\beta E_k)^{-1}$ for $k = h, v$, the relation (3.11.95) can be expressed as $(1-z_h^2)(1-z_v^2) = 4z_h z_v$. This is the same as $(1-z_h z_v)^2 = (z_h + z_v)^2$ or $1 - z_h z_v = z_h + z_v$. The latter yields $1 = z_h^{-1}(1-z_v)/(1+z_v) = \alpha_2$. This now implies that

$$\alpha_1 = z_h^2 \alpha_2 = z_h^2 < 1. \tag{3.11.99}$$

Hence, by (3.11.98), $\phi(\theta)$ in (3.11.59) becomes (3.11.96). □

To obtain the asymptotic behavior of the determinants S_n and $\widetilde{S}_n$ as $n \to \infty$, we will apply the so-called *strong Szegö limit theorem*. This theorem concerns the limiting behavior, as $n \to \infty$, of the $n \times n$ Toeplitz determinant

$$D_n = \det \begin{pmatrix} c_0 & c_{-1} & \dots & c_{-n+1} \\ c_1 & c_0 & \dots & c_{-n+2} \\ \vdots & \vdots & \ddots & \vdots \\ c_{n-1} & c_{n-2} & \dots & c_0 \end{pmatrix}, \tag{3.11.100}$$

where the entries c_m are the Fourier coefficients of a function C on the unit circle; that is,

$$c_m = \frac{1}{2\pi}\int_0^{2\pi} d\theta\, e^{-im\theta} C(e^{i\theta}).$$

Thus, $C(e^{i\theta}) = \sum_{m=-\infty}^{\infty} c_m e^{i\theta m}$ is the Fourier transform of the sequence $\{c_m\}$.[20] In the case of interest here, $C(e^{i\theta})$ is chosen to be either $\phi(\theta)$ or $\widetilde{\phi}(\theta)$.

The classical strong Szegö limit theorem supposes that $C(e^{i\theta})$ is "smooth" and allows one to conclude that

$$D_n \sim E(C)\mu^n, \tag{3.11.101}$$

[20] The notation $C(e^{i\theta})$ instead of $C(e^{-i\theta})$ found in Appendix A.1.1 is used to be consistent with the referenced literature.

as $N \to \infty$, where μ and $E(C)$ are two constants given by

$$\mu = \frac{1}{2\pi}\int_0^{2\pi} d\theta \ \log C(e^{i\theta}), \quad E(C) = \exp\left\{\sum_{m=1}^{\infty} m(\log C)_m (\log C)_{-m}\right\},$$

and $(\log C)_m$ denotes the mth Fourier coefficient of the function $\log C(e^{i\theta})$. See Szegö [939]. A typical assumption for (3.11.101) to hold is that $C(\xi)$ be continuous on the unit circle $|\xi| = 1$. Note that this is not the case with the functions $\widetilde{\phi}(\theta)$ and $\phi(\theta)$ in (3.11.97) and (3.11.96). For example, note that $\widetilde{\phi}(0) = ie^{-i0} = i$ and $\widetilde{\phi}(2\pi) = ie^{-i\pi} = -i$. In the case when $C(\xi)$ is not continuous, the function $C(e^{i\theta})$ and the determinant D_N are said to have a *singularity*.

Though conjectured in the past by Fisher and Hartwig [365], the asymptotic behavior of determinants with singularities has been established only recently in Böttcher and Silbermann [176], Ehrhardt and Silbermann [347], Deift, Its, and Krasovsky [298]. We state next a corollary of Theorem 2.5 in Ehrhardt and Silbermann [347] which will be sufficient for our purposes.[21] Suppose the function $C(e^{i\theta})$ can be expressed as

$$C(e^{i\theta}) = b(e^{i\theta})t_\beta(e^{i\theta}), \tag{3.11.102}$$

where

$$t_\beta(e^{i\theta}) = e^{i\beta(\theta-\pi)}, \quad \beta \in \mathbb{R}.$$

Note that the function $t_\beta(e^{i\theta})$ satisfies $t_\beta(e^{i0}) = e^{-i\beta\pi}$, $t_\beta(e^{i2\pi}) = e^{i\beta\pi}$, and hence is a discontinuous function on the unit circle (unless β is an integer). The function $b(e^{i\theta})$ will be assumed to be continuous on the unit circle. We will also need the Wiener-Hopf factorization of b, namely,

$$b(e^{i\theta}) = b_+(e^{i\theta})g(b)b_-(e^{i\theta}),$$

where the factors are defined as

$$b_\pm(e^{i\theta}) = \exp\left\{\sum_{m=1}^{\infty} e^{\pm i\theta m}(\log b)_{\pm m}\right\}, \quad g(b) = \exp((\log b)_0) \tag{3.11.103}$$

and where $(\log b)_m$ denotes the mth Fourier coefficient of the function $\log b$.

Theorem 3.11.7 *(Strong Szegö limit theorem for determinants with singularity.) Suppose that the function $C(e^{i\theta})$ can be expressed as (3.11.102) where $b(e^{i\theta})$ is infinitely differentiable on the unit circle. Then, the determinant D_n in (3.11.100) satisfies, as $n \to \infty$,*

$$D_n \sim E\, g(b)^n n^{-\beta^2}, \tag{3.11.104}$$

where $g(b)$ is defined in (3.11.103), and

$$E = E(b)b_+(1)^\beta b_-(1)^{-\beta} G(1+\beta)G(1-\beta). \tag{3.11.105}$$

[21] The result we use follows the statement of Theorem 2.5 in Ehrhardt and Silbermann [347].

In (3.11.105), $b_+(1)$, $b_-(1)$ are defined in (3.11.103),

$$E(b) = \exp\Big(\sum_{m=1}^{\infty} m(\log b)_m(\log b)_{-m}\Big)$$

and

$$G(1+z) = (2\pi)^{z/2}e^{-(z+1)z/2-\gamma z^2/2}\prod_{k=1}^{\infty}\Big\{\Big(1+\frac{z}{k}\Big)^k e^{-z+z^2/2k}\Big\} \tag{3.11.106}$$

stands for the so-called Barnes G–function (γ is Euler's constant).

3.11.5 Long-Range Dependence at Critical Temperature

Applying Theorem 3.11.7 to the determinants S_n and $\widetilde{S}_n$ in (3.11.92) and (3.11.93) with the respective functions $\phi(\theta)$ and $\widetilde{\phi}(\theta)$ in (3.11.59) and (3.11.94) at the critical temperature yields the following result which concludes the proof of Theorem 3.11.1.

Theorem 3.11.8 *Let S_n and $\widetilde{S}_n$ be the determinants in (3.11.92) and (3.11.93). At the critical temperature (3.11.95), as $n \to \infty$,*

$$S_n = \lim_{L_h,L_v\to\infty} \mathbb{E}\sigma_{0,0}\sigma_{0,n} \sim \Big(\frac{1+\alpha_1}{1-\alpha_1}\Big)^{1/4} A n^{-1/4}, \tag{3.11.107}$$

$$\widetilde{S}_n = \lim_{L_h,L_v\to\infty} \mathbb{E}\sigma_{0,0}\sigma_{n,n} \sim A n^{-1/4}, \tag{3.11.108}$$

where α_1 is defined in (3.11.57),

$$A = G(1/2)G(3/2) = 0.6450024\ldots$$

and G is the Barnes G–function in (3.11.106).

Proof Consider first the determinant S_n with the function $\phi(\theta)$ in (3.11.96). We will apply Theorem 3.11.7. Note that $\phi(\theta)$ can be written as in (3.11.102),

$$\phi(\theta) = b(e^{i\theta})t_\beta(e^{i\theta}),$$

where

$$b(e^{i\theta}) = \Big(\frac{1-\alpha_1 e^{i\theta}}{1-\alpha_1 e^{-i\theta}}\Big)^{1/2}, \quad t_\beta(e^{i\theta}) = e^{i\beta(\theta-\pi)}, \quad \beta = -1/2. \tag{3.11.109}$$

The factors (3.11.103) of b in its Wiener–Hopf factorization are

$$b_+(e^{i\theta}) = (1-\alpha_1 e^{i\theta})^{1/2}, \quad b_-(e^{i\theta}) = \frac{1}{(1-\alpha_1 e^{-i\theta})^{1/2}}, \quad g(b) = 1. \tag{3.11.110}$$

In particular,

$$b_+(1) = (1-\alpha_1)^{1/2}, \quad b_-(1) = 1/(1-\alpha_1)^{1/2}$$

and hence $b_+(1)^{-1/2}b_-(1)^{1/2} = (1-\alpha_1)^{-1/2}$. For $b(e^{i\theta})$ in (3.11.109), we have

$$\log b(e^{i\theta}) = \frac{1}{2}\log(1-\alpha_1 e^{i\theta}) - \frac{1}{2}\log(1-\alpha_1 e^{-i\theta}) = -\frac{1}{2}\sum_{k=1}^{\infty}\frac{\alpha_1^k e^{i\theta k}}{k} + \frac{1}{2}\sum_{k=1}^{\infty}\frac{\alpha_1^k e^{-i\theta k}}{k}$$

and hence the Fourier coefficients of the function $\log b$ are given by

$$(\log b)_m = \frac{1}{2\pi}\int_0^{2\pi} d\theta e^{-im\theta}\log b(e^{i\theta}) = \begin{cases} -\frac{\alpha_1^m}{2m}, & \text{if } m > 0, \\ 0, & \text{if } m = 0, \\ \frac{\alpha_1^{|m|}}{2|m|}, & \text{if } m < 0, \end{cases} \tag{3.11.111}$$

so that

$$\begin{aligned} E(b) &= \exp\Big(\sum_{m=1}^{\infty} m(\log b)_m(\log b)_{-m}\Big) = \exp\Big(-\frac{1}{4}\sum_{m=1}^{\infty}\frac{(\alpha_1^2)^m}{m}\Big) \\ &= \exp\Big(\frac{1}{4}\log(1-\alpha_1^2)\Big) = (1-\alpha_1^2)^{1/4}. \end{aligned} \tag{3.11.112}$$

In view of (3.11.112), (3.11.105) and (3.11.110), we deduce (3.11.107). The result (3.11.108) for the determinant $\widetilde{S}_N$ can be obtained similarly. □

Remark 3.11.9 The asymptotic behavior (3.11.108) of $\widetilde{S}_n$ was obtained in McCoy and Wu [703] directly without using the strong Szegö limit theorem as follows. In view of (3.11.97), the elements $\widetilde{a}_n$ of $\widetilde{S}_n$ can be evaluated as

$$\widetilde{a}_m = \frac{1}{2\pi}\int_0^{2\pi} d\theta\, e^{-im\theta} i e^{-i\theta/2} = \frac{2}{\pi(2m+1)}.$$

Hence, by (3.11.93),

$$\widetilde{S}_n = (2\pi^{-1})^n \det\left(\frac{1}{2m_1 - 2m_2 + 1}\right)_{0\le m_1, m_2 \le n-1}. \tag{3.11.113}$$

The latter determinant is the special case of the so-called Cauchy determinant,

$$\widetilde{D}_n = \det\left(\frac{1}{\mu_{m_1} + \nu_{m_2}}\right)_{0\le m_1, m_2\le n-1}, \tag{3.11.114}$$

where $\mu_{m_1} = 2m_1 + 1$ and $\nu_{m_2} = -2m_2$. It is known (for example, McCoy and Wu [703], theorem on p. 261) that

$$\widetilde{D}_n = \frac{\prod_{0\le m_1 < m_2 \le n-1}(\mu_{m_1} - \mu_{m_2})(\nu_{m_1} - \nu_{m_2})}{\prod_{m_1=0}^{n-1}\prod_{m_2=0}^{n-1}(\mu_{m_1} + \nu_{m_2})}. \tag{3.11.115}$$

Using this fact, (3.11.113) becomes

$$\begin{aligned} \widetilde{S}_n &= (2\pi^{-1})^n \frac{\prod_{0\le m_1<m_2\le n-1}(2(m_1 - m_2))(2(m_2 - m_1))}{\prod_{m_1=0}^{n-1}\prod_{m_2=0}^{n-1}(2(m_1 - m_2) + 1)} \\ &= (2\pi^{-1})^n \prod_{0\le m_1<m_2\le n-1} \frac{-4(m_1 - m_2)^2}{1 - 4(m_1 - m_2)^2} \end{aligned}$$

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$$= \left(\frac{2}{\pi}\right)^n \prod_{0 \le m_1 < m_2 \le n-1} \left(1 - \frac{1}{4(m_1 - m_2)^2}\right)^{-1}$$

$$= \begin{cases} \frac{2}{\pi}, & n = 1, \\ \left(\frac{2}{\pi}\right)^n \prod_{l=1}^{n-1} \left(1 - \frac{1}{4l^2}\right)^{l-n}, & n \ge 2. \end{cases}$$

Now use the identity (Gradshteyn and Ryzhik [421], p. 45, Formula 1.431.1)

$$\frac{\sin(\delta\pi)}{\delta\pi} = \prod_{l=1}^{\infty} \left(1 - \frac{\delta^2}{l^2}\right)$$

to write

$$\frac{2}{\pi} = \prod_{l=1}^{\infty} \left(1 - \frac{1}{4l^2}\right).$$

We get that

$$\begin{aligned}
\log \widetilde{S}_n &= n \sum_{l=1}^{\infty} l \log\left(1 - \frac{1}{4l^2}\right) + \sum_{l=1}^{n-1} (l-n) \log\left(1 - \frac{1}{4l^2}\right) \\
&= \sum_{l=1}^{n-1} l \log\left(1 - \frac{1}{4l^2}\right) + n \sum_{l=n}^{\infty} \log\left(1 - \frac{1}{4l^2}\right) \\
&= -\sum_{l=1}^{n-1} \frac{1}{4l} + \sum_{l=1}^{n-1} l\left(\log\left(1 - \frac{1}{4l^2}\right) + \frac{1}{4l^2}\right) + n \sum_{l=n}^{\infty} \log\left(1 - \frac{1}{4l^2}\right) \\
&= -\frac{1}{4}\log(n-1) - \frac{1}{4}\left(\sum_{l=1}^{n-1} \frac{1}{l} - \log(n-1)\right) \\
&\quad + \sum_{l=1}^{n-1} l\left(\log\left(1 - \frac{1}{4l^2}\right) + \frac{1}{4l^2}\right) + n \sum_{l=n}^{\infty} \log\left(1 - \frac{1}{4l^2}\right) \\
&= -\frac{1}{4}\log(n-1) - \frac{\gamma}{4} + \overline{A} - \frac{n}{4}\int_n^{\infty} \frac{dl}{l^2} + o(1) \\
&= -\frac{1}{4}\log(n-1) - \frac{\gamma}{4} + \overline{A} - \frac{1}{4} + o(1), \qquad (3.11.116)
\end{aligned}$$

where

$$\gamma = \lim_{n\to\infty} \left(\sum_{l=1}^{n} l^{-1} - \log n\right) = 0.5772157\ldots$$

is Euler's constant, and

$$\overline{A} = \sum_{l=1}^{\infty} l\left(\log\left(1 - \frac{1}{4l^2}\right) + \frac{1}{4l^2}\right).$$

Thus, as $n \to \infty$,

$$\widetilde{S}_n \sim A n^{-1/4}, \qquad (3.11.117)$$

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where the constant A is numerically about $A = 0.6450024\ldots$ as stated in Theorem 3.11.8.

Proof of Corollary 3.11.2 Suppose $E_h = E_v = E$ and set $c = \cosh \beta E$, $s = \sinh \beta E$, $c_2 = \cosh 2\beta E$ and $s_2 = \sinh 2\beta E$. We have

$$\frac{1+\alpha_1}{1-\alpha_1} = \frac{c^2 - s^2 + 2sc}{c^2 + s^2} = \frac{1+s_2}{\sqrt{1+s_2^2}},$$

where the last equality follows from the relations $c^2 - s^2 = 1$, $2cs = s_2$ and $c^2 + s^2 = c_2 = \sqrt{1+s_2^2}$. But (3.11.10) yields $s_2 = 1$, and therefore

$$\frac{1+\alpha_1}{1-\alpha_1} = \sqrt{2}. \qquad \square$$

3.12 Stochastic Heat Equation

The classical heat equation

$$\frac{\partial u}{\partial t} = \frac{1}{2}\frac{\partial^2 u}{\partial x^2}, \quad u(0,x) = u_0(x), \tag{3.12.1}$$

has the solution $u(t,x)$, $t \geq 0$, $x \in \mathbb{R}$, given by

$$u(t,x) = \frac{1}{\sqrt{2\pi t}} \int_{\mathbb{R}} u_0(y) e^{-\frac{(x-y)^2}{2t}} dy =: (P_t u_0)(x). \tag{3.12.2}$$

When the forcing term $f = f(t,x)$ is added to (3.12.1) as

$$\frac{\partial u}{\partial t} = \frac{1}{2}\frac{\partial^2 u}{\partial x^2} + f, \quad u(0,x) = u_0(x), \tag{3.12.3}$$

the solution can be expressed as

$$\begin{aligned} u(t,x) &= (P_t u_0)(x) + \int_0^t (P_{t-s} f(s,\cdot))(x) ds \\ &= \frac{1}{\sqrt{2\pi t}} \int_{\mathbb{R}} u_0(y) e^{-\frac{(x-y)^2}{2t}} dy + \int_0^t \frac{1}{\sqrt{2\pi(t-s)}} \int_{\mathbb{R}} e^{-\frac{(x-y)^2}{2(t-s)}} f(s,y) dy ds \end{aligned} \tag{3.12.4}$$

for a large class of functions (distributions) f.

In a *stochastic heat equation*, the forcing f is taken as a space-time white noise. The equation is written as

$$\frac{\partial u}{\partial t} = \frac{1}{2}\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 W}{\partial t \partial x} \quad \text{or} \quad \frac{\partial u}{\partial t} = \frac{1}{2}\frac{\partial^2 u}{\partial x^2} + \dot{W}, \tag{3.12.5}$$

where $W = \{W(t,x), t \geq 0, x \in \mathbb{R}\}$ is a two-parameter Brownian motion; that is, W is a zero mean Gaussian process with covariance

$$\mathbb{E} W(t,x) W(s,y) = (t \wedge s) \begin{cases} (|x| \wedge |y|), & \text{if } \operatorname{sign}(x) = \operatorname{sign}(y), \\ 0, & \text{if } \operatorname{sign}(x) \neq \operatorname{sign}(y), \end{cases} \tag{3.12.6}$$

since $W(t,x)$ is independent of $W(s,y)$ when x and y have different signs. Stochastic heat equation is one of the basic and canonical models of stochastic partial differential equations.

Assuming $u_0(y) \equiv 0$ for simplicity and by using (3.12.4), the solution to the stochastic heat equation (3.12.5) is given by

$$u(t,x) = \int_0^t \int_{\mathbb{R}} \frac{1}{\sqrt{2\pi(t-s)}} e^{-\frac{(x-y)^2}{2(t-s)}} W(ds,dy), \tag{3.12.7}$$

where W is a Gaussian random measure on $(0,\infty)\times\mathbb{R}$ with the control measure $dsdy$. Since the integrand is square integrable, the process $u(t,x)$ is a well-defined, mean zero Gaussian process.

Observe that, for fixed $x \in \mathbb{R}$, the covariance function of $u(t,x)$ can be computed as

$$\begin{aligned}
\mathbb{E}u(t_1,x)u(t_2,x) &= \frac{1}{2\pi}\int_0^{t_1\wedge t_2}\int_{\mathbb{R}} \frac{1}{\sqrt{(t_1-s)(t_2-s)}} e^{-\frac{(x-y)^2}{2(t_1-s)}-\frac{(x-y)^2}{2(t_2-s)}} dsdy \\
&= \frac{1}{2\pi}\int_0^{t_1\wedge t_2}\int_{\mathbb{R}} \frac{1}{\sqrt{(t_1-s)(t_2-s)}} e^{-\frac{(x-y)^2}{2}(\frac{1}{t_1-s}+\frac{1}{t_2-s})} dsdy \\
&= \frac{1}{\sqrt{2\pi}}\int_0^{t_1\wedge t_2} (t_1+t_2-2s)^{-1/2} ds \\
&= \frac{1}{\sqrt{2\pi}}\Big((t_1+t_2)^{1/2} - (t_1+t_2-2(t_1\wedge t_2))^{1/2}\Big) \\
&= \frac{1}{\sqrt{2\pi}}\Big((t_1+t_2)^{1/2} - |t_1-t_2|^{1/2}\Big).
\end{aligned} \tag{3.12.8}$$

Thus, for fixed $x \in \mathbb{R}$, the process $\{u(t,x)\}_{t\geq 0}$ is a bifractional Brownian motion with indices $H = 1/2$ and $K = 1/2$ introduced in Section 2.6.2.

3.13 The Weierstrass Function Connection

The complex-valued Weierstrass function

$$W^{(0)}(t) = \sum_{n=0}^{\infty} a^n e^{ib^n t}, \tag{3.13.1}$$

where $t \in \mathbb{R}, 0 < a < 1$ and $ab \geq 1$, is a well-known example of a continuous but nowhere differentiable function. It can be modified and randomized to FBM. When b is an odd integer and $ab > 1 + (3\pi/2)$, the proof of non-differentiability of $W^{(0)}(t)$ was first given by Weierstrass [1000] (or Weierstrass [1001]). It was later generalized to the condition $ab \geq 1$ by Hardy [454]. To see its relation with FBM, set

$$a = r^{-H}, \quad b = r$$

and note that

$$0 < a < 1, \quad ab > 1 \quad \Leftrightarrow \quad r > 1, \quad 0 < H < 1.$$

We then obtain the function

$$W^{(1)}(t) = \sum_{n=0}^{\infty} r^{-Hn} e^{ir^n t},$$

which was originally discussed by Mandelbrot [679] (pp. 388–390). He noted that its frequency spectrum (r^{-Hn} at frequency r) and its cumulative energy spectrum (see below) are similar to those of FBM with index H.

It is convenient, in this context, to represent FBM as

$$B_H(t) = c\,\Re \int_0^\infty (e^{itx} - 1)x^{-H-1/2}(dB^1(x) + i dB^2(x)), \tag{3.13.2}$$

where B^1 and B^2 are two independent standard Brownian motions and c is some constant. The representation is equivalent to that when replacing $dB^1(x)$ and $dB^2(x)$ above by independent Gaussian random measures $B^1(dx)$ and $B^2(dx)$ on $\mathbb{R}$ having control measures dx. But working with Brownian motions will be more advantageous in some of our arguments.

It is can be checked that (3.13.2) defines an H–SSSI Gaussian process and hence is a representation of FBM (Corollary 2.6.3). The function $(x^{-H-1/2})^2 = x^{-2H-1}$ is viewed as its frequency spectrum, and its cumulative spectrum at frequencies $x \geq a$ is $\int_a^\infty x^{-2H-1}dx = c\,a^{-2H}$, where c is some other constant.

There are three major differences, however, between the Weierstrass function and FBM:

- the spectrum of the Weierstrass function is located at frequencies $\{r^n\}_{n\geq 0}$, whereas the spectrum of FBM is supported on $(0, \infty)$,
- the spectrum of the Weierstrass function is discrete, whereas that of FBM is continuous,
- the Weierstrass function is deterministic whereas FBM is a random process.

To deal with the first difference Mandelbrot extended the lower frequency bound $f = r^0 = 1$ to $f = 0$ (the case of FBM) by introducing the modified Weierstrass function

$$W^{(2)}(t) = \sum_{n=-\infty}^{\infty} (e^{ir^n t} - 1)r^{-Hn}, \tag{3.13.3}$$

which converges for $0 < H < 1$. Replacing $e^{ir^n t}$ by $e^{ir^n t} - 1$ is necessary to ensure convergence as $n \to -\infty$; that is, at frequencies close to the origin, and hence to prevent the infrared (low-frequency) divergence.

Observe also the similarities between (3.13.3) and (3.13.2). $W^{(2)}$ is still nowhere differentiable, has a discrete frequency spectrum located at $\{r^n\}_{n\in\mathbb{Z}}$ and, also, is self-similar with index H, i.e., $W^{(2)}(ct) = c^H W^{(2)}(t)$, for any $t \in \mathbb{R}$, but self-similarity is restricted to factors $c = r^m$, $m \in \mathbb{Z}$. Notice that, when r tends to 1, these factors become more clustered on $\mathbb{R}_+$. Were the function $W^{(2)}(t)$ random and r close to 1, one would expect it to approximate FBM (Mandelbrot [679]).

Randomization is achieved by multiplying each term of the series $W^{(2)}(t)$ by complex random variable $\xi_n + i\eta_n$ where $\{\xi_n\}_{n\in\mathbb{Z}}$ and $\{\eta_n\}_{n\in\mathbb{Z}}$ are two independent sequences of i.i.d. real-valued random variables; that is, by considering

$$W_r^{(3)}(t) = \sum_{n=-\infty}^{\infty} (e^{ir^n t} - 1)r^{-Hn}(\xi_n + i\eta_n), \tag{3.13.4}$$

which we call the *Weierstrass–Mandelbrot process*. We suppose that ξ_n, η_n have finite variance.

Our objective is to study the convergence of the Weierstrass–Mandelbrot process $W^{(3)}$, suitably normalized as r tends to 1. To gain some insight on the choice of normalization, consider the variance of the Weierstrass–Mandelbrot process $W^{(3)}$. Observe that if $\mathbb{E}\xi_n^2 = \mathbb{E}\eta_n^2 = 1$, then

$$\begin{aligned}
\mathbb{E}|\Re W_r^{(3)}(t+s) - \Re W_r^{(3)}(s)|^2 &= \frac{1}{2}\mathbb{E}|W_r^{(3)}(t+s) - W_r^{(3)}(s)|^2 \\
&= 2\sum_{n=-\infty}^{\infty}(1-\cos r^n t)r^{-2Hn} = \frac{2}{r-1}\cdot\sum_{n=-\infty}^{\infty}(1-\cos r^n t)(r^n)^{-2H-1}(r^{n+1}-r^n) \\
&= \frac{2}{r-1}\cdot\int_0^\infty \sum_{n=-\infty}^{\infty}(1-\cos r^n t)(r^n)^{-2H-1}\, 1_{[r^n,r^{n+1})}(u)\, du.
\end{aligned}$$

Since, for fixed $t \geq 0$, as $r \to 1$,

$$\sum_{n=-\infty}^{\infty}(1-\cos r^n t)(r^n)^{-2H-1}\, 1_{[r^n,r^{n+1})}(u) \to (1-\cos ut)u^{-2H-1}\ ,\ \forall u > 0,$$

it follows from the dominated convergence theorem that, as $r \to 1$,

$$(r-1)\sum_{n=-\infty}^{\infty}(1-\cos r^n t)r^{-2Hn} \to \int_0^\infty (1-\cos ut)u^{-2H-1}du = t^{2H}\int_0^\infty \frac{(1-\cos u)}{u^{2H+1}}du.$$

The correct normalization is therefore $r-1$, or equivalently, $\log r$, since $\frac{\log r}{r-1} \to 1$, as $r \to 1$.

Another way of looking at the normalizer $r-1$ (or $\log r$) is as follows. Consider the modified Weierstrass function $W^{(2)}$ given in (3.13.3). The cumulative energy, i.e., the amplitude (coefficient of $(e^{ir^n t}-1)$) squared, in the frequencies $f \geq r^m$, for some $m \in \mathbb{Z}$, is given by

$$r^{-2Hm} + r^{-2H(m+1)} + \cdots = \frac{r^{-2Hm}}{1-r^{-2H}},$$

which explodes as $r \to 1$. The necessary correction $1 - r^{-2H}$ behaves (up to a constant) like $1-r$ in the limit.

We are then led to consider

$$W_r(t) = (\log r)^{1/2}W_r^{(3)}(t) = (\log r)^{1/2}\sum_{n=-\infty}^{\infty}(e^{ir^n t}-1)r^{-Hn}(\xi_n + i\eta_n). \tag{3.13.5}$$

When studying the limit behavior of (3.13.5), we want to be able to apply the central limit theorem which involves a normalization of the form $a^{1/2}$, $a \to \infty$. To make this normalization appear, express r^n as $e^{n\log r}$ and set $\log r = 1/a$. We then get

$$\begin{aligned}
W_r(t) &= (\log r)^{1/2}\sum_{n=-\infty}^{\infty}(e^{ie^{n\log r}t}-1)e^{-Hn\log r}(\xi_n + i\eta_n) \\
&= \sum_{n=-\infty}^{\infty}(e^{ie^{n/a}t}-1)e^{-Hn/a}\frac{1}{a^{1/2}}(\xi_n + i\eta_n) = \sum_{n=-\infty}^{\infty} f_t\Big(\frac{n}{a}\Big)\frac{1}{a^{1/2}}(\xi_n + i\eta_n),
\end{aligned} \tag{3.13.6}$$

where

$$f_t(u) = (e^{ie^u t} - 1)e^{-Hu}, u \in \mathbb{R}. \tag{3.13.7}$$

Notice that as r tends to 1, a tends to infinity.

As $a \to \infty$, we expect (3.13.6) to converge to $\int_{\mathbb{R}} f_t(u)(dB^1(u) + idB^2(u))$, where B^1 and B^2 are Brownian motions. The following theorem provides sufficient conditions on the sequence $\{(\xi_n, \eta_n)\}_{n\in\mathbb{Z}}$ and on the functions f_t to imply the required convergence. Set

$$f_{t,a}(u) = \sum_{n=-\infty}^{\infty} f_t\left(\frac{n}{a}\right) 1_{[\frac{n}{a}, \frac{n+1}{a})}(u).$$

Theorem 3.13.1 *Suppose that $\{\xi_n, \eta_n, n \geq 1\}$ are independent random variables with $\mathbb{E}\xi_n^2 = \mathbb{E}\eta_n^2 = 1$, $\mathbb{E}\xi_n = \mathbb{E}\eta_n = 0$. Let $f_t : \mathbb{R} \to \mathbb{C}$ be functions such that $f_t, f_{t,a} \in L^2(\mathbb{R})$ and that*

$$\|f_t - f_{t,a}\|_{L^2(\mathbb{R})} \to 0, \quad \text{as } a \to \infty. \tag{3.13.8}$$

Then, $\sum_{n=-\infty}^{\infty} f_t(n/a)(\xi_n + i\eta_n)$ is well-defined in the $L^2(\Omega)$–sense and, as $a \to \infty$,

$$\sum_{n=-\infty}^{\infty} f_t\left(\frac{n}{a}\right) \frac{1}{a^{1/2}}(\xi_n + i\eta_n) \xrightarrow{fdd} \int_{\mathbb{R}} f_t(u)(dB^1(u) + idB^2(u)), \tag{3.13.9}$$

where $\{B^1(u)\}_{u\in\mathbb{R}}$ and $\{B^2(u)\}_{u\in\mathbb{R}}$ are two independent standard Brownian motions and $\xrightarrow{fdd}$ means the convergence in the sense of the finite-dimensional distributions.

Proof It is enough to prove the result for fixed t and *real-valued* f. For the simplicity of notation, we drop the index t from all functions f. The assumption $f_a \in L^2(\mathbb{R})$ implies that $\|f_a\|^2_{L^2(\mathbb{R})} = \sum_{n=-\infty}^{\infty} f(n/a)^2/a < \infty$. Hence, for example for $k_2 > k_1 \geq 1$,

$$\mathbb{E}\Big|\sum_{n=k_1+1}^{k_2} f\left(\frac{n}{a}\right)(\xi_n + i\eta_n)\Big|^2 = 2\sum_{n=k_1+1}^{k_2} f\left(\frac{n}{a}\right)^2 \to 0,$$

as $k_1, k_2 \to \infty$. This shows that $\sum_{n=-\infty}^{\infty} f_t(n/a)(\xi_n + i\eta_n)$ is well-defined in the $L^2(\Omega)$–sense. Set

$$X_a = \sum_{n=-\infty}^{\infty} f\left(\frac{n}{a}\right) \frac{1}{a^{1/2}}(\xi_n + i\eta_n), \quad X = \int_{\mathbb{R}} f(u)(dB^1(u) + idB^2(u)).$$

Since $f \in L^2(\mathbb{R})$, there is a sequence of elementary functions[22] f^j such that $\|f - f^j\|_{L^2(\mathbb{R})} \to 0$, as $j \to \infty$. Set also

$$X_a^j = \sum_{n=-\infty}^{\infty} f^j\left(\frac{n}{a}\right) \frac{1}{a^{1/2}}(\xi_n + i\eta_n), \quad X^j = \int_{\mathbb{R}} f^j(u)(dB^1(u) + idB^2(u)).$$

By Theorem 3.1 in Billingsley [154], the series X_a converges in distribution to X if

[22] A function f is elementary if it has a form $f(u) = \sum_{k=1}^{n} f_k 1_{[a_k,b_k)}(u)$.

Step 1: $X^j \xrightarrow{d} X$, as $j \to \infty$,

Step 2: for all $j \geq 1$, $X_a^j \xrightarrow{d} X_j$, as $a \to \infty$,

Step 3: $\limsup_j \limsup_a \mathbb{E}|X_a^j - X_a|^2 = 0$.

Step 1 follows since X^j and X are integrals with respect to Brownian motion and $\|f - f^j\|_{L^2(\mathbb{R})} \to 0$, as $j \to \infty$. For Step 2, observe that

$$X_a^j = \int_{\mathbb{R}} f^j(u) dB_a(u),$$

where

$$B_a(u) = \begin{cases} \frac{1}{a^{1/2}} \sum_{j=1}^{[au]} (\xi_j + i\eta_j), & u \geq 0, \\ -\frac{1}{a^{1/2}} \sum_{j=[au]+1}^{0} (\xi_j + i\eta_j), & u < 0. \end{cases}$$

By the assumptions on ξ_n, η_n and the central limit theorem, B_a converges to $B^1 + iB^2$ in the sense of finite-dimensional distributions. Since f^j is an elementary function, the integral X^j depends on the process B_a through a finite number of time points only. It then converges in distribution to X^j, as $a \to \infty$. Finally, for Step 3, observe that $\mathbb{E}|X_a^j - X_a|^2 = 2\|f_a^j - f_a\|_{L^2(\mathbb{R})}^2$, where

$$f_a^j(u) = \sum_{n=-\infty}^{\infty} f^j\left(\frac{n}{a}\right) 1_{[\frac{n}{a}, \frac{n+1}{a})}(u).$$

For fixed j, by using the dominated convergence theorem and the assumption (3.13.8), we have $\|f_a^j - f_a\|_{L^2(\mathbb{R})}^2 \to \|f^j - f\|_{L^2(\mathbb{R})}^2$ as $a \to \infty$. Then, $\limsup_a \mathbb{E}|X_a^j - X_a|^2 = 2\|f^j - f\|_{L^2(\mathbb{R})}^2$, which tends to 0 as $j \to \infty$. □

The function f_t in (3.13.7) characterizing the Weierstrass–Mandelbrot process satisfies the assumptions of Theorem 3.13.1. Indeed, $f_t \in L^2(\mathbb{R})$ since f_t is continuous and $|f_t(u)| \leq Ce^{-Hu}$, $|f_t(u)| \leq Ce^{(1-H)u}$. These bounds also imply that $f_{t,a} \in L^2(\mathbb{R})$. Finally, $\|f_t - f_{t,a}\|_{L^2(\mathbb{R})} \to 0$ by using the dominated convergence theorem. Consequently, applying Theorem 3.13.1 to the Weierstrass–Mandelbrot process, we get the following result.

Theorem 3.13.2 *As r tends to* 1, *the normalized Weierstrass–Mandelbrot process (3.13.6) converges in the sense of finite-dimensional distributions to the complex-valued fractional Brownian motion*

$$X_H(t) = \int_0^{\infty} (e^{ixt} - 1) x^{-H-\frac{1}{2}} (dB^1(x) + i dB^2(x)). \tag{3.13.10}$$

The limit of the normalized Weierstrass–Mandelbrot process (3.13.6) given in relation (3.13.9) of Theorem 3.13.1 is, in fact,

$$X'_H(t) = \int_{\mathbb{R}} (e^{ie^u t} - 1) e^{-Hu} (dB^1(u) + i dB^2(u)).$$

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The processes $X'_H(t)$ and $X_H(t)$, however, have identical finite-dimensional distributions because they are both Gaussian, have mean zero and same covariance functions. One can also use the fact that, if $B(u)$, $u \in \mathbb{R}$, is a standard real-valued Brownian motion, then so is $\mathcal{B}(x) = \int_{-\infty}^{\log x} e^{u/2} dB(u)$, $x \geq 0$, and then make a change of variables. Observe finally that $\{\Re X_H(t)\}_{t\in\mathbb{R}}$ and $\{\Im X_H(t)\}_{t\in\mathbb{R}}$ are both fractional Brownian motions, but they are not independent.

3.14 Exercises

The symbols * and ** next to some exercises are explained in Section 2.12.

Exercise 3.1 Show that FARIMA$(0, d, 0)$ series can be viewed as an aggregated series with the mixing density (3.1.14); that is, its spectral density $f_X(\lambda)$ in (3.1.13) satisfies the relation (3.1.8) with $F_a(dx)$ having the density $f_a(x)$ given by (3.1.14). *Hint:* Use the identity $|1 - xe^{-i\lambda}|^2 = (1-x)^2(1+2x)(1-\cos\lambda)/(1-x)^2)$ and a change of variables $y^{1/d} = 2x/(1-x)^2$. See also Celov et al. [208], Proposition 5.1.

Exercise 3.2 Similarly to Proposition 3.1.1 suppose that the distribution $F_a(dx)$ has a density

$$f_a(x) = (1+x)^{1-2d} l(1+x)\phi(x)1_{(-1,0]}(x),$$

where $0 < d < 1/2$, $l(y)$ is a slowly varying function at $y = 0$ and $\phi(x)$ is a bounded function on $(-1, 0]$ and continuous at $x = -1$ with $\phi(-1) \neq 0$. Show that, as $\lambda \to \pi$,

$$f_X(\lambda) \sim |\lambda - \pi|^{-2d} l(\lambda - \pi) \frac{\sigma_\epsilon^2 \phi(-1)}{2\pi} \int_0^\infty \frac{s^{1-2d}}{1+s^2} ds,$$

where $f_X(\lambda)$ appears in (3.1.4) and (3.1.8).

Exercise 3.3** Exercise 3.1 provides an aggregation result for FARIMA$(0, d, 0)$ series. What about fractional Gaussian noise (FGN)? Can FGN be viewed as an aggregated series? A question related to this is whether fractional Brownian motion, which is not stationary but has stationary increments, can be viewed as an aggregated process in a suitable sense. The suitable sense here could mean aggregation of the integrals of continuous-time Ornstein–Uhlenbeck (OU) processes with random coefficients, since OU processes can be thought as continuous-time analogues of AR(1) series.

Exercise 3.4 Prove Theorem 3.2.2. *Hint:* Use $\gamma_\epsilon(h) = (2p-1)^h$ and follow the proof of Theorem 3.2.3.

Exercise 3.5 (i) Show that in the case $H = 1/2$, the limit in Theorem 3.2.3 is Brownian motion but the normalization N^H should be replaced by $\sqrt{N \log N}$. (ii) Suppose that μ is the uniform probability on $[1/2, 1]$. In Theorem 3.2.3, let $c_H = 1/\sqrt{2}$ and replace N^H by $\sqrt{N \log N}$. Show that the limit in Theorem 3.2.3 is then Brownian motion. (iii) Show that if one reverses the order of the limits in Theorem 3.2.3, then the first limit as $N \to \infty$ would yield 0, since the correct normalization would be $\sqrt{N}$.

Exercise 3.6 Prove the representation (3.3.19)–(3.3.20) for the workload process W_λ^*.

Exercise 3.7 Prove that the integrand $I_\lambda(s, u, r)$ is dominated by an integrable function in the proof of Theorem 3.3.12.

Exercise 3.8 Show that Proposition 3.4.4 and Theorem 3.4.5 hold when the function g in Definition 3.4.2 is such that $g(y) = 0$, $y < 0$, and $g(y) = (1 + y)^{-\beta} L_g(y)$, where L_g is a slowly varying function at infinity which is positive, and locally bounded away from 0 and infinity in $[0, \infty)$. That is, the shots $g(y)$ need not diverge like a power function $y^{-\beta}$ around $y = 0$.

Exercise 3.9* For the infinite source Poisson model in Section 3.3, various regimes of $\lambda \to \infty$ were considered and led to different limiting processes. Are there regimes of $\lambda \to \infty$ in the power-law shot noise model which lead to other limiting processes than fractional Brownian motion?

Exercise 3.10 Prove the results (3.6.9), (3.6.10), (3.6.11) and (3.6.27) for the Markov chain in the proof of Proposition 3.6.2.

Exercise 3.11 Consider a stationary Gaussian series $X_n^N = \phi_N X_{n-1}^N + \epsilon_n$, $n = 1, \ldots, N$, with i.i.d. $\mathcal{N}(0, \sigma_\epsilon^2)$ variables ϵ_n and $\phi_N = 1 - cN^{-d}$, where $0 < c < 1$ and $0 < d < 1/2$. As for the Markov switching model (see (3.6.31) and Proposition 3.6.2) show that

$$\mathrm{Var}(X_1^N + \cdots + X_N^N) \sim CN^{2d+1}, \qquad \frac{1}{N^{d+1/2}} \sum_{n=1}^{[Nt]} X_n^N \xrightarrow{fdd} B(t),$$

where B is Brownian motion. Other similar models can be found in Diebold and Inoue [312]. *Hint:* Use the arguments in the proof of Proposition 3.6.2 and the fact that $\mathbb{E} X_h^N X_0^N = \sigma_\epsilon^2 (\phi_N)^h / (1 - (\phi_N)^2)$.

Exercise 3.12** The aggregation result in Proposition 3.1.1, Theorem 3.2.3 on correlated random walks and Theorem 3.3.8, (a), for the infinite source Poisson model are associated with LRD series and FBM with the self-similarity parameter $H > 1/2$. How can these models be modified, especially according to some common mechanism, to be associated with antipersistent series and FBM with the self-similarity parameter $H < 1/2$? *Hint:* For the correlated random walks, a modification is suggested in Enriquez [351].

Exercise 3.13 Prove the convergence of the finite-dimensional distributions in (3.7.10).

Exercise 3.14 Prove the first equality in (3.7.11). *Hint:* Suppose $0 \le s < t$. Then $n(t) - n(s) = N(A^+(s,t)) - N(A^-(s,t))$, where

$$A^+(s,t) = \{(x, \xi) : x + \xi(s) \le 0, x + \xi(t) > 0\},$$
$$A^-(s,t) = \{(x, \xi) : x + \xi(s) > 0, x + \xi(t) \le 0\}.$$

Since $A^+(s,t) \cap A^-(s,t) = \emptyset$, $N(A^+(s,t))$ and $N(A^-(s,t))$ are independent Poisson with mean

$$\int \int_{\mathbb{R}} 1_{\{-\xi(t)<x\leq -\xi(s)\}} \rho dx d\mathbb{P}_\xi = \rho \mathbb{E}(\xi(t) - \xi(s))_+ = \frac{\rho}{2}\mathbb{E}|\xi(t) - \xi(s)|.$$

Exercise 3.15 Prove the relation (3.8.4) for the simple symmetric exclusion model. *Hint:* Argue as in (3.7.5)–(3.7.8).

Exercise 3.16 Prove the relation (3.8.10) for the simple symmetric exclusion model.

Exercise 3.17 Prove that the second term in (3.8.16) converges to 0 as $t \to \infty$.

Exercise 3.18 Prove the relation (3.10.9) for the local time.

Exercise 3.19* Following Section 3.11.1, consider the Ising model in one dimension having the Boltzmann (Gibbs) distribution $e^{-\beta\mathcal{E}}/Z_{L_h}$, where

$$\mathcal{E} = - \sum_{k=-L_h+1}^{L_h} E^h \sigma_k \sigma_{k+1}.$$

Show that, in the thermodynamic limit,

$$\lim_{L_h \to \infty} \mathbb{E}\sigma_0\sigma_n = (\tanh(\beta E^h))^n, \quad n \geq 0.$$

Thus, in one dimension, the Ising model exhibits short-range dependence for all inverse temperatures $\beta > 0$. *Hint:* This is a classical result for one-dimensional Ising model whose proof can be found in many sources, e.g., McCoy and Wu [703], Chapter III.

3.15 Bibliographical Notes

Section 3.1: Analysis and statistical inference for AR(1) series with random coefficients go back at least to Robinson [851]. Aggregation of short-range dependent series as one explanation of long-range dependence was put forward by Granger [422]. In particular, Granger [422] focused on the mixing density (3.1.11) and LRD was shown through the argument (3.1.12). The ideas of Granger were studied and extended by many authors. More thorough mathematical analysis of aggregation can be found in Gonçalves and Gouriéroux [417], Lippi and Zaffaroni [637], Zaffaroni [1025], Oppenheim and Viano [780], Dacunha-Castelle and Fermín [275]. This includes aggregation of series other than AR(1), and considering other densities than (3.1.11). Disaggregation problem is discussed in Dacunha-Castelle and Oppenheim [276], Dacunha-Castelle and Fermín [274], Celov et al. [208]. Estimation of the underlying mixing density from an aggregated series is considered in Leipus, Oppenheim, Philippe, and Viano [605], Beran, Schützner, and Ghosh [126]. Long-range dependence in volatility models is explained through aggregation in Zaffaroni [1026]. The superposition of Ornstein–Uhlenbeck-type processes, which can be viewed as the analogues of AR(1) series in the continuous time, was considered by Barndorff-Nielsen and Leonenko [101]. Some other works related to aggregation are Chambers [212], Leipus and Surgailis [604], Davidson and Sibbertsen [289], Leonenko and Taufer [611], Leipus, Philippe, Puplinskaitė, and Surgailis [606], Candelpergher, Miniconi, and Pelgrin [200].

Aggregation of short-range dependent series has been geared mostly towards applications in economics. In this regard, it should be noted that a more general model than (3.1.1) is often considered, given by

$$Y_n^{(j)} = a^{(j)} Y_{n-1}^{(j)} + Z_n + \epsilon_n^{(j)}, \quad n \in \mathbb{Z}, \tag{3.15.1}$$

where the series $\{Z_n\}_{n\in\mathbb{Z}}$ is common across all $j \geq 1$ and is referred to as a macroeconomic shock. See, for example, Lewbel [621], Altissimo, Mojon, and Zaffaroni [23], Mayoral [700] to name but a few.

Section 3.2: Theorem 3.2.3 is due to Enriquez [351]. He also shows that if $M(N)$ is made to be a function of N, then the convergence to fractional Brownian motion holds as $N \to \infty$ as long as $M(N)/N^{1-2H}$ tends to ∞. Enriquez considered not only the case $H > 1/2$ but also the case $H < 1/2$. To ensure that the sum of the covariances $\sum_{n=-\infty}^{\infty} r(n)$ equals 0 when $H < 1/2$, Enriquez introduces an alternating correlated random walk, which also moves by jumps of $+1$ and -1, but where the probability of making the same jump alternates between p and 0. The account here is partly based on a presentation of this material by Shuyang Bai.

Section 3.3: A number of other models related to the infinite source Poisson model with heavy tails have been considered in the literature including the renewal-reward processes and ON/OFF models (Mandelbrot [676], Levy and Taqqu [617], Taqqu and Levy [947], Pipiras, Taqqu, and Levy [822], Gaigalas and Kaj [380], Kaj [545], Leland, Taqqu, Willinger, and Wilson [607], Taqqu, Willinger, and Sherman [952], Mikosch, Resnick, Rootzén, and Stegeman [719], Abry, Borgnat, Ricciato, Scherrer, and Veitch [12], Dombry and Kaj [323]). The asymptotic results for these related models run in parallel to those for the infinite source Poisson model. But these results in many cases preceded and inspired the work on the infinite source Poisson model, especially the analysis of the ON/OFF model with heavy tails in Mandelbrot [676] who pioneered the use of self-similar and long-range dependent models in Economics and with Leland et al. [607] which pioneered the use of self-similar and long-range dependent models in telecommunication networks.

The presentation of this section follows Kaj and Taqqu [546]. This is just one work in the otherwise vast literature on the infinite source Poisson model and its variants. The notable works include Cioczek-Georges and Mandelbrot [244], Kurtz Kurtz [585], Heath, Resnick, and Samorodnitsky [462], Konstantopoulos and Lin [576], Mikosch et al. [719], Rosenkrantz and Horowitz [862], Barakat, Thiran, Iannaccone, Diot, and Owezarski [94], Guerin et al. [438],Çağlar [196]. Variants and extensions of the model consider random transmission rates (Maulik, Resnick, and Rootzén [699], D'Auria and Resnick [288]), cluster point processes (Hohn, Veitch, and Abry [479], Faÿ, González-Arévalo, Mikosch, and Samorodnitsky [356], Fasen [354]), random fields (Kaj, Leskelä, Norros, and Schmidt [547], Biermé and Estrade [148], Kuronen and Leskelä [584]), limit theorems under control (Budhiraja, Pipiras, and Song [193]) and quite general formulations (Mikosch and Samorodnitsky [717]). Moment measures of the models are studied in Dombry and Kaj [324], Antunes, Pipiras, Abry, and Veitch [39]. This is also not to mention numerous other papers in the networking literature, where long-range dependent models are used – several books in this area are Park and Willinger [794], Sheluhin, Smolskiy, and Osin [903].

Long-range dependence aside, Araman and Glynn [41] consider related models leading to FBM with the self-similarity parameter $H < 1/2$.

Section 3.4: Power-law shot noise models go back to at least Lowen and Teich [647] (see also Chapter 9 in their monograph [648]), who also refer to earlier work by Schönfeld [891], Van der Ziel [970] for special cases related to noise in electronic devices (where, as a matter of fact, shot noise models have originated). Lowen and Teich call them fractal shot noise models and also do not consider limit laws. The presentation here is somewhat inspired by the works of Klüppelberg and Mikosch [569], Klüppelberg, Mikosch, and Schärf [570], Klüppelberg and Kühn [568], where some financial applications are considered. See also Lane [591]. A field version of the model was considered by Baccelli and Biswas [66].

Section 3.5: The presentation of this section follows Cox [265], who also refers to his earlier work in Cox and Townsend [266]. More recent related discussions can be found in Huh, Kim, Kim, and Suh [500], Militkỳ and Ibrahim [720].

Section 3.6: The material of this section is based on Diebold and Inoue [312], Baek et al. [71]. From a related practical perspective, LRD series are well known to be easily confused with series exhibiting several changes in local mean level. Indeed, apparent changes in the local mean level across a range of larger scales is a characteristic feature of LRD series. Conversely, a series composed of several changes in mean superimposed by a SRD series will exhibit LRD when using common estimators of the LRD parameter. This confusion between LRD and nonstationary-like models, such as the model involving changes in mean, has been documented well and is raised in almost all applications of LRD series. See, for example, Klemeš [563], Boes and Salas [165] in hydrology, Roughan and Veitch [869], Veres and Boda [989] in teletraffic, Diebold and Inoue [312], Granger and Hyung [423], Mikosch and Stărică [718], Smith [913] in economics and finance, Mills [721] in climatology.

A number of statistical procedures have also emerged aiming at distinguishing LRD and non-stationary models. Jach and Kokoszka [526] showed that wavelet-based, maximum likelihood tests for SRD are robust to the presence of nonstationarities. Ohanissian, Russell, and Tsay [778] used temporal aggregation to test against nonstationary models. Iacone [513], McCloskey and Perron [701] employ common estimators of the LRD parameter in the Fourier domain by selecting carefully the range of frequencies to be used in estimation. Tests for distinguishing LRD and changes in mean models are proposed in Berkes, Horváth, Kokoszka, and Shao [130], Yau and Davis [1023], Baek and Pipiras [69]. See also Qu [837].

Section 3.7: We have described the history of the elastic collision problem. Harris [455] started the investigation by looking at colliding Brownian motions, followed by Spitzer [919] who considered Newtonian particles and then Gisselquist [407] who looked at infinite variance stable particles. According to Gisselquist [407], page 235, it is Frank Spitzer, who identified in a private communication the limiting covariance in the case considered by Harris, showing that it is the covariance of FBM with $H = 1/4$. The paper of Dürr et al. [339] also deals with weak convergence. Our account is partly based on a presentation of this material by Shuyang Bai.

Section 3.8: The results presented in the section concern the simple symmetric nearest-neighbor exclusion. What happens with the motion of the tagged particle for other cases of the transition probabilities $p(x, y)$ and the dimension $d \geq 1$ of the state space $\mathbb{Z}^d$? The answer here is almost complete. Other symmetric exclusion processes on $\mathbb{Z}^d$, $d \geq 1$, were studied in Kipnis and Varadhan [560]. The asymmetric but centered exclusion processes on $\mathbb{Z}^d$, $d \geq 1$, were considered in Varadhan [973]. The asymmetric, non-centered, nearest-neighbor exclusion processes on $\mathbb{Z}$ were studied by Kipnis [559], and on $\mathbb{Z}^d$, $d \geq 3$, by Sethuraman, Varadhan, and Yau [900]. The general case of asymmetric, non-centered exclusion processes on $\mathbb{Z}$ and $\mathbb{Z}^2$ is not fully resolved, with some preliminary work reported in Sethuraman [899]. While the limit was FBM in Section 3.8, all the works referenced above involve the usual BM in the limit.

All the works discussed above concern the so-called equilibrium case (that is, when the initial configuration is in equilibrium). For related work on nonequilibrium dynamics, see Jara [530], Jara and Landim [532], Jara, Landim, and Sethuraman [533].

Renormalization and the macroscopic structure of other interacting particle systems, e.g., the voter model, were studied by Bramson and Griffeath [182], Holley and Stroock [481], among others. See also the monograph by Liggett [632].

Section 3.9: The presentation of this section follows Hammond and Sheffield [449]. The model is used in the so-called seed bank application in Blath, González Casanova, Kurt, and Spanò [162]. A fields version of the model is considered in Biermé, Durieu, and Wang [153], and an extension of the model that yields fractional Brownian motion with the self-similarity parameter $H \in (0, 1/2)$ can be found in Durieu and Wang [338].

Section 3.10: What happens in Theorem 3.10.1 if $0 < \alpha < 1$? The random walk S_N is transient in this case and Kesten and Spitzer [553] show that $N^{-1/\beta} \sum_{j=1}^{N} \xi(S_j)$ merely converges to a Lévy stable motion of index β. This makes also physical sense since the transience of S_N ensures that it stays a finite amount of time in any neighborhood instead of returning to it again and again.

Cohen and Samorodnitsky [254] expand the class of limits in (3.10.10) by taking $l_t(x)$ to be the local time of fractional Brownian motion $B_H(t)$, $0 < H < 1$. See also Jung and Markowsky [539]. As noted in the section, an even more general setting of random walks in random scenery is considered by Dombry and Guillotin-Plantard [322], allowing one to analyze a random walk and a random scenery separately. See also the references in Dombry and Guillotin-Plantard [322] for other related work. In connection to superdiffusions, similar models have been considered in Physics literature, for example, in Bouchaud, Georges, Koplik, Provata, and Redner [178].

A self-similar process arising from a random walk with random environment in random scenery is studied in Franke and Saigo [370].

Section 3.11: The Ising model was first studied in the 1920s by Ernst Ising in his Ph.D. thesis, supervised by Wilhelm Lenz.[23] Ising considered the model in one dimension and found that it does not exhibit a phase transition. Although Ising conjectured that no phase transition occurs in higher dimension, further studies pointed to the contrary, culminating in the seminal work of Onsager [779] who provided an explicit calculation for the so-called *free energy* of the two-dimensional Ising model. The Ising model has since played a special role

[23] The model is sometimes referred to as the Lenz–Ising model.

in Statistical Mechanics and other areas, as a source of new ideas and as one of few realistic and exactly solvable models. A nice account of the historical developments concerning the Ising model can be found in Niss [752, 753].

New research on the Ising model – as well as on other lattice and growth models – has recently explored their behavior when the edges of the lattice become infinitely small. These scaling limits are generally characterized by conformal invariance, and involve the Schramm–Loewner Evolution (SLE), the Conformal Loop Ensemble (CLE) and related probabilistic objects (see Schramm [892], Werner [1004], Lawler [598]). For example, the *interface* between the $+1$ and -1 spins in the Ising model at critical temperature has long been conjectured to converge to the SLE model with parameter $\kappa = 16/3$. This is studied rigorously by Smirnov [912].

Section 3.11 first appeared in Pipiras and Taqqu [821]; it appears here, with minor changes, with permission of the publisher. Recent studies concerning correlations, the use of dimers and large scales asymptotics in the Ising and related models include Sakai [876], Hara [453], Boutillier and de Tilière [180], Schrenk, Posé, Kranz, van Kessenich, Araújo, and Herrmann [893], Camia, Garban, and Newman [198, 199], Chen and Sakai [217].

Section 3.12: The stochastic heat equation considered in the section is one of the most basic stochastic partial differential equations (SPDEs). A number of extensions of the model have been considered. For example, the closest in spirit and its relation to self-similar processes is a stochastic heat equation with fractional white noise (in the space variable) – see a nice review paper by Balan [92] and references therein, as well as Torres, Tudor, and Viens [960], Tudor and Xiao [966] for more recent work. Other generalizations of the basic model involve a multiplicative white noise $\sigma(u)\dot{W}$ instead of $\dot{W}$ in (3.12.5). See, for example, Hairer [445], Borodin and Corwin [175], Conus, Joseph, Khoshnevisan, and Shiu [260], Hu, Huang, Nualart, and Tindel [494] for a snapshot of recent work in the linear case $\sigma(u) = u$ (also known as the parabolic Anderson model) and references therein. Yet more general SPDE models have been considered in this vast and still expanding area of research, with a large number of introductory textbooks and lecture notes available for interested readers.

Section 3.13: The Wierstrass function in the context of fractional Brownian motion was discussed originally in Mandelbrot [679], pp. 388–390. See also Berry and Lewis [135]. The material of this section is based on Pipiras and Taqqu [814], Szulga and Molz [941]. An extension to heavy-tailed and stable random processes can be found in Pipiras and Taqqu [816]. A multivariate extension of the Weierstrass–Mandelbrot function is studied in Ausloos and Berman [56].

Other notes: We have not touched upon several other topics related to physical models for long-range dependence and self-similarity. These include:

- $1/f$ noise and its origins: see, for example, Ward and Greenwood [998] and references therein.
- Anomalous diffusion, especially when related to stable laws: see, for example, Klafler and Sokolov [561]], Bouchaud and Georges [177], Uchaikin and Zolotarev [967]. Related models known as continuous time random walks were studied by Becker-Kern, Meerschaert, and Scheffler [112], Meerschaert and Scheffler [711], Meerschaert, Nane, and

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Xiao [713], Jara and Komorowski [531], Magdziarz, Weron, Burnecki, and Klafter [671], Chen, Xu, and Zhu [226]. See also the notes at the end of Section 7.4.
- Particle system perspective of many self-similar processes in Bojdecki, Gorostiza, and Talarczyk [170, 171], Bojdecki and Talarczyk [169], Bojdecki, Gorostiza, and Talarczyk [172].
- Constructions involving the so-called β–regular Markov chains in Owada and Samorodnitsky [784], Jung, Owada, and Samorodnitsky [541].
- Asymptotics in multinomial occupancy schemes: see Gnedin, Hansen, and Pitman [410].
- Other miscellaneous constructions in, for example, Koutsoyiannis [578], Alvarez-Lacalle, Dorow, Eckmann, and Moses [24], Chevillon and Mavroeidis [233].

The collection of articles in Rangarajan and Ding [841] provides a number of other examples and models in physical sciences where long-range dependence is observed.